A Real-Time Global Scenario System for Policy Analysis: An EM–Bayesian GVAR with an Endogenous United States

(Working draft)

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Abstract

Policy analysts often need to answer a simple question, and answer it quickly: if one thing changes, what happens to the rest of the world? We build a tool that answers it. The user fixes a few variables—a policy rate, a credit spread, a path for the budget—and the tool reports how growth, inflation, interest rates, and public debt respond across 50 economies. It runs in a web browser and updates in a fraction of a second, so the user can try one scenario after another during a meeting.

Behind the tool is a single model: a set of country vector autoregressions, estimated with Bayesian shrinkage and tied together by trade. Three choices make it work in practice. First, we fill in the missing data—short samples and the pandemic quarters—*inside* the estimation, rather than cleaning the data beforehand. Second, we let the United States respond to the rest of the world, instead of holding it fixed as an anchor the way the standard global VAR does. Third, we use one simple fact: once the model is estimated, a scenario is a linear problem. That lets us compute the answer to any scenario once and reuse it, which is why the tool is fast.

We ask the model two questions. Does it forecast well? In a real-time contest over roughly 2013–2024 against eight competitors, it does not win on point accuracy—a factor model and a simple autoregression do—but on growth, inflation, and interest rates it belongs to the set of best models by a formal test (the Model Confidence Set), and it significantly beats, on growth and inflation, the textbook global VAR it generalizes. Its probability bands come out too narrow, which we report openly. What does it add as a scenario tool? A US interest-rate scenario spreads to every economy in a single solve, and because the US now responds to the rest of the world, the model captures a feedback onto the US that the fixed-anchor version cannot. The contribution is the whole system: one model that both forecasts and supports transparent, fast, multi-country scenario analysis.

1 Introduction

Suppose the Federal Reserve is about to raise interest rates. An analyst needs to know what this does to growth and inflation in the euro area, in Mexico, in the rest of the world—and what it does back to the United States. Or suppose corporate credit spreads jump, or a government loosens its budget. Questions like these come up every day in central banks, finance ministries, and international organizations. Answering one well needs a model with four properties at once: it is global, so it covers many countries together; it is disciplined, so it is small enough to estimate reliably; it is honest about messy data; and it is fast enough to re-run while the discussion is still going. No standard tool has all four.

This paper builds and documents a tool that does. The user fixes a few variables—say, a path for the US policy rate—and the tool returns the response of every variable in every one of

50 economies, consistent with the estimated model and with how the economies are linked. It runs in an ordinary web browser and re-solves in well under a second, so the user can explore one scenario after another. The tool is live and in use; this paper explains what is inside it and shows that it works.

Inside the tool is a single model. Its starting point is standard: a vector autoregression for each economy, estimated with Bayesian shrinkage (Litterman, 1986; Giannone et al., 2019), with the economies linked through trade-weighted averages of their partners' variables, as in the global VAR literature (Pesaran et al., 2004; Dees et al., 2007). Three choices make it work in practice. First, we fill in the missing data—short emerging-market samples and the pandemic quarters—as part of the estimation, using a Kalman smoother, rather than cleaning the data beforehand (Durbin and Koopman, 2012; Lenza and Primiceri, 2022). Second, we let the United States respond to the rest of the world, instead of holding it fixed as an anchor, which is how classical global VARs treat the dominant economy. Third, we add a small fiscal block for six large economies, so a scenario carries its public-debt implications automatically.

The step that turns this model into an interactive tool is a piece of algebra. Once the model is estimated, imposing a scenario is a linear problem: the response of the whole system is a fixed, known function of the numbers the user pins down. We solve that linear problem once, offline, and store the answer, so that during use each new scenario is a single matrix multiplication. This is why a 50-economy model can re-solve in a browser in a fraction of a second.

We then ask whether the model is any good, and we are strict about it. In a real-time forecasting contest over roughly 2013–2024 against eight competitors—from a random walk up to the textbook global VAR—the model does not win on point accuracy: a factor model forecasts growth best, and a simple autoregression forecasts inflation and interest rates best. But by a formal test of forecast accuracy it is statistically among the best models on growth, inflation, and interest rates, and it significantly beats the textbook global VAR it generalizes on growth and inflation, so the extra machinery earns its place. One weakness is visible and we report it plainly: the model's probability bands come out too narrow.

Finally we show what the tool adds. A US interest-rate scenario spreads coherently to every economy in a single solve, and the tool shows the debt consequences next to the growth-and-inflation response. Because we let the United States respond to the rest of the world, the model also captures a feedback from abroad back onto the US that a fixed-anchor global VAR cannot produce; we measure its size, while being clear that this is a scenario projection, not a causal impulse response.

The contribution is the system as a whole. None of the individual ingredients is new; carrying all of them in one estimated model that both forecasts credibly and answers scenario questions in real time, from a single reproducible set of files, is. The rest of the paper follows the tool. Section 2 sketches the architecture; Section 3 explains how the model is estimated; Section 4 explains how it is solved and used in real time; Section 5 describes the data and is candid about its limits; Section 6 is the forecasting test; Section 7 is the scenario application; Section 8 discusses what the tool enables and where it falls short; Section 9 concludes.

2 The system at a glance

The tool has three parts (Figure 1): a set of country models, a layer that links them through trade, and a layer that solves and conditions the linked system in real time. The first two parts are about estimating the model; the third is what turns it into an interactive tool. The paper is organized the same way—Section 3 covers the first two parts, Section 4 the third—and this section gives the map.

The first part is a separate model for each economy. Each is a vector autoregression—a set of equations in which this quarter's variables depend on the last three quarters'—covering output, prices, the labour market, interest rates, credit, the exchange rate, and, for 42 economies, the

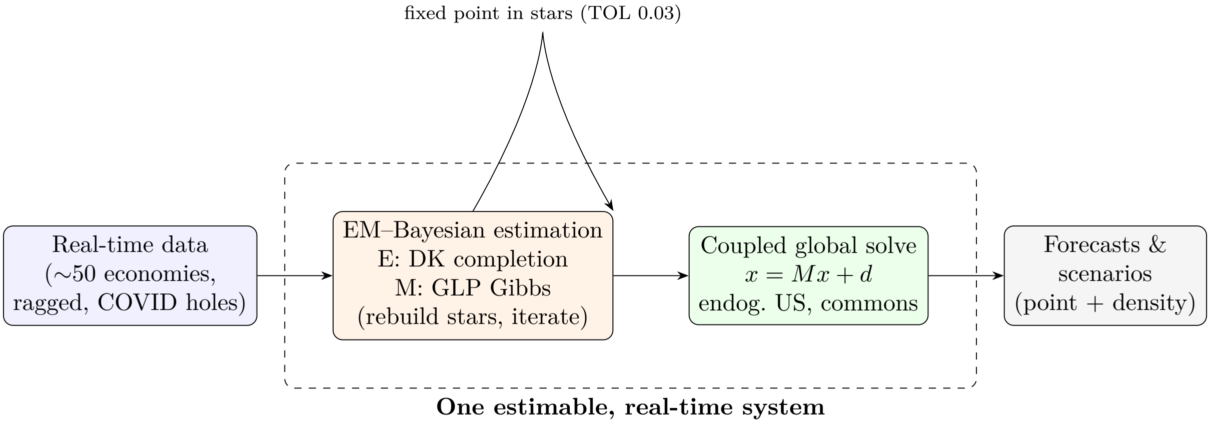

Figure 1: The three layers: real-time data → EM–Bayesian estimation (layers one and two, coupled and completed inside one fixed point) → the coupled global solve $x = Mx + d$ (layer three) → forecasts and scenarios. The dashed box is the single estimable object; the solve layer is what makes it an interactive tool.

public finances. The models range from 16 variables (a small emerging market) to 40 (the United States, which also carries the globally-traded prices and its own foreign-feedback terms); across the 50 economies there are 1,182 estimated series in all. Each model is small enough to estimate reliably yet detailed enough to answer a policy scenario.

The second part links the economies through trade. Each economy sees three summaries of the rest of the world—its trading partners' growth, inflation, and interest rate, each partner weighted by how much the economy trades with it—together with four globally-traded prices set mainly in US markets: oil, the US long-term interest rate, a US corporate borrowing spread, and US equity. Two features matter. Summarizing the rest of the world in three trade-weighted numbers, rather than tracking every foreign variable, is what keeps a 50-economy system estimable. And, unlike the standard global VAR, the United States is one of the linked economies rather than an outside anchor, so a shock abroad can feed back onto it.

The third part solves and conditions the linked system in real time. Because a scenario is a linear problem once the model is estimated (Section 4), the answer can be pre-computed once and reused, so any constraint the user imposes on any economy re-solves all 50 at once, instantly. This is the part that makes the model a tool rather than a batch forecaster.

We use the following notation throughout. There are 33 trade-linked economies plus 17 euro-area members estimated on their own (they take the euro-area interest rate and related series as external inputs); six economies (US, EA, JP, UK, CA, CN) carry a public-finance block. Each model uses three lags ($p = 3$) and is forecast 20 quarters ahead ($H = 20$). Stacked together, the linked system has about 2,060 dimensions and is stable, in the sense that shocks die out rather than build up (its largest eigenvalue is 0.947, below one). We write $y_{i,t}$ for economy i's own variables and $y^*_{i,t}$ for its three foreign summaries, and leave the details of each part to the sections that follow.

3 The EM–Bayesian GVAR

This section details the engine: how the fused object of layers one and two is estimated as a single fixed point. We build it up in the order the estimator runs—the per-bloc Bayesian VAR and its prior (§3.1), the trade-weighted coupling (§3.2), the EM data-completion that makes ragged and pandemic-scarred panels usable (§3.3), the endogenous dominant unit (§3.4), the fiscal block (§3.5)—and close with the fixed point that ties them together (§3.6).

3.1 Per-country Bayesian VARs and the GLP prior

Each economy is a reduced-form VAR whose coefficients are shrunk by a hierarchical Minnesota/GLP prior with data-driven tightness. Stacking economy i's domestic variables $y_{i,t}$ with its three trade-weighted foreign stars $y^*_{i,t}$ into $x_{i,t} = [\,y'_{i,t}\;\; y^{*\prime}_{i,t}\,]'$,

$$x_{i,t} = c_i + \sum_{\ell=1}^{p} A_{i,\ell}\, x_{i,t-\ell} + u_{i,t}, \qquad p = 3, \quad u_{i,t} \sim \mathcal{N}(0, \Sigma_i). \tag{1}$$

Variables enter in one of two spaces: levels in $100\log$ for real quantities and price indices (whose displayed growth is $4\,\Delta$), and raw levels for rates and ratios; the "factor of four, not four hundred" quarterly-annualised convention is fixed once and used everywhere.

A bloc with up to 40 variables and three lags has too many coefficients to estimate freely on a short sample, so we discipline them with a prior. The prior pulls each equation toward a simple benchmark—a random walk, in which each variable's best forecast is its current value—and lets the data pull away from that benchmark only where the evidence is strong. How hard it pulls is not chosen by hand but learned from the data: this is the hierarchical prior of Giannone et al. (2019) and Giannone et al. (2015), governed by two numbers (an overall tightness λ, prior mode 0.2, and a sum-of-coefficients weight μ, prior mode 1). This learned shrinkage is what makes even the short-sample emerging-market blocs estimable without dropping the cross-country links.

The hyperparameters are not fixed but sampled, which matters under missing data. Within each bloc's Gibbs sampler, (λ, μ) are updated by a random-walk Metropolis–Hastings step on the *completed-data* marginal likelihood, initialised at the numerical optimum and with a proposal covariance taken from the prior's inverse Hessian; the shrinkage therefore adapts to the imputed panel rather than to a truncated balanced window. Because the same sampler also draws the missing cells (§3.3), estimation of the prior, the coefficients, and the data completion all condition on one another.

A small but consequential prior choice concerns the fiscal ratios. Most variables carry the random-walk prior mean natural to macro levels, but the estimated fiscal ratios—revenue and primary expenditure as shares of GDP, and the stock–flow adjustment—carry a *white-noise* (stationary) prior mean of zero own-persistence, reflecting that a revenue-to-GDP ratio mean-reverts rather than drifts. The effective interest rate, by contrast, retains the random-walk prior. This keeps the fiscal drivers' long-run behaviour economically sensible and, as Section 3.5 shows, keeps the derived debt path well-behaved.

3.2 Cross-country coupling through trade-weighted stars

The link between economies runs through three "foreign" variables per economy, each built from its trading partners rather than taken from outside data. For economy i we form a foreign growth, a foreign inflation, and a foreign interest rate by averaging its partners' values, weighting each partner by how much i trades with it:

$$y^*_{i,t} \;=\; \sum_{j \neq i} w_{ij,t}\, z_{j,t}, \qquad w_{ij,t} = \frac{\bar{w}_{ij}\, \mathbf{1}\{j \in \mathcal{P}_{i,t}\}}{\sum_{k \neq i} \bar{w}_{ik}\, \mathbf{1}\{k \in \mathcal{P}_{i,t}\}}, \tag{2}$$

where $z_{j,t}$ is partner j's growth, inflation, or interest rate and $\bar{w}_{ij}$ is the trade weight. Following the global-VAR literature we call these three foreign averages the *stars*. Summarizing the entire rest of the world in three numbers per economy, rather than carrying every foreign variable, is the step that keeps a 50-economy model estimable and, later, quick to solve.

We recompute the weights each quarter over the partners that actually have data that quarter—the set $\mathcal{P}_{i,t}$ in (2)—which is what lets the model run on an unbalanced panel. If a partner's series is missing (a country not yet in the sample, or one with no comparable interest

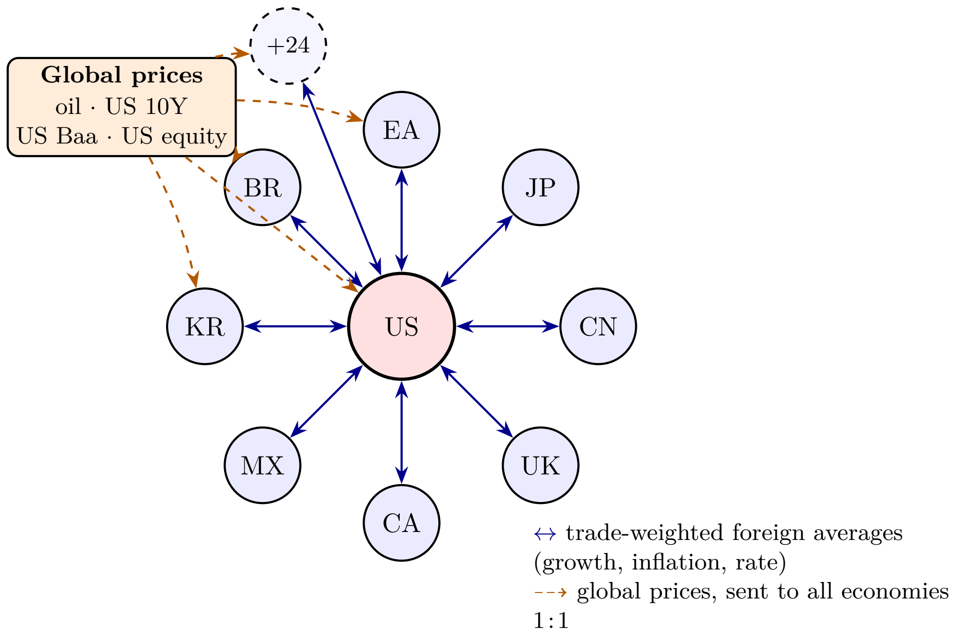

Figure 2: How the economies are linked. A central United States bloc is tied to the 32 other economies (eight labelled, plus 24 more) through two-way, trade-weighted "foreign average" links carrying each partner's growth, inflation, and interest rate; four globally-traded prices are broadcast to every economy at once. Crucially the United States is a full member of the system, not a fixed anchor—it both drives the rest of the world and responds to it. The resulting cross-country system is stable: the feedback across economies has spectral radius $\rho(M) = 0.947 < 1$ (against 0.70 for the fixed-anchor version), so the coupled solve of §4.1 has a unique answer.

rate) it simply drops out of that quarter's average for that variable. The growth and inflation averages can draw on a wider set of partners than the interest-rate average, which leaves out economies without a comparable rate or with runaway inflation that would swamp it.

Alongside these three, four globally-traded prices enter every economy in the same way: oil, the US ten-year interest rate, a US corporate borrowing spread, and US equity. For these a single world price, set mostly in US markets, is the natural object, so we carry one series each rather than a separate copy per economy. This saves parameters and imposes the sensible restriction that the oil price a scenario feeds into Germany is the same one it feeds into Chile.

Figure 2 is the picture to keep in mind. Every economy is a small model in its own right; the trade-weighted foreign averages (the two-way arrows) tie the economies together; and the four globally-traded prices (the dashed arrows) reach all of them at once. The one feature that sets this construction apart from the textbook global VAR is at the centre: the United States is a bloc like any other, both sending shocks out to the world *and* taking feedback back in, rather than sitting outside the system as a fixed anchor.

3.3 EM data-completion under ragged and pandemic missingness

Our data have two kinds of holes, and we fill both as part of the estimation rather than patching them beforehand. The first kind is ragged edges: economies enter the sample at different dates, so at any quarter some series are simply not yet available. The second is the 2020 pandemic, whose swings are so large that leaving them in would distort the estimated dynamics. The usual fixes—cut every series back to the shortest common sample, or delete the pandemic quarters—either throw away years of data or leave a hole that the cross-country links then spread everywhere. Instead we treat each hole as missing data and fill it with its best model-based estimate given everything else that is observed.

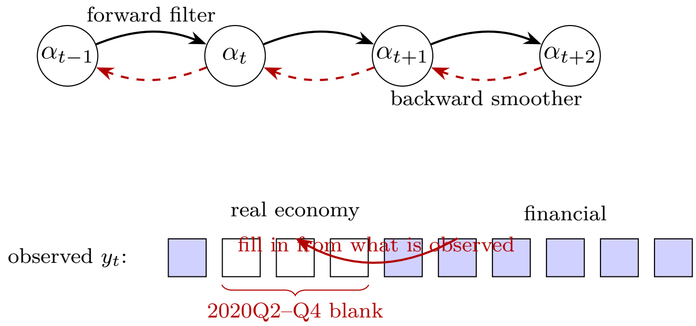

Figure 3: Filling in the missing data. The model is cast in state-space form; a Kalman filter runs forward (solid) and a smoother runs backward (dashed), so each estimate uses information from both directions. In the observation row, over the three pandemic quarters the real-economy cells (output, prices, and the real-economy foreign averages) are left blank while the financial cells (rates, spreads, equity) are kept, so the blanks are reconstructed from the financial signal that markets kept providing. The same procedure fills the future when a scenario is imposed (Section 4.2): a pinned value is an observed cell, a free value a blank.

For the pandemic we make one specific choice, and it does most of the work. Over the three quarters 2020Q2–Q4 we treat the real-economy variables—output, prices, and the two real-economy foreign averages—as missing, but keep the financial variables—interest rates, spreads, equity—as observed. The reason is that financial markets kept pricing through the pandemic and so carry genuine information about it. A Kalman smoother (Durbin and Koopman, 2012) then reconstructs the missing real-economy path from that financial signal, instead of mechanically rescaling the pandemic's unusual volatility (Lenza and Primiceri, 2022; Carriero et al., 2015). This reconstruction is the "fill-in" step of the estimation loop described in §3.6.

Figure 3 shows how the fill-in works. The model is written in state-space form, and a Kalman filter runs forward through time while a smoother runs backward, so every estimate of the missing cells uses information from both before and after. Over the three pandemic quarters the real-economy cells are left blank (hollow) and the financial cells are kept (filled); the smoother reads across the row and reconstructs the blanks from what it can see. The same machine does double duty: in Section 4.2 it is exactly what fills in the *future* when a user imposes a scenario—a pinned future value is just another observed cell, a free future value just another blank.

We fix the pandemic window in advance rather than letting the model find it. If we asked the estimator to locate each economy's "crisis" quarters on its own, it would often latch onto the wrong episode—a 2008 trough for one economy, a 2015 commodity bust for another—so we pin the window to the three quarters that are unambiguously the pandemic for every economy. The cost is a little rigidity for economies whose pandemic timing differed slightly; the benefit is that the treatment is transparent, the same everywhere, and does not quietly absorb ordinary business-cycle movements into the fill-in.

3.4 The endogenous dominant unit

The standard global VAR treats the United States as an outside anchor: it is estimated on its own, it feeds into every other economy, but nothing feeds back into it (Pesaran et al., 2004; Dees et al., 2007). So a shock that starts abroad, or a US shock that bounces back through trading partners, never reaches the US model. For a scenario tool this is a real limitation, because the feedback onto the United States is often exactly the question being asked.

We remove the limitation by giving the United States the same structure as everyone else: its own three foreign averages and its own row in the trade-weight matrix, so it both feeds and is fed by the rest of the world. This raises the US model from 37 variables to 40. The estimator starts it from a quick no-feedback fit and then re-estimates it with its foreign averages, exactly like the other economies, so the global channel closes back onto the economy that dominates it.

The payoff is precisely the channel a scenario tool needs. A US tightening now spreads abroad, is partly reflected back through trading partners' demand and financial conditions, and adjusts the US path in turn; and a shock that starts abroad can move the United States. Section 7 measures this channel directly, by running the same US-rate scenario with and without the feedback and taking the difference—which is, by construction, exactly what the anchor version leaves out.

3.5 The fiscal stock–flow block

Six economies carry a public-finance block, and here we split the work the way fiscal analysts do (Escolano, 2010; International Monetary Fund, 2013): the model forecasts the pieces of the budget, and the debt ratio then follows by arithmetic rather than from a separate regression. This keeps the debt path consistent with the rest of the forecast, instead of adding a debt equation that might disagree with it.

The block forecasts four things—government revenue and primary spending (both as shares of GDP), the average interest rate the government pays on its debt, and a residual adjustment—and then rolls the debt ratio forward by the standard identity,

$$d_t = d_{t-1}\,\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^{\text{nom}}/100} - \frac{pb_t}{4} + \frac{\text{sfa}_t}{4}, \tag{3}$$

where d_t is debt as a share of GDP, pb_t the primary balance (revenue minus primary spending), i_t^{eff} the average interest rate, g_t^{nom} nominal GDP growth, and the last term the residual adjustment (Bohn, 1998). In words: debt grows with the interest rate, shrinks with nominal growth, and is reduced by running a primary surplus. Interest is charged on last quarter's debt and the ratio is scaled by this quarter's growth, following the standard official definitions. Figure 4 traces the flow: the four forecast drivers on the left feed the pieces of the budget in the middle, which roll the debt ratio forward on the right, quarter after quarter.

Because the debt path is an explicit formula in these four pieces, we can also run it backwards. Given a target for debt, we solve (3) for the primary balance that would hit it and impose that as a constraint (§4.2). So the same block answers both the forward question—"given this fiscal stance, where does debt go?"—and the backward one—"what fiscal adjustment stabilises debt here?"

3.6 The estimable fixed point and convergence

The three pieces above—the country VARs, the trade links, and the data completion—are not estimated separately; they are solved together, by repeating a simple loop until it settles. There is a chicken-and-egg problem: to complete a country's missing data we need its partners' data, but those partners have missing data of their own. We break it by iterating. Start with a rough guess of the foreign averages; fill in each country's missing data given everything currently observed; rebuild the foreign averages from the filled-in data; re-estimate each country's VAR; and repeat until the foreign averages stop changing. This is exactly the expectation–maximization (EM) idea: the "missing" pieces are the foreign series that tie the countries together, the completion step is the E-step, and the re-estimation step is the M-step. Algorithm 1 states it precisely.

Algorithm 1: EM–Bayesian GVAR fixed-point estimation (`pipeline/export_bvar_gvar24_loop.m`)

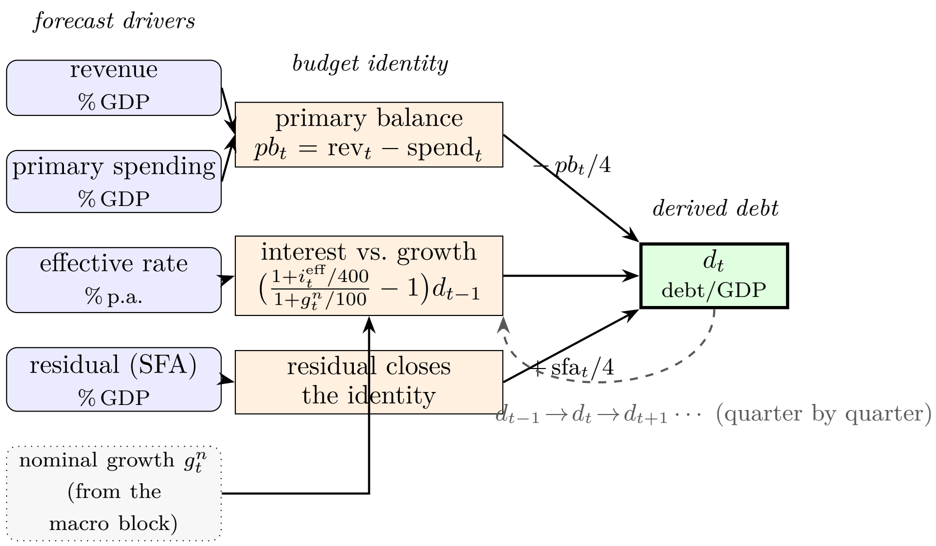

Figure 4: The public-finance block. The model forecasts the four drivers on the left; these feed the pieces of the government budget in the middle—revenue minus spending is the primary balance, the effective rate against nominal growth is the interest-versus-growth term, and the residual closes the identity—which roll the debt ratio d_t forward on the right by the formula (3), with last quarter's debt fed back each period (dashed). Debt is computed by arithmetic, never estimated.

1. **Bootstrap.** Form an initial guess of each foreign star from raw partner data; mask the COVID window 2020Q2–Q4, setting *macro only* to missing (domestic macro + ROWDEM + ROWINF) and keeping all financial columns and ROWFIN observed.
2. **E-step (DK completion).** For each bloc, run the Durbin–Koopman smoother inside the per-bloc sampler to impute the macro holes conditional on the observed financials, giving a completed panel $\widehat{X}_i = \mathbb{E}[X_i \mid \text{observed}]$.
3. **Rebuild stars.** Recompute $y^*_{i,t}$ from the completed partner panels via (2), renormalising weights over present partners.
4. **M-step (Bayesian estimation).** Re-estimate each bloc (1) by GLP Gibbs on $[\,\text{domestic} \mid 3 \text{ stars}\,]$, updating $(\widehat{A}_i, \widehat{\Sigma}_i)$ and the hyperparameters.
5. **Convergence.** Stop when $\max_i \|\Delta y^*_i\|_\infty < \text{TOL} = 0.03$ (evaluated from the second sweep and excluding the COVID-imputed cells, whose Monte-Carlo churn floors around 0.13); else return to Step 2, capped at MAXIT = 12.

Figure 5 draws the loop. A quick no-feedback fit of the United States seeds the first set of foreign averages; from there each pass rebuilds the foreign averages from partners' filled-in data (top), fills in the missing cells and re-estimates every country model (the E- and M-steps on the right and bottom), and checks whether the foreign averages have stopped moving (left). When they have, the system has found its joint solution and is exported.

The sweep is run as a Jacobi update, which is order-independent and reaches the same fixed point as a sequential Gauss–Seidel pass. All stars are built from the completions frozen at the start of a sweep, then every bloc is re-estimated in parallel; this parallelism is what keeps a 50-economy, 1000-draw estimation tractable. The convergence metric deliberately excludes the COVID-imputed star cells, whose quarter-to-quarter movement is imputation Monte-Carlo noise rather than genuine non-convergence, so the tolerance measures the systematic drift of the stars, not the sampler's variance.

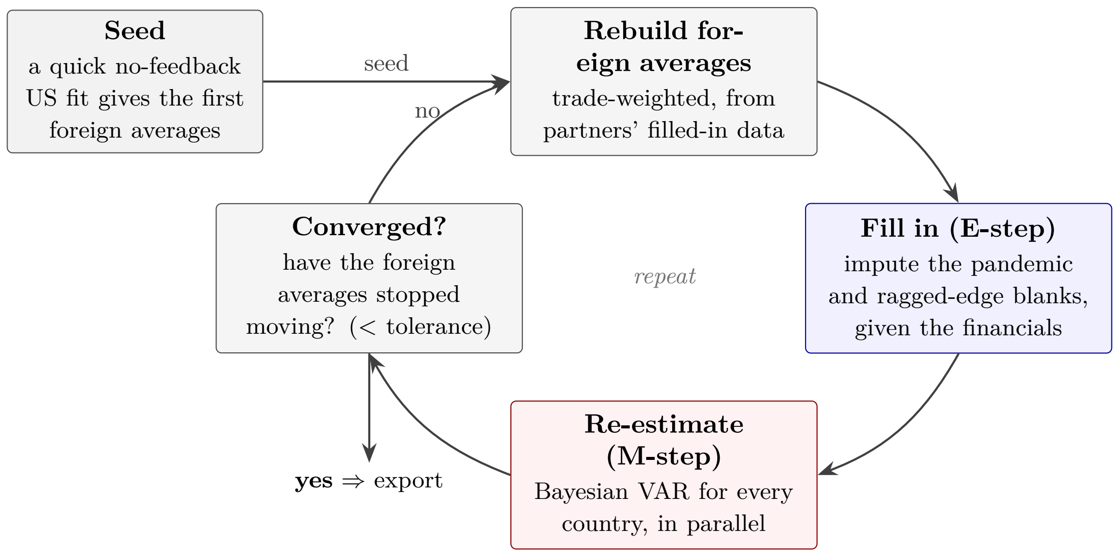

Figure 5: The estimation loop. A no-feedback fit of the United States seeds the first foreign averages. Each pass then rebuilds those averages from partners' filled-in data, fills in the missing cells and re-estimates every country's model (the fill-in and re-estimation are the two steps of one sampler), and checks whether the foreign averages have stopped changing. When they have, the joint solution is reached and exported; the countries are re-estimated in parallel, which keeps a 50-economy fit tractable.

The resulting system is stationary and, as Section 4 exploits, contractive. At the deployed configuration the coupled operator has spectral radius $\rho(M) = 0.947 < 1$, comfortably below the stability boundary, so the cross-country fixed point is unique and the forecasts do not explode at the $H = 20$ horizon. Setting the iteration cap to one recovers a single-country mode (no star re-convergence), which we use in Section 6 as an ablation that isolates exactly the value of the EM fixed point.

4 Real-time solution and the scenario engine

This section is the heart of the paper: it shows how the estimated model of Section 3 becomes an interactive tool. The idea is simple. A scenario is a linear problem, so it can be solved once and reused (§4.1); a scenario is then just a forecast that respects the pinned values, which a Kalman smoother produces (§4.2); users can also pin a variable's long-run value, and the forecast glides to it (§4.3); and because each solve is a single matrix multiplication, the whole thing runs in a browser in milliseconds (§4.4).

4.1 The affine fixed point and the pre-shipped inverse

Once the model is estimated, imposing a scenario is a linear problem, and this is what makes the tool fast. Hold the estimated coefficients fixed. Then each country's conditional forecast is a linear function of the values the user pins down: doubling the size of a pinned change doubles every response, and the *pattern* of responses is unchanged. Stacking all countries together, the scenario response x solves

$$x = Mx + d \qquad \Longrightarrow \qquad x = (I - M)^{-1}d, \tag{4}$$

where d collects the user's own pinned changes and M is the map that carries one country's movements into its partners through the trade links (`app/src/lib/compute/gvarSolve.ts`). The countries are tied together in *deviations* from the baseline, so M never has to be rebuilt from one scenario to the next.

The key point is that M depends only on the estimated model and on which variables are pinned—not on the numbers themselves. So the answer to *any* scenario that pins the same set of variables is $x = (I - M)^{-1}d$, and we need the inverse $(I - M)^{-1}$ only once. We compute it offline, store it, and ship it with the tool. This matters because the system is large: stacked across the 33 coupled economies (three lags, twenty quarters, plus the shared US-driven series) the problem has about 2,060 dimensions, and inverting a matrix that size inside a browser would take close to an hour. Doing it once, offline, turns each scenario into a single matrix multiplication that runs in milliseconds. A stored fingerprint of the model guards the file: if the estimated model ever changes, the fingerprint no longer matches and the tool falls back to a slower but always-correct step-by-step solve. The pre-computed inverse is therefore not a speed trick added on top of the tool—it is what makes an interactive global tool possible at all.

The scenario always has a unique, finite answer because the system is stable. The map M keeps all of its dynamics safely inside the unit circle: its largest eigenvalue is 0.947, comfortably below one, so the feedback across countries dies out rather than building up, and $(I - M)^{-1}$ is well defined.

4.2 Conditional forecasting via the smoother

A scenario is just a set of values the user fixes for the future, and the job is to fill in everything else consistently. We do this with a standard trick: treat the future the same way we treat missing past data (Durbin and Koopman, 2012; **?**). A value the user pins is like an observation we happen to have for a future quarter; a value left free is like a missing observation. Running the Kalman filter and smoother then produces the best estimate of the *whole* future path consistent with the pinned values, so fixing one variable moves all the others together rather than one at a time. As a check, a scenario that fixes nothing reproduces the plain forecast exactly.

The user can pin a value hard or soft, which is a way of saying how sure they are. A hard pin is imposed exactly; a soft pin is imposed only partially, so the model is allowed to disagree with it. Soft pins are how outside forecasts—a published inflation projection, a market-implied rate path—enter as *ingredients* rather than as facts: a “trust” dial runs from full trust (a hard pin) to no trust (no effect) (**?**). The user can also pin a *combination* of variables, which is what the fiscal block uses to hit a debt target by constraining revenue minus spending.

A scenario crosses borders through the same linear system of §4.1. Pinning a variable moves that country’s trade-weighted averages away from baseline; those movements enter its partners through M; each partner’s smoother passes them into its own variables; and the linear solve settles all of this at once. Because the whole thing is linear, changing the size of a pinned value re-solves all 50 economies instantly, with no filter to re-run—which is what lets a user drag a scenario and watch every country move in real time. For genuine policy counterfactuals the tool also offers a stricter mode that treats the pinned rate as a policy *shock* at the constrained quarters, in the spirit of **?** and **?**, and lets the remaining quarters follow the model’s own dynamics.

4.3 Long-run anchoring: a reactive steady-state prior

The user can also fix where a variable settles in the long run—say, inflation returning to two percent—and the forecast will glide there, without re-estimating anything. This is useful because the estimated dynamics are very persistent, so the far end of a forecast can drift to a level the user does not believe. The textbook remedy is a prior on the long-run mean (**?**), but that means re-estimating the model; a tool needs the same effect instantly, on top of the shipped model.

The idea is to nudge the forecast’s long-run drift by just enough to land on the chosen value. Writing the forecast as $Y_{T+h} = T^h Y_T + S_h\, c$, where $S_h = \sum_{k=0}^{h-1} T^k$ accumulates the model’s dynamics out to horizon h, a change δc in the constant c shifts the path by exactly $S_h\,\delta c$; we pick δc so the long-run forecast hits the target. We deliberately do not use the model's exact long-run mean $(I - \sum_\ell A_\ell)^{-1}c$, because with such persistent dynamics that number is unstable and unreliable; the finite-horizon quantity S_H is well behaved and does the job.

One detail makes the result look right. If we applied the whole correction from the first forecast quarter, the near term would jump—a growth variable could be pushed into a spurious recession on impact. Instead we phase the correction in gradually, at the speed the variable's own estimated dynamics converge: the fraction applied by horizon h is

$$w_j(h) = \frac{S_h[j,j]}{S_H[j,j]}, \qquad w_j(1) \approx 0, \quad w_j(H) = 1, \tag{5}$$

zero at first and one at the end. So the forecast starts where the data leave it and drifts smoothly to the chosen long-run value. The correction is added in the same way to the central forecast and to every uncertainty band, so the bands keep their width—the anchor moves the central view without pretending to add certainty.

4.4 From estimation apparatus to interactive tool

Because each new scenario is a single matrix multiplication, the whole 50-economy model runs inside the browser and answers in a fraction of a second. The files it needs—the estimated country models, the pre-computed inverse, and the response tables—are shipped with the web page; imposing or dragging a scenario triggers the multiplication and a quick pass of the smoother, with no re-estimation and no trip to a server. An analyst can therefore explore a scenario interactively, tightening or loosening a constraint and watching all 50 economies move.

Shipping the model this way also makes it reproducible. Every number the user sees comes from a shipped file and a documented calculation, and the files carry a fingerprint of the estimated model, so anyone can check that the tool is running the model this paper describes. The forecasting test of Section 6 and the scenarios of Section 7 use these same files, so the paper's evidence and the live tool are one and the same.

5 Data

The system runs on a harmonised quarterly panel of 50 economies and 1,182 series drawn from official statistical sources. The 33 GVAR-coupled blocs span the United States, the euro area as an aggregate, the other major advanced economies, and a broad set of emerging markets; a further 17 euro-area members are estimated as standalone models that carry the common euro-area instruments as external regressors. Appendix B lists the full roster with each bloc's variable count and sample.

The variable set per bloc spans a macro core, a financial/credit block, and—for 42 economies—a fiscal block. The macro core is real GDP and its expenditure components, price deflators and consumer prices, unemployment, a short rate, a long yield, and a nominal effective exchange rate; the financial block adds a corporate spread, equity, a house-price index, and non-financial-corporate credit; the fiscal block adds the four drivers of Section 3.5. Sources are the standard official providers—Eurostat, the OECD, the BIS, the Federal Reserve's FRED, the ECB, the IMF, and national government-finance statistics—documented per series in the shipped coverage manifest.

Estimation samples are genuinely ragged, and the system uses each series' true span rather than a truncated common window. Advanced economies reach back furthest (the United States to 1998, the euro-area members to around 2000), while several emerging markets begin only in the late 2000s or 2010s; the EM data-completion of §3.3 is precisely what lets these unequal spans coexist in one coupled system. All blocs share a common last estimation quarter, and the coupling aligns them by projection step rather than by calendar date, so a bloc whose data ends a quarter early is completed by one coupled conditional solve rather than by dropping a quarter from every other bloc.

We are explicit about data provenance and its limits, and scope the empirical claims accordingly. A small number of blocs carry a model-extended tail (the last few quarters filled by a temporal-disaggregation step) or a proxy for a variable that is not directly available at quarterly frequency; these are documented per bloc in the project's data audit. Accordingly, Section 6 draws its headline conclusions from the clean anchor blocs (US, EA, JP, UK, CA) whose panels are genuinely observed throughout, and reports the full 33-bloc panel in Appendix C with the model-extended blocs flagged. The pandemic window is carried as missing data rather than removed or dummied, consistent with §3.3, and is therefore not a provenance caveat but a modelling choice.

6 Does it forecast? The out-of-sample credibility floor

Before we ask the model to support scenarios, we check that it forecasts credibly, so that the scenarios in Section 7 rest on more than a nice picture. We set the bar as a floor, not a leaderboard: the question is not whether the model wins, but whether it holds its own against the best available models and clearly beats the textbook global VAR it generalizes.

6.1 Design

We test the model the way it would actually have been used: at a sequence of past dates, using only the data available at the time, and comparing its forecasts against those that later came true. We run the experiment from 2010Q1 to 2024Q4, re-estimating every model on the data up to each date and forecasting to $h \in \{1, 2, 4, 8\}$ quarters ahead; the four 2020 dates are dropped as an outlier the exercise is not about. One honest limitation sets in at the early dates: the model carries a credit block, and credit history is short, so before about 2013 the credit-augmented model cannot be estimated reliably and simply does not enter—the effective evaluation window is roughly the eleven years 2013–2024. This is a genuine feature of the model, not a bug, and it is why we compare all models only on the dates they all share. We report the clean anchor blocs (US, EA, JP, UK, CA) for growth, inflation, and the short interest rate, and put the full 33-economy panel in Appendix C. The pandemic-as-missing treatment still applies within each estimation, so the model faces the same ragged, pandemic-scarred data in the test as it does in use.

The single most important design choice is that the trade weights are recomputed at each date from data up to that date only. This is the main guard against letting the model peek at the future: a forecast made in 2012 never uses the trade patterns of 2021. The trade weights shipped with the tool, which average a recent period, are not allowed in the test. Three further guards go with it—each model sees only data up to the forecast date, the cross-country loop re-converges on that data alone, and nothing is adjusted after the fact—so the test genuinely mimics how the model would have been run in real time.

The set of competitors runs from the very simple to the textbook global VAR, and includes two stripped-down versions of our own model that show what each ingredient adds (Table 1). A random walk is the universal reference; an autoregression is a strong single-variable benchmark; a single-country Bayesian VAR removes the trade links, so the comparison with it shows what *linking the countries* buys; the textbook global VAR is the literature's standard, so beating it shows what the Bayesian estimation and data-completion buy; a single-pass version of our model (one iteration of the loop) shows what *iterating to convergence* buys; and a dynamic factor model asks whether the global VAR structure is needed at all. Against these stand the two versions of our model—with the United States responding to the world (the deployed one)

Table 1: The nine-model out-of-sample suite: two proposed variants, six benchmarks, and a climatology sibling of the random walk. The two ablations (`cgvar_b`, `sbvar`) isolate the EM fixed point and the coupling respectively.

Model (id)	Cross-country structure	Role / ablation
Random walk (`rw`, + `rw_clim`)	none	universal reference
AR(p) (`arp`)	none	strong univariate dynamic
Single-country BVAR (`sbvar`)	none (per-bloc GLP)	value of *coupling*
Textbook OLS GVAR (`cgvar_a`)	fixed weights, exog. US, OLS	value of the Bayesian+EM
Single-pass Bayesian GVAR (`cgvar_b`)	coupled, MAXIT = 1	value of the *EM star fixed*
Dynamic factor model (`dfm`)	static factors	“is the GVAR structure n
EM–Bayesian GVAR, frozen (`emgvar_frozen`)	coupled, exog. US	legacy anchor
EM–Bayesian GVAR, endog. (`emgvar_endo`)	coupled, endog. US	the deployed system

and with it held as a fixed anchor.

6.2 Scoring and inference

We judge the models in two ways. For point accuracy we use the root-mean-squared forecast error, reported relative to the random walk, so a number below one means “better than the naive benchmark.” For the full probability forecast we use the continuous ranked probability score, or CRPS, a standard measure of how well a predicted distribution matches what actually happened (Gneiting and Raftery, 2007). Our main way of ranking models is the Model Confidence Set (Hansen et al., 2011): for each variable and horizon it returns the group of models that are statistically tied for best, so a model earns credit not for winning outright but for belonging to that group.

For the individual comparisons we use the tests that are valid for this setup—forecasts from models that often nest one another, re-estimated on a growing sample. Giacomini–White (Giacomini and White, 2006) carries the main comparisons; Clark–West (Clark and West, 2007) is used when one model is a special case of another; Diebold–Mariano (Diebold and Mariano, 1995; Harvey et al., 1997) for models that do not nest; and a Romano–Wolf step-down (Romano and Wolf, 2005) keeps the overall error rate under control when we make many comparisons at once. We also check whether the probability bands are the right width, using the probability-integral transform.

One scoring choice is forced on us, and we are open about it. The natural score for a probability forecast, the log-score, turns out to be numerically unusable here: our forecasts are clouds of simulation draws, and scoring them with a fitted bell curve blows up on the fat-tailed pandemic-era draws. Following the density-forecast literature (**?**) we therefore rely on the CRPS and the Model Confidence Set, which do not suffer from this problem, and set the log-score aside.

6.3 Results

We report three results. First, the model does not win on point accuracy, and we say so plainly. On squared error, the factor model forecasts growth best at every horizon, and the autoregression forecasts inflation and the short interest rate best at the shorter horizons. This is what one expects of a large linked model: its value is a consistent picture across many economies, not the smallest error on any one series.

Second, once we judge the whole probability forecast, the model is in the tied-for-best group almost everywhere. By the Model Confidence Set (scored on the density, the CRPS), our model is statistically indistinguishable from the best available for growth at every horizon, for inflation

Table 2: 90% Model Confidence Set membership (density/CRPS), anchor blocs, over the estimable window. ✓ = statistically indistinguishable from the best. The sets are wide on this window (several models survive per cell), so this establishes that the model is *among* the best rather than the uniquely best; the pairwise test against the textbook global VAR (text) is the sharper comparison.

	horizon h			
Target (→ our model)	1	2	4	8
Growth	✓	✓	✓	✓
Inflation	✓	✓	✓	✓
Short rate	✓	✓	—	—

at every horizon, and for the short rate up to one year ahead; it drops out only for the short rate at the longest horizons (Table 2). We are careful not to over-read this. On the window the credit-augmented model can be estimated (about eleven years, discussed in §6.1), the confidence sets are wide, so several models—including the textbook global VAR—survive in many cells. The Model Confidence Set tells us our model is not distinguishable from the best; the sharper question is whether it beats the specific model it generalizes, which we take up next.

Third, against the textbook OLS global VAR—the model ours generalizes—the extra machinery pays off on the real side. A direct accuracy test on the density (Giacomini–White on the CRPS) shows our model significantly beats the textbook version on growth (decisively one quarter ahead, and again at one and two years) and on inflation at the longer horizons; the one place the textbook version holds its own is the short interest rate at longer horizons. So the Bayesian estimation, the pandemic-as-missing completion, and the cross-country loop earn their place chiefly on growth and inflation—a more measured claim than an across-the-board win, and the honest one.

A separate comparison concerns our own two versions. With the United States responding to the world, and with it held as a fixed anchor, the two are statistically indistinguishable as *forecasters*: endogenizing the United States does not sharpen the forecast. Its value is the scenario feedback of Section 7, which the fixed anchor cannot produce at all—so the two variants are a wash here and part company only when the model is used as a scenario tool.

One weakness is clear and we report it openly: the model's probability bands are too narrow. Its 90% bands actually cover the outcome about 74–82% of the time, short of the 90% they claim, and the gap is worst one quarter ahead. The reason is that the forecast carries the usual shock and parameter uncertainty but leaves out the extra uncertainty from having filled in the pandemic quarters. We therefore lean on the CRPS and the Model Confidence Set, which are not thrown off by this, and we flag the fix—feeding the fill-in uncertainty into the bands—in Section 8. These are sweep-quality results; a final pass with more posterior draws should tighten the bands further.

6.4 What the test establishes

The point is not a win but a floor. Linking the countries, shrinking the estimates, and filling in the missing data together deliver forecasts that sit inside the best group on growth, inflation, and the short rate, and that significantly beat the textbook global VAR on growth and inflation. That is what the scenario application needs: a model whose forecasts we can trust well enough to believe the way it propagates a scenario. We turn to that next.

7 What it enables: global scenario analysis

The point of the tool is the analysis it makes easy: fast, consistent what-if projections across all the economies at once. Section 6 asked whether the model forecasts; this section shows what using it as a tool adds—scenarios that hang together across 50 economies, are quick enough to explore live, and carry their public-debt consequences automatically.

7.1 A scenario is just a conditional forecast

Here a scenario is not a set of paths adjusted by hand. It is a forecast of the whole 50-economy system that respects a few values the user fixes. The user fixes what the scenario says—a policy rate held above its baseline, a credit spread widened, a budget loosened—and the engine of §4.2 reports the implied path of every other variable in every economy, consistent with the estimated dynamics and the trade links. Nothing is touched by hand afterward, so two analysts who set the same values get the same global picture.

We draw our scenarios from a documented library of policy-relevant shocks. Four are the mainstays: a jump in corporate borrowing spreads with a flight to safe bonds, a global energy-supply shock, a budget loosening that widens deficits, and a monetary policy that falls behind the curve. Each is written as a small set of fixed values with a sourced rationale, and each runs through the same engine, so the library is a set of inputs to one model rather than a collection of separate models.

7.2 A global transmission exhibit

A US interest-rate rise reaches every economy in one solve. Fixing the US policy rate above its baseline tightens US financial conditions, which pass through the trade links and the globally-traded prices to every partner: the euro area and the other advanced economies tighten alongside the United States, emerging markets with strong dollar links tighten more, and the resulting pattern of growth and inflation across economies is the consistent global picture the tool draws in real time (Figure 6). Re-run with the rate raised or lowered, the same solve updates every economy at once—which is what makes the tool usable in a meeting.

7.3 The feedback onto the United States

Letting the United States respond to the rest of the world adds a feedback that the fixed-anchor version cannot produce, and we measure it by running the very same US-rate scenario both ways. We hold the US policy rate 100 basis points above baseline for the whole horizon and solve twice from identical inputs: once with the United States responding to the world, and once with it held as a fixed anchor (which we obtain by removing the three US foreign-average terms, leaving everything else byte-for-byte the same, so the difference reflects only the feedback). With the anchor, the US path is driven purely by its own dynamics; with the feedback on, the rest of the world loops back onto the US.

The gap between the two, endogenous minus frozen, is the feedback the frozen GVAR structurally omits (Table 3). With the anchor, the United States gets no feedback, so its growth and inflation move only through its own dynamics; letting it respond adds the loop from the rest of the world, and the difference—about 1 to 2 points at the peak, and of mixed sign across the six economies—is the size of that loop. We are careful about these numbers for two reasons. First, holding the US rate 100 basis points high for the whole horizon is a scenario projection, not a one-off policy shock or an impulse response (Pesaran and Shin, 1998): because high interest rates tend to go together with strong activity in the data, as much as they cause weak activity, some of the individual responses come out with a counter-intuitive sign—which is exactly why we do not call this a shock. Second, the *difference* is cleaner than

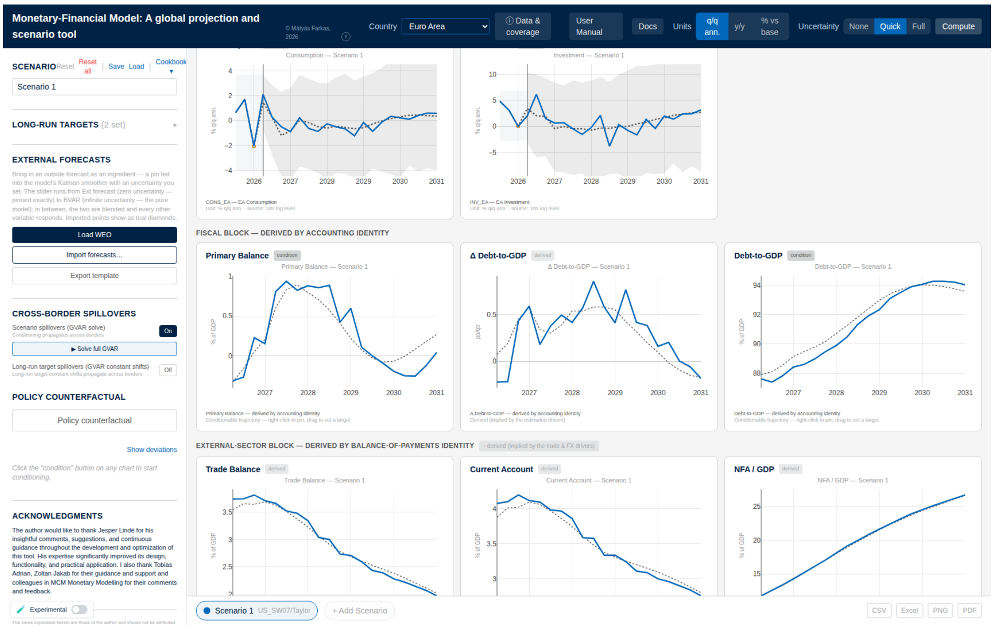

Figure 6: The euro-area view after a US tightening scenario is solved through the linked system: the domestic paths (central line and uncertainty band) respond to the US rate through the trade links and the globally-traded prices. Drawn in the browser from the pre-computed files.

the individual responses, because it cancels the own-dynamics that both versions share, but its sign and pattern still depend on the estimated trade links. So the qualitative message is solid—letting the United States respond adds a real feedback the fixed anchor cannot—while pinning down the numbers needs a proper impulse-response and connectedness calculation (Diebold and Yilmaz, 2012), which we leave to future work. The run also serves as a check on the solver: the fast (pre-computed-inverse) solve matches the slow step-by-step solve to within 10^{-4} for every economy, and the fixed-anchor version is stable (largest eigenvalue 0.70, against 0.947 for the full linked system).

7.4 The fiscal/debt response

A scenario's debt consequences come for free, because the debt path is computed from the forecast. When a scenario moves growth, inflation, and the interest rate the government pays, the debt formula (3) turns them into a debt path with no extra modelling: a tightening that raises the interest rate and slows growth pushes debt up, and the tool draws the resulting debt path next to the growth and inflation charts (Figure 7). Run backwards, the same formula answers the other question a policymaker asks—if we want debt to stabilize here, what fiscal adjustment does that take?—so both the "what happens" and the "what would it take" come out of one scenario.

8 What the system enables; limitations

The two parts of the paper together make one point: a single model does both jobs, in real time, from one reproducible set of files. The forecasting test shows the model sits among the best on growth, inflation, and interest rates and significantly beats the textbook global VAR on growth and inflation; the scenario application shows the same model produce consistent what-if projections across 50 economies and a genuine feedback onto the United States that a fixed anchor cannot. Neither the forecasting nor the scenario side is new on its own; carrying both in one model that anyone can inspect and re-run is.

Table 3: US +100 bp conditional projection on the deployed system: peak (max-abs) scored-space response by bloc and topology, and the endogenous−frozen gap. The gap is the Rest-of-World→US feedback channel, absent under the frozen anchor. Individual own-responses inherit the reduced-form projection confound and are not causally interpretable; the gap differences it out. Scored space is 4Δ(level) for growth/inflation.

Bloc	Var	endo	frozen	gap
US	growth	+1.80	+2.15	−1.97
	infl	+0.71	+1.11	−1.09
EA	growth	+1.53	+1.32	−1.31
	infl	+0.91	+0.55	−1.06
JP	growth	+1.91	+1.50	−1.52
	infl	+1.22	+0.69	+1.14
UK	growth	+1.57	+1.89	+1.63
	infl	+0.77	+0.74	−0.97
CA	growth	+1.02	+1.10	+0.81
	infl	+2.05	−0.97	+1.74
CN	growth	+3.12	+2.21	−1.66
	infl	−0.59	+1.20	−1.50

Having one model for both lowers the cost of coherent global policy analysis. An institution that would otherwise run a forecasting model and a separate scenario setup—and reconcile the two by hand—can instead fix a few values in one model and read a globally consistent answer, with the debt implications attached. And because everything ships as files with a documented calculation behind each number, the analysis can be audited.

We are open about the current limitations. The probability bands are too narrow, because the forecast leaves out the uncertainty from filling in the pandemic quarters, so we read the density results through the CRPS and the Model Confidence Set rather than at face value. The feedback onto the United States is a scenario projection, not a causal impulse response, and pinning down its size needs the impulse-response calculation we have not yet done. And a few economies carry model-extended tails or proxy series, which is why the headline forecasting test uses the clean economies and flags the rest.

Each limitation has a clear fix. Feeding the fill-in uncertainty into the forecast would widen the bands to the right size; adding an impulse-response and connectedness calculation would turn the US feedback from a scenario projection into a causal statement; and re-estimating the few model-extended economies on clean data would extend the forecasting results to the whole panel. None of these changes the design of the model; each sharpens one piece.

9 Conclusion

Policy institutions need to answer multi-country what-if questions quickly and consistently, but the tools to do it—Bayesian shrinkage, cross-country linking, conditional forecasting, and a careful treatment of missing data—have existed only separately. This paper puts them together in one model, an EM–Bayesian global VAR in which the United States responds to the rest of the world, and turns that model into a tool by using one fact: once the model is estimated, a scenario is a linear problem, so its answer can be pre-computed and reused, and any scenario re-solves all 50 economies in a browser in a fraction of a second.

The evidence comes in two parts, and we report it honestly. Out of sample the model does not win on point accuracy, but on growth, inflation, and interest rates it is statistically among the best, and it significantly beats the textbook global VAR it generalizes on growth

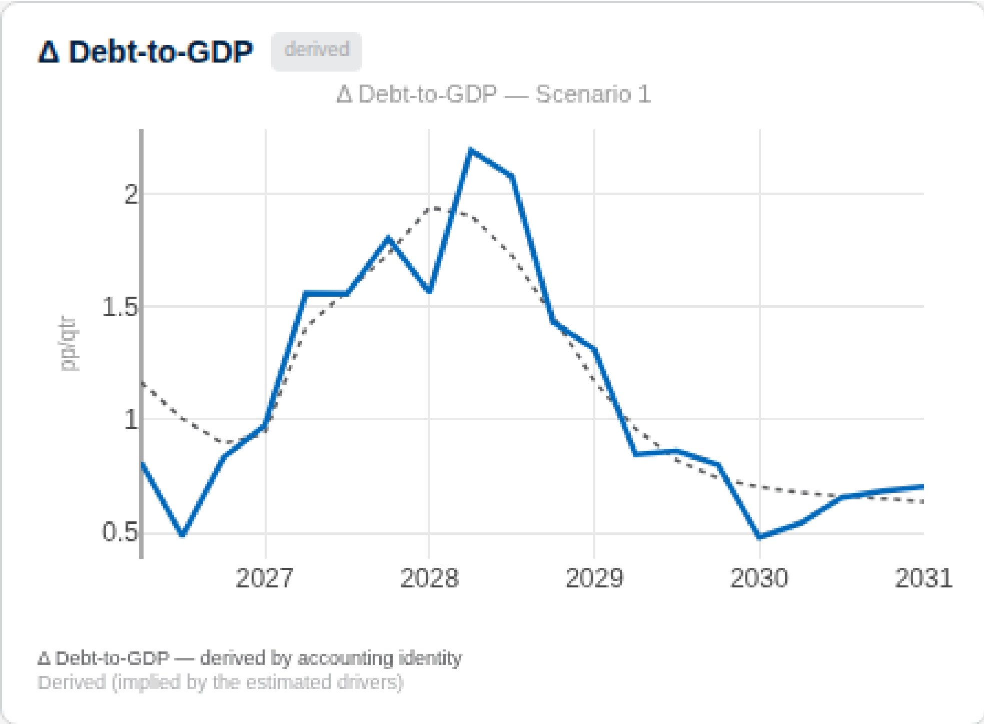

Figure 7: The derived debt block responding to a tightening scenario: the debt ratio follows from the budget identity (3) given the scenario's growth, inflation, and effective-rate paths, so the debt-sustainability implication is automatic rather than separately modelled.

and inflation—though its probability bands come out too narrow. Letting the United States respond to the world does not sharpen the forecast (the two variants tie); its payoff is elsewhere. As a scenario tool the model spreads global shocks consistently across every economy, carries their debt consequences automatically, and captures a feedback onto the United States that a fixed anchor cannot—which is where making the United States endogenous earns its keep. The contribution is the whole system: one model, fast and reproducible, that both forecasts credibly and supports coherent multi-country scenario analysis.

A Estimation and solution details

The per-bloc sampler, the Metropolis–Hastings update of the shrinkage hyperparameters, the convergence diagnostics of the EM sweep, and the offline precomputation of the coupling inverse are documented in full in the technical and user manual attached at the end of this document, and reproduced by the shipped pipeline (`pipeline/export_bvar_gvar24_loop.m`, `app/scripts/precompute_gvar_co`

B Data roster and per-bloc variable tables

Table 4 lists every economy in the panel: its variable count, whether it is a coupled GVAR bloc, whether it carries a fiscal block, whether it carries the shared external instruments, and its estimation sample. The table is generated directly from the shipped model specifications, so it is accurate by construction; the per-bloc variable tables (identifier, description, transformation, prior, block) are produced by the same script (`docs/generate_manual_tables.py`) and are available in the online supplement.

Table 4: The deployed panel: 50 economies, 1,182 estimated series. "Coupled" marks the 33 GVAR blocs; the other 17 are euro-area members estimated as standalone models with the euro-area instruments as external regressors. "Fiscal" marks the 42 economies with a fiscal block; "External" marks the three blocs (US, EA, CN) that carry shared instruments used by other blocs.

Code	Economy	n	Coupled	Fiscal	External	Sample
AR	Argentina	20	✓			2004Q1–2025Q4
AU	Australia	25	✓	✓		1999Q1–2025Q4
BR	Brazil	24	✓	✓		2012Q2–2025Q4
CA	Canada	27	✓	✓		1998Q1–2025Q4
CL	Chile	25	✓	✓		2000Q1–2025Q4
CN	China	22	✓	✓	✓	1999Q1–2025Q4
CO	Colombia	26	✓	✓		2005Q1–2025Q4
CR	Costa Rica	20	✓	✓		2010Q3–2025Q4
CZ	Czechia	25	✓	✓		2000Q1–2025Q4
DK	Denmark	26	✓	✓		2000Q1–2025Q4
EA	Euro area	25	✓	✓	✓	1999Q1–2025Q4
HU	Hungary	25	✓	✓		2000Q1–2025Q4
IS	Iceland	21	✓			2000Q2–2025Q4
IN	India	21	✓			2011Q4–2025Q4
ID	Indonesia	24	✓	✓		2005Q3–2025Q4
IL	Israel	25	✓	✓		2000Q1–2024Q4
JP	Japan	25	✓	✓		2000Q1–2025Q4
KR	Korea	25	✓	✓		2000Q1–2025Q4
MY	Malaysia	21	✓			2015Q1–2025Q4
MX	Mexico	26	✓	✓		2001Q4–2025Q4
NZ	New Zealand	25	✓	✓		2000Q1–2025Q4
NO	Norway	24	✓	✓		2000Q1–2025Q4
PL	Poland	25	✓	✓		2000Q1–2025Q4
RU	Russia	17	✓			2003Q1–2025Q4
SA	Saudi Arabia	18	✓			2010Q1–2025Q4
ZA	South Africa	25	✓	✓		2002Q1–2025Q4
SE	Sweden	26	✓	✓		2000Q1–2025Q4
CH	Switzerland	20	✓			2000Q1–2025Q4
TR	Türkiye	23	✓	✓		2000Q1–2025Q4
TW	Taiwan	20	✓			1997Q1–2025Q4
TH	Thailand	26	✓	✓		2003Q1–2025Q4
UK	United Kingdom	27	✓	✓		1997Q1–2025Q4
US	United States	40	✓	✓	✓	1998Q3–2025Q4
AT	Austria	23		✓		2000Q1–2025Q4
BE	Belgium	23		✓		2000Q1–2025Q4
DE	Germany	22		✓		2000Q1–2025Q4
EE	Estonia	23		✓		2000Q1–2025Q4
ES	Spain	23		✓		2000Q1–2025Q4
FI	Finland	23		✓		2000Q1–2025Q4
FR	France	23		✓		2000Q1–2025Q4
GR	Greece	22		✓		2000Q1–2025Q4

... continued

Code	Economy	n	Coupled	Fiscal	External	Sample
`IE`	Ireland	23		✓		2000Q1–2025Q4
`IT`	Italy	23		✓		2000Q1–2025Q4
`LT`	Lithuania	23		✓		2000Q1–2025Q4
`LU`	Luxembourg	23		✓		2000Q1–2025Q4
`LV`	Latvia	22		✓		2000Q1–2025Q4
`NL`	Netherlands	23		✓		2000Q1–2025Q4
`PT`	Portugal	23		✓		2000Q1–2025Q4
`SI`	Slovenia	23		✓		2000Q1–2025Q4
`SK`	Slovakia	23		✓		2000Q1–2025Q4

C Full 33-bloc evaluation panel

The headline evaluation of Section 6 is scoped to the clean anchor blocs; this appendix reports the median and inter-quartile range of relative RMSE and the MCS membership across all 33 coupled blocs, with the model-extended blocs explicitly flagged.

D Replication

The evaluation harness is a separate repository that references the main tool without copying estimator code; given the shipped artifacts and the trade-weight caches it reproduces the horse-race and the Tier-A spillover, and the model signature ties the reproduced results to the deployed system.

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Appendix: The Technical and User Manual

The remainder of this document reproduces the system's technical and user manual. It gives, in full, the detail behind every summary in the body: the exact equations of the estimator and the real-time solver, the state-space and conditioning machinery, the fiscal accounting, the data sources and transformations for every economy, the numerical constants of the deployed engine, and a guided tour of the tool's interface. The paper is self-contained without it; the manual is attached so that every modelling and implementation choice is documented and reproducible in one place.

A guide to the Monetary-Financial-Fiscal Macromodel:

conditional forecasts, global scenarios, and policy counterfactuals in the browser[1]

Mátyás Farkas

International Monetary Fund

July 17, 2026

Source code: https://github.com/matyasfarkas/BVARScenarioTool

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Abstract

This document describes the Monetary-Financial-Fiscal Macromodel, a browser-based tool for quarterly macroeconomic projection and scenario analysis covering about fifty economies. Each economy is a Bayesian vector autoregression estimated with the hierarchical prior of Giannone et al. (2015) under a COVID-as-missing data-augmentation treatment; thirty-three of them are coupled into a global system with an endogenous United States, solved as a linear fixed point at display time. The user conditions any subset of variables on hard or soft paths — a mouse drag, an imported outside forecast with a chosen degree of trust — and a Durbin–Koopman smoother (Durbin and Koopman, 2002) propagates the constraints to every other variable and, on request, across borders. Public debt and the external position are derived from the estimated drivers by their accounting identities; long-run anchors shift the companion drift exactly and instantly; and monetary or fiscal counterfactuals are evaluated with impulse responses from the Macroeconomic Model Data Base (Wieland et al., 2012). We state the methods with their exact implemented conventions, show how each is operated from the interface, and document the data, the out-of-sample evaluation, and the replication path. All computation runs client-side; there is no server.

Keywords: BVARs, Conditional Forecasting, GVAR, Missing Values, Scenario Analysis, Kalman Filter, Policy Counterfactuals, DSGE Models

JEL classification: C32, C53, E37, E52, E58, F47

Email: `mfarkas@imf.org`. I am grateful to Jesper Lindé for comments, advice, and continued support throughout the development of this tool, and to Tobias Adrian for his support. The views and any remaining errors are my own.

Contents

1 Introduction

The tool described in this document is a single web page. It loads the estimated models of about fifty economies into the browser and lets the user read, bend, and export their quarterly forecasts. There is nothing to install and no server-side computation: every number on the screen is produced client-side, so the tool can be hosted as a static website and used offline once loaded. The source code, the estimation pipeline, and the data files are in one repository,

https://github.com/matyasfarkas/BVARScenarioTool

with the deployed application served from its `app/` static export. The tool comes as is; it is a research platform, not an official forecast of any institution.

The economic machinery is organized in three layers, applied in order:

(i) **Baseline.** Each economy is an n-variable Bayesian vector autoregression (BVAR) estimated on quarterly data with the hierarchical Minnesota prior of Giannone et al. (2015), treating the pandemic quarters as missing data (Section 3). Thirty-three economies are additionally coupled into a global VAR (GVAR) with a fully endogenous United States (Section 5). The baseline forecast is computed once, offline, and shipped with the model.

(ii) **Scenario.** The user conditions any subset of variables on desired paths. Conditioning is a partial-observation problem in the companion-form state space, solved by a Kalman filter and Durbin–Koopman smoother (Bańbura et al., 2015; Durbin and Koopman, 2002); a pinned value can be *hard* (exact) or *soft* (an outside forecast held with a chosen degree of trust), and it propagates to the other variables — and, on request, to the other economies — through the estimated covariance and trade structure (Sections 4 and 5).

(iii) **Counterfactual.** A monetary or fiscal policy counterfactual is overlaid on the scenario using impulse responses from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and re-conditioned through the BVAR so the remaining variables stay consistent with the estimated co-movements (Section 9).

On top of these, two accounting blocks derive what should not be estimated freely: public debt from the budget identity (Section 6) and the net external position from the balance-of-payments identity (Section 7); and a long-run anchor mechanism shifts the far end of any forecast onto a user-chosen value, exactly and instantly (Section 8).

Three display rules govern everything on screen, and we state them here because the rest of the document relies on them. *Data are data*: observed history is drawn solid; any model-filled value — a not-yet-published quarter estimated from everything already observed — is drawn in amber and labelled, never disguised as an observation. *The model is frozen within the year*: quarterly data updates advance the forecast origin but never silently change the estimated coefficients; re-estimation happens once a year on a complete final year of data (Section 10). *Judgment is visible*:

whenever the user moves a forecast away from the model — a scenario pin, a long-run target, a counterfactual — the chart says so with a badge.

This document is both the user guide and the methodological documentation. It is organized so that each section states a method with its exact implemented conventions — the equations below were checked against the code, and file names are cited where the correspondence matters — and then shows how that method is operated from the interface, in numbered Examples that name the on-screen controls exactly. Section 2 gets a new user from a blank page to a first scenario. Sections 3–4 cover the estimation and the conditional-forecasting engine, Section 5 the cross-country system, Sections 6–7 the accounting blocks, Section 8 the long-run anchors, and Section 9 the policy counterfactuals. Section 10 documents the data and the vintage policy, Section 11 the out-of-sample evaluation, Section 12 the implementation and the replication path, and Section 13 concludes with limitations and known issues. Notation is collected in Appendix A; the per-economy variable tables — generated directly from the deployed data files — are Appendix B; a glossary is Appendix C.

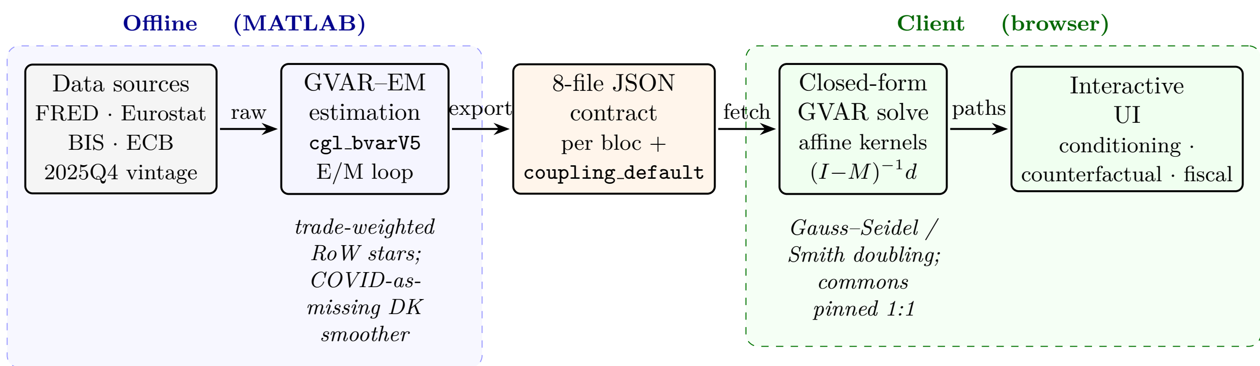

Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Harmonized macro-financial data (FRED, Eurostat, BIS, ECB; common 2025Q4 *estimation* vintage, with the observed panel subsequently refreshed to 2026Q1 with no re-estimation, §10) feed the *offline* MATLAB GVAR–EM estimator: per-bloc Bayesian VARs (`cgl_bvarV5`) are re-fit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world "star" variables (ROWDEM/ROW-INF/ROWFIN) from partners' Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to $\max|\Delta \text{star}| < 0.03$ (cf. §5.2). Each converged bloc is serialized to the eight-file JSON contract (`spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations`; plus `fiscal_meta` and the precomputed `coupling_default.json.gz` coupling artifact). In the *client* (browser), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§6).

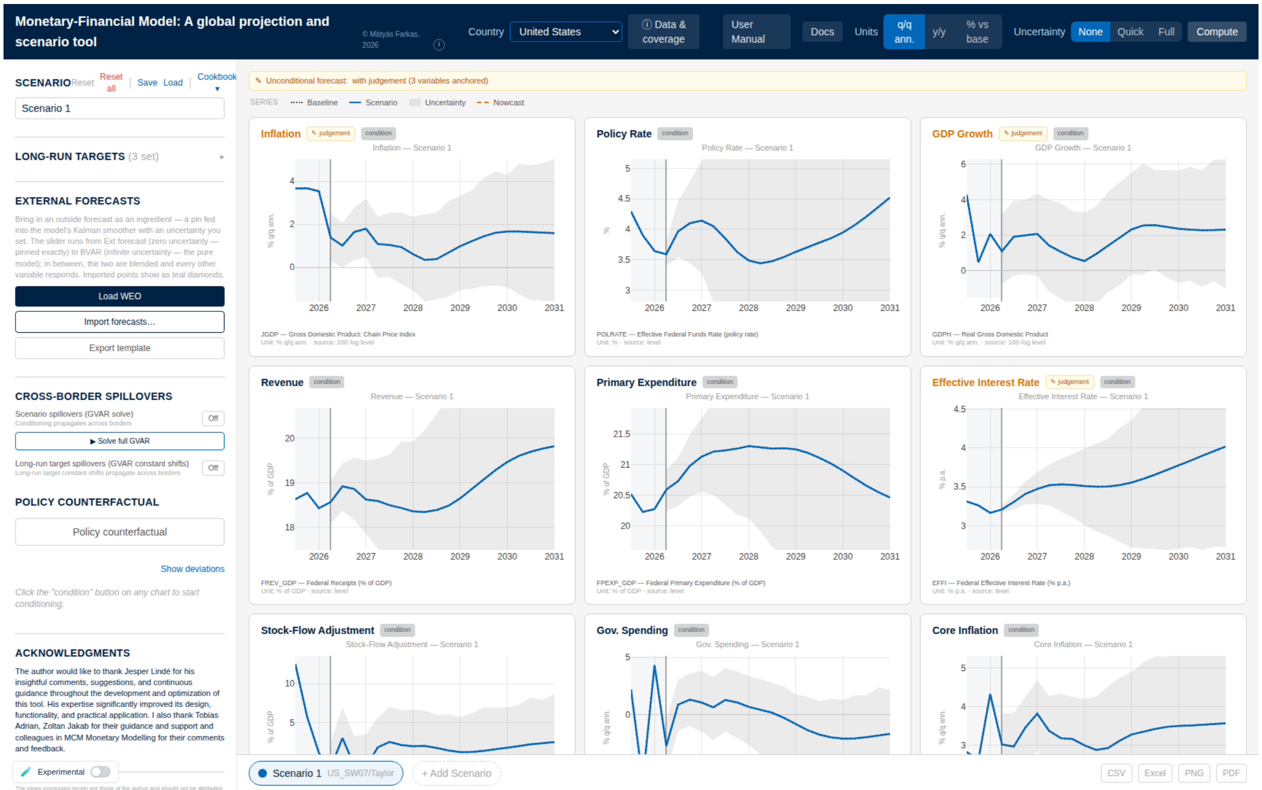

Figure 2: The screen at load (United States). The top bar holds the country selector, the *Data & coverage* provenance panel, the display-unit toggles (q/q annualized, y/y, % vs base), the uncertainty setting (None/Quick/Full), and Compute. The left rail holds the scenario manager, long-run targets, external forecasts (§4.5), cross-border spillovers (§5.5), and the policy counterfactual (§9). Each chart shows history, the dotted baseline, the solid scenario path, and the predictive band; judgement chips mark anchored variables (§8) and every chart carries a condition chip (§4.4).

2 Getting started

2.1 The screen

Open the tool's address in a modern browser and give the first visit a few seconds; the model files load into the page. The screen has four parts:

- the *top bar*: the **Country** dropdown, the display **Units** toggle, the **Uncertainty** switch (**None**/**Quick**/**Full**), the **Compute** button, the **Data & coverage** dialog, and links to this manual, the online documentation (**Docs**), and the release-impact **Monitor**;
- the *left control panel*: the **Scenario** block (with **Reset**, **Reset all**, **Save**, **Load**, and the **Cookbook** of ready-made scenarios), **Long-run targets**, **External forecasts**, **Cross-border spillovers**, the **Policy Counterfactual** block, and — once a variable is conditioned — the quarter-by-quarter conditioning list;
- the *chart grid*: one chart per variable; this is where the user clicks and drags;

- the *scenario bar* along the bottom: one tab per scenario (up to four, **+ Add Scenario**) and the **CSV**, **Excel**, **PNG**, and **PDF** export buttons.

The **Country** dropdown has two groups. Economies under **GVAR** belong to the coupled cross-country system of Section 5; a scenario built there can reach every other member. Economies under **Other countries** — including the individual euro-area member states, which receive shocks through the euro-area aggregate (Section 5.6) — are forecast on their own. Switching country never deletes work: each economy keeps its scenario settings until reset.

2.2 Reading one chart

Every chart follows the same grammar, and reading it correctly is most of the tool:

- *History* sits on a lightly shaded background; the solid line there is observed data. If a series' latest quarters have not been published, its history ends in an *amber dashed* tail with open circles — model-filled values (a *nowcast*, Section 10.2), never disguised as data. Hovering one reads *Model-implied nowcast (imputed)*.

- The *dotted grey line* is the **Baseline**: the model's own forecast, untouched by the user. The pale grey fan around it is the model's predictive uncertainty, shipped with the estimation (Section 3.4); it is there before anything is computed.

- The *solid coloured line* is the active scenario. Until the user conditions something it lies exactly on the baseline.

- The *header* carries a **condition** button (click to make the variable steerable; it turns into **CONDITIONED**), a **derived** badge on charts computed by an accounting identity, and an amber **judgement** badge whenever a long-run target has moved the variable off the model's estimate.

- The *footer* gives the variable's definition and unit.

The **Units** toggle changes the lens, not the numbers: **q/q ann.** is the quarter-on-quarter annualized rate (the model's native display unit, equation (2) below); **y/y** the change on the same quarter a year earlier; **% vs base** the *difference* between scenario and baseline, centred at zero — a read-only impact view in which a second toggle, **Level** versus **Rate (IRF)**, selects between the cumulative level effect and the quarter-by-quarter response (Section 9.3). Charts that are already rates (the policy rate, unemployment) ignore the toggle, and a handful of index series (oil, exchange-rate, equity, and house-price indices) are always drawn as levels, because a trade-weighted index quoted q/q-annualized is not an informative number.

Example 1 (A first scenario in five minutes)**.** *Open the tool; the* ***Country*** *dropdown reads* ***United States****, with* ***Units*** *on* ***q/q ann.*** *and* ***Uncertainty*** *on* ***None****. (1) On the* ***Policy Rate*** *chart click* ***condition****; the button turns blue and reads* ***CONDITIONED****. (2) Press the mouse on the* *forecast region about four quarters out and drag half a percentage point above the dotted baseline; release. A dot marks the pinned quarter, and the solid scenario line passes through it — only that quarter is imposed, and the model fills in the most plausible route through the point. (3) Scroll the grid: every chart has updated, not just the rate — growth cools, inflation eases with a lag — because the smoother re-draws the whole forecast consistently with the pin (Section 4). Other countries have not moved; cross-border transmission is off until requested (Section 5.5). (4) Set* ***Uncertainty*** *to* ***Quick*** *and press* ***Compute****: after a few seconds a band in the scenario's colour wraps the solid line — the middle two-thirds of plausible outcomes — and collapses onto the line at the pinned quarter, because an imposed value carries no uncertainty (Section 4.6). (5) Clean up: in the left panel click* ***Reset*** *(clears the active scenario for this country only) and set* ***Uncertainty*** *back to* ***None****. A right-click on an unconditioned chart is the shortcut for steps (1)–(2): it conditions the variable and plants the pin at the cursor in one gesture.*

Remark 1 (When to press Compute). The central scenario path updates live with every pin; **Compute** is never needed to see it. The button exists for the two calculations that are too heavy to run on every mouse move: the scenario uncertainty bands (Section 4.6) and the complete cross-country solve (Section 5.5). It is greyed out until either **Quick**/**Full** uncertainty is selected or a long-run target is set. Bands are a snapshot — change a pin afterwards and the band no longer matches until **Compute** is pressed again.

3 The country models

3.1 Reduced form and data spaces

Each economy ("bloc") c is an n-variable VAR with p lags in reduced form,

$$y_t = c + \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ collects the bloc's observables, $c \in \mathbb{R}^n$ is the intercept, $B_\ell \in \mathbb{R}^{n\times n}$ are the autoregressive matrices, and $\Sigma \in \mathbb{R}^{n\times n}$ is the innovation covariance. Throughout, $p = 3$ and the forecast horizon is $H = 20$ quarters. The deployed dimensions run from $n = 16$ (Costa Rica) to $n = 40$ (the United States); Appendix B tabulates every bloc. The estimated coefficients are stored as a $(1+np)\times n$ matrix $\hat\beta$ whose *first* row is the intercept and whose remaining np rows stack the lag coefficients $[B_1; \ldots; B_p]$ — a convention the reader needs only twice, for the companion construction (5) and the model signature of Section 5.3.

Variables enter the VAR in one of two spaces, recorded per variable in the bloc's specification (`spec.json`) and never changed at run time:

- `log`: the stored series is $\ell_t = 100 \ln Y_t$. Displayed growth is quarter-on-quarter annualized,

$$g_t \;=\; 4\,(\ell_t - \ell_{t-1}) \;=\; 400\,(\ln Y_t - \ln Y_{t-1}), \tag{2}$$

and year-on-year growth is $\ell_t - \ell_{t-4}$. The factor of 4 on the stored $100 \ln$ space — never 400, which would re-apply the 100 — is the single most load-bearing convention in the tool; every pin, band, and cross-country link below respects it.

- `lin`: the series enters in levels (interest rates in percent, unemployment, ratios to GDP). No growth conversion is applied.

Display units are chosen per variable for readability: activity and price series are shown as growth and inflation via (2); rates and ratios as levels in percent; and index series (the trade-weighted dollar, equity, house prices, oil, commodity indices) as the index level $\exp(\ell_t/100)$, with conditioning still performed in the stored $100 \ln$ space. Unemployment-rate bands are floored at zero; policy-rate bands are not, since negative nominal rates occur.

3.2 The hierarchical prior

Estimation follows the hierarchical approach of Giannone et al. (2015), henceforth GLP: the Minnesota prior (Litterman, 1986; Doan et al., 1984) with its tightness treated as a hyperparameter and selected by maximizing the marginal likelihood. The prior centers each equation on a univariate own-lag model, with a sum-of-coefficients component (Doan et al., 1984) shrinking toward "a unit increase in every level raises its own forecast one for one." Two hyperparameters are sampled — the overall tightness λ and the sum-of-coefficients weight μ — by a random-walk Metropolis step on the marginal likelihood of the *completed* data (Section 3.3), initialized at the `csminwel` optimum; the remaining prior settings are held at their GLP defaults. The estimator is the bundled MATLAB implementation (`BVAR/MFBVAR/bvarGLPmf.m`, called through the COVID-aware wrapper `cgl_bvarV5.m`); no external MATLAB packages are required.

Own-lag prior mean: random walk or white noise. The prior mean on variable i's own first lag is δ_i, with $\delta_i = 1$ (random walk, `RW`) the default: for (near-)integrated series — log activity, log prices, nominal yields — "the best forecast of the level is the current level." For a genuinely *stationary* series, a mean-reverting ratio that fluctuates around a constant, the random-walk center is the wrong null: it shrinks toward a unit root and lets the projection drift at long horizons. Such variables carry `prior:"WN"` in the specification and are estimated with $\delta_i = 0$ (white noise): "the best forecast is the unconditional mean." The mechanism is one line in the prior construction — the own-lag prior-mean diagonal is zeroed at the `WN` indices (`diagb(pos)=0` in `bvarGLPmf.m`, with `pos` built from the per-variable tags) — and it moves only the prior *center*, not a hard constraint: a persistent series whose data favour a near-unit root still obtains a large posterior own-lag coefficient at the estimated tightness. In the deployed panel the `WN` tag is carried by the three fiscal GDP-ratios (Section 6); every other variable is `RW` (Appendix B).

3.3 COVID as missing data

The 2020 pandemic quarters are extreme outliers that would otherwise contaminate both the autoregressive coefficients and Σ. Rather than dummying or rescaling them, the estimator treats the pandemic window as *missing* and imputes it inside the estimation, by a data-augmentation Gibbs sampler in the sense of Tanner and Wong (1987): each sweep alternates

(i) a Durbin–Koopman simulation-smoother draw (Durbin and Koopman, 2002) of the missing cells given the current coefficients, treating them as `NaN` observations in the companion-form state space of Section 4.1 (Algorithm 1);

(ii) a Metropolis–Hastings update of (λ, μ) on the completed-data marginal likelihood; and

(iii) a draw of (β, Σ) from the completed data.

The surrounding quarters stay in the sample and inform the imputation, so the pandemic history is preserved while the outlier does not drive the dynamics.

Two implementation choices matter. First, the production window is the *fixed* three quarters 2020Q2–Q4 for every bloc. (A legacy single-bloc variant self-calibrates the window to the GDP trough searched within 2019Q4–2021Q1 — restricting the search matters, since an unrestricted trough lands on the wrong crisis for some economies, e.g. Korea's 2008Q4 — but the deployed GVAR estimator uses the fixed window.) Second, the masking is *macro-only*: over the window the domestic macro columns and the two macro star variables of Section 5.1 are set missing, while the financial columns (rates, yields, equity, house prices, spreads; the `isFinancial` tag in Appendix B) and the financial star stay *observed*. The smoother therefore imputes the pandemic macro holes conditional on financial series that carry genuine signal through 2020. The same device handles non-pandemic artefacts: the UK effective-rate spike of 2008–09, a mechanical consequence of inflation uplift on index-linked gilts being recorded as accrued interest, is NaN'd and imputed identically (Section 6).

3.4 What ships: posterior output and predictive bands

For the deterministic engine the posterior is summarized by the means $(\hat{\beta}, \hat{\Sigma})$. For uncertainty the chain is thinned to $D = 500$ retained draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$, shipped per bloc in `companion_draws.json` and loaded lazily — only when the user first presses **Compute** (the US file is the largest at roughly 34 MB).

The shipped baseline carries *predictive* bands, not parameter-only bands: the Gibbs sampler emits a forecast density that integrates over parameter uncertainty, future innovations, and the missing-data imputation simultaneously, and the exported quantiles are

$$\big(q_{05},\, q_{16},\, q_{50},\, q_{84},\, q_{95}\big)_{T+h} \;=\; \text{Quantile}\big\{\, \tilde{y}^{(d)}_{T+h} \,\big\}_{d=1}^{D}, \tag{3}$$

drawn on the charts as the grey fan (the 68% band (q_{16}, q_{84}) shaded, the 90% band (q_{05}, q_{95}) fainter). The *central* path remains the deterministic forecast from the posterior means, so an untouched

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep; `cgl_bvarV5.m`).

Input: system matrices T, Z, Q of Section 4.1; partial observations $\{y_t^{(o)}\}$ (COVID-masked and ragged cells absent); current draw of (β, Σ).
Output: one draw $\tilde{\alpha}_{1:T}$ from $p(\alpha_{1:T} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for the missing cells.

1: ▷ (1) Forward simulation of an unconditional path.
2: Draw α_0^+ from the initial law; **for** $t = 1, \ldots, T$: draw shocks and set $\alpha_t^+ = T\alpha_{t-1}^+ + \eta_t$, $y_t^+ = Z\alpha_t^+$.
3: Form synthetic partial observations $y_t^{+(o)}$ on the *same* missing pattern as the data.
4: ▷ (2) Kalman filter (Section 4.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$, processing only the observed rows at each date.
5: ▷ (3) Backward smoothing recursion (equation (10) of Section 4.2) ⇒ smoothed discrepancy $\hat{\alpha}_t$.
6: ▷ (4) Mean correction. $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed cells off $\tilde{y}_t = Z\tilde{\alpha}_t$.

screen shows the stored baseline exactly. Scenario bands — computed in the browser, conditional on the user's pins — are constructed differently and are described with the conditioning engine (Section 4.6).

Every bloc is serialized to one eight-file JSON contract — `spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations` — fiscal blocs adding `fiscal_meta` and external blocs `external_meta`, with every float compacted to six significant figures. The deployed application reads only this contract and never re-estimates (Section 12).

4 Conditional forecasting

Conditioning a forecast on a chosen path is, in state-space form, the same problem as filtering with missing data: a pinned cell is an *observation* of the future, a free cell is a missing one (Bańbura et al., 2015; Waggoner and Zha, 1999). This section states the state space, the filter and smoother exactly as implemented (`app/src/lib/compute/kalman.ts`), the display construction, and the two kinds of pin — hard and soft — the interface exposes.

4.1 Companion-form state space

Stack the current and lagged observables into the state $\alpha_t = (y_t', y_{t-1}', \ldots, y_{t-p+1}')' \in \mathbb{R}^{np}$. The VAR (1) is then the first-order system

$$\alpha_t = T\,\alpha_{t-1} + c_\alpha + \eta_t, \qquad y_t = Z\,\alpha_t + \varepsilon_t, \tag{4}$$

with

$$T = \begin{pmatrix} B_1 & B_2 & \cdots & B_{p-1} & B_p \\ & I_{n(p-1)} & & & 0 \end{pmatrix}, \qquad c_\alpha = \begin{pmatrix} c \\ 0 \end{pmatrix}, \qquad Z = \left(I_n \mid 0_{n\times n(p-1)}\right), \tag{5}$$

$\eta_t \sim \mathcal{N}(0, Q)$ where Q carries Σ in its top-left $n \times n$ block and zeros elsewhere, and $\varepsilon_t \sim \mathcal{N}(0, R)$ with

$$R = \sigma_\varepsilon^2 I_n, \qquad \sigma_\varepsilon^2 = 10^{-12}. \tag{6}$$

The top block of T is filled from the stored coefficients as $T_{ij} = \hat{\beta}_{j+1,\,i}$ (the transpose of the lag rows, skipping the intercept row), and c_α from the intercept row. The measurement variance (6) is the *hard-pin* noise: small enough that a pinned value is honoured to machine precision, nonzero for numerical stability. The state is initialized at the last p observed rows, $\alpha_0 = (y_T', \ldots, y_{T-p+1}')'$, with $P_0 = 10^{-7} I_{np}$.

4.2 Filter and smoother with partial observations

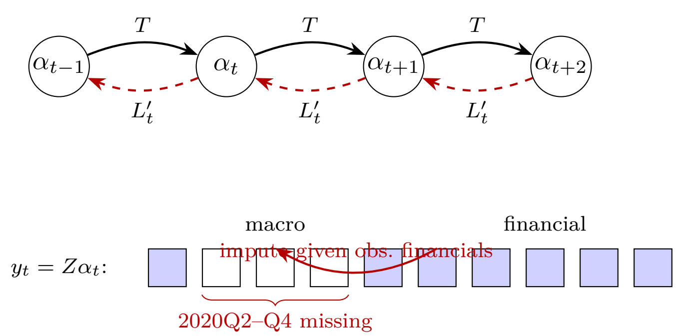

Figure 3: Companion-form state space with the Durbin–Koopman filter/smoother under the COVID-as-missing treatment. States α_t (§4.1) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (7)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L_t' r_t$ (dashed, Eq. (10)). In the observation row $y_t = Z\alpha_t$ (§4.1), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

Collect the user's pins in a matrix $Y^* \in \mathbb{R}^{H \times n}$ with $Y^*_{t,j}$ the pinned value of variable j at forecast quarter t and `NaN` where free. At each t the filter processes only the observed rows: let Z_t stack the rows of Z for the pinned cells (plus any linear-combination rows, below) and R_t the corresponding diagonal of measurement variances. For $t = 1, \ldots, H$:

$$a_{t|t-1} = T\, a_{t-1|t-1} + c_\alpha, \qquad P_{t|t-1} = T\, P_{t-1|t-1}\, T' + Q, \tag{7}$$

$$v_t = y_t^* - Z_t\, a_{t|t-1}, \qquad F_t = Z_t\, P_{t|t-1}\, Z_t' + R_t, \tag{8}$$

$$K_t = P_{t|t-1}\, Z_t'\, F_t^{-1}, \qquad a_{t|t} = a_{t|t-1} + K_t\, v_t, \qquad P_{t|t} = P_{t|t-1} - K_t\, Z_t\, P_{t|t-1}, \tag{9}$$

with $a_{t|t} = a_{t|t-1}$, $P_{t|t} = P_{t|t-1}$ at quarters with no pins. F_t^{-1} is computed by LU decomposition with partial pivoting (singularity threshold 10^{-15}), and the covariance updates are symmetrized.

The Durbin–Koopman fixed-interval smoother then runs backward with $r_H = 0$:

$$L_t = T\big(I - P_{t|t-1} Z_t' F_t^{-1} Z_t\big), \qquad r_{t-1} = Z_t' F_t^{-1} v_t + L_t' \, r_t, \tag{10}$$

($r_{t-1} = T' r_t$ at pin-free quarters), and the smoothed state is $\hat{\alpha}_{t|H} = a_{t|t-1} + P_{t|t-1} \, r_{t-1}$. The smoothed *variance* is delivered by the companion N-recursion,

$$N_{t-1} = Z_t' F_t^{-1} Z_t + L_t' \, N_t \, L_t, \qquad V_t = P_{t|t-1} - P_{t|t-1} \, N_{t-1} \, P_{t|t-1}, \tag{11}$$

with $N_H = 0$; V_t is the state covariance conditional on *all* pins, so uncertainty about a quarter *preceding* a future pin is correctly reduced by it. The conditional forecast is the observation block of $\hat{\alpha}_{t|H}$; the display shading of the impact view (Section 4.6) reads V_t.

Linear-combination pins. A constraint on a linear combination $h' y_t = v$ — used by the fiscal block to pin the primary balance, $h_{\text{rev}} = +1$, $h_{\text{pexp}} = -1$ (Section 6) — enters the same machinery as one extra observation row $(h'Z, \, v)$ with measurement variance $\sum_k h_k^2 \, \sigma_\varepsilon^2$. No separate variable is created.

4.3 The display: exact deltas on the stored baseline

To guarantee that an untouched screen reproduces the shipped baseline exactly, the tool displays

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \big(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}}\big), \tag{12}$$

the stored baseline plus the smoother *delta* between a conditional and an unconditional pass (the latter cached and reused). Zero pins give a zero delta and hence the stored baseline to the last digit. For `log` variables the delta is computed in the $100 \ln$ level space and scaled to displayed growth by the factor 4 of (2); a dragged growth pin is correspondingly cumulated into the level pin, one quarter's worth (deviation/4) per active quarter.

4.4 Operating hard pins

A pin is set on the chart (drag on a **CONDITIONED** chart; right-click to condition and plant in one gesture; right-click again to toggle a quarter's pin off) or in the conditioning list, which shows one row per forecast quarter, labelled **NC**, **t+1**, . . . , **t+19**. **NC** — the nowcast — is the current, not-yet-published quarter (Section 10.2). Each row has an on/off toggle and a value slider; **Show deviations/Show levels** switches between entering distances from baseline and absolute values. Pins are what-you-set-is-what-you-see: a pinned quarter sits exactly at its number, always; the quarters left off are filled by the model. Multiple variables stack, and all pins are imposed *jointly* — the smoother finds the single most plausible forecast through every pin at once, so mutually strained pins show their cost in the unconditioned charts. Up to four scenarios can be held side by

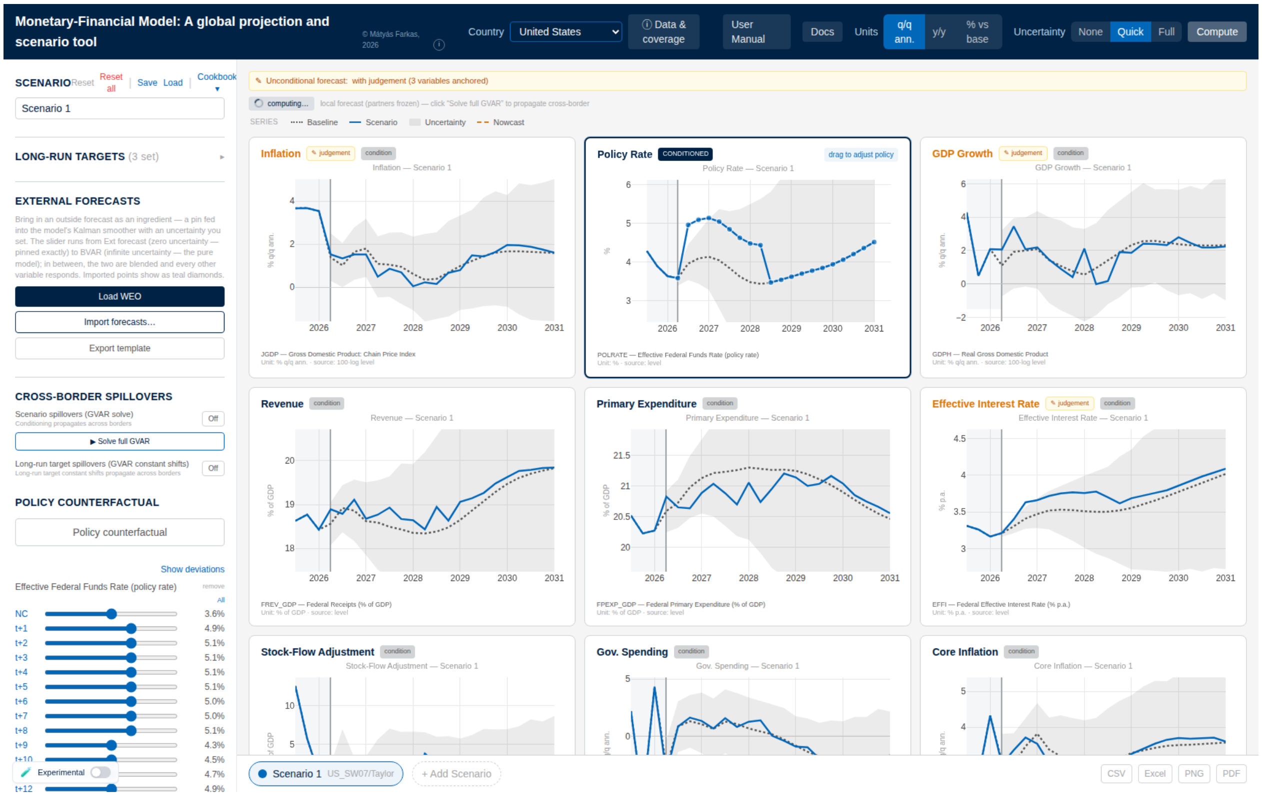

Figure 4: Hard conditioning in action. The policy-rate chart is put in CONDITIONED mode and the path is raised by +100 bp for eight quarters — by dragging the dots on the chart or the per-quarter sliders that appear in the left rail (here $t{+}1, \ldots, t{+}8$). After Compute, every panel redraws the smoothed scenario (solid) against the frozen baseline (dotted): output growth dips, inflation eases, the effective interest rate on debt climbs, and the fiscal ratios respond — all through the estimated cross-correlations of §4.2; nothing is re-estimated.

side (the bottom bar); **Reset** clears the active scenario for the current country, **Reset all** everything everywhere, and **Save/Load** round-trip one country's conditioning to an Excel file.

Example 2 (A ready-made scenario, and what identification buys)**.** *The* ***Cookbook*** *(left panel,* ***Scenario*** *header) loads documented, source-calibrated scenarios; each recipe offers* ***Load naive*** *— pin only the headline variable — and* ***Load + restriction*** *— add the short-run pins that shape the shock into its economically meaningful form. The credit-crunch recipe (calibrated to the Federal Reserve's 2020 severely-adverse scenario) illustrates on the United States what the extra pins buy.* Naive*: pin only the corporate bond yield +400 bp. The model reads a corporate-yield spike as generic rates tightening — the whole curve rises (at the 2026Q1 vintage, the policy rate peaks around* $7\frac{1}{2}$ *%, the 10-year near* $8\frac{1}{2}$ *%) and inflation moves* up *on impact: comovement alone cannot tell a credit-supply dislocation from a broad rate shift.* Restricted*: add the documented crunch signature — safe 10-year yield* down *(flight to safety), output down, inflation down, equity down, the policy rate cut. The same +400 bp spread now produces a deep recession (GDP trough near −6 % q/q ann.), disinflation, and easing — the policy rate and the safe yield move in the* opposite *direction from* *the naive read. Loading a recipe* replaces *the active scenario's pins; run the two variants on two scenario tabs to compare.*

4.5 Soft pins: outside forecasts with a trust dial

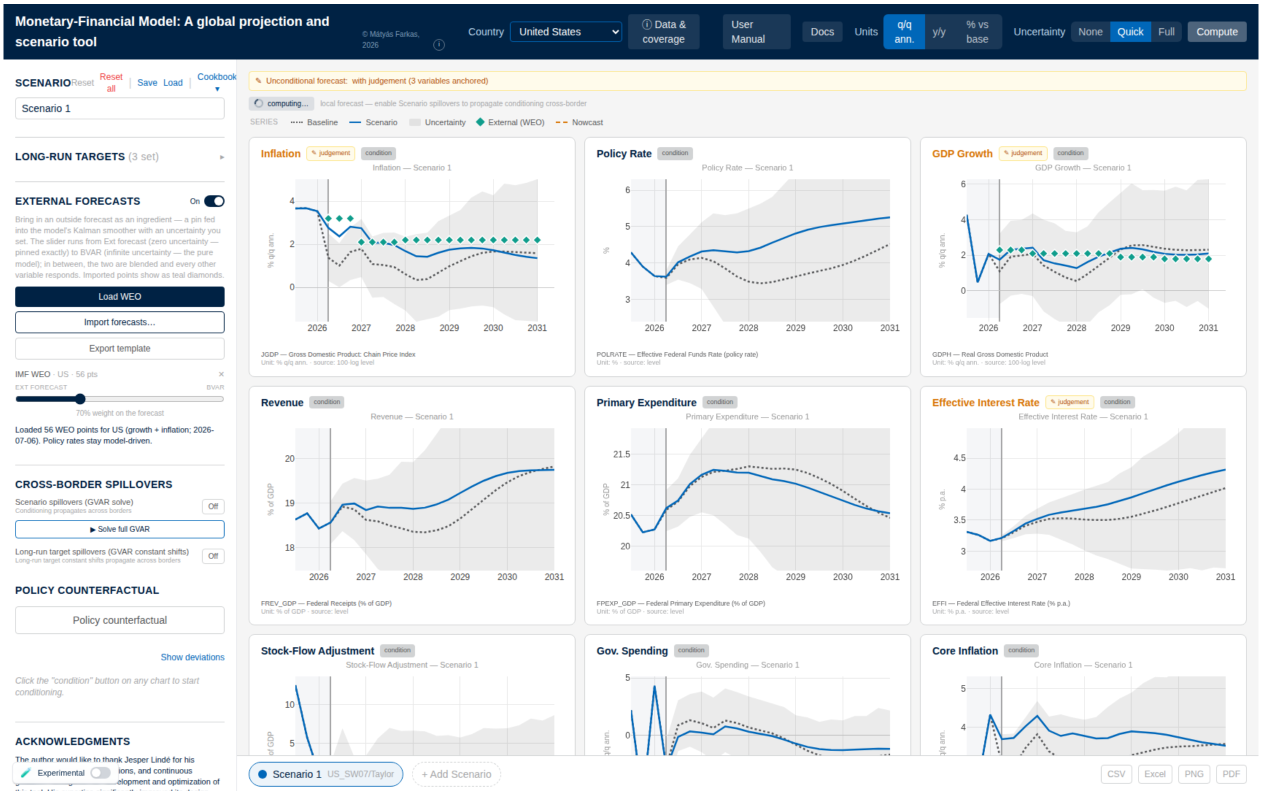

Figure 5: An outside forecast as an ingredient (Load WEO). The teal diamonds are the imported WEO growth and inflation points; the trust dial (left rail, here at 70% weight on the forecast) maps to the measurement-error variance of Eq. (13), so the scenario blends the outside path with the model instead of pinning it; policy rates stay model-driven.

An imported outside forecast — IMF WEO, a consensus survey, desk numbers — enters as a *soft* pin: an observation with a measurement variance set by the user's trust in the source, so the model reconciles the outside number with everything else rather than being overridden by it. For a cell of variable j at horizon h held with trust $\tau \in (0, 1)$, the measurement variance in (8) is

$$R^{\text{ext}}_{h,j} = \widehat{\text{Var}}_{h,j}\,\frac{1-\tau}{\tau}, \qquad \widehat{\text{Var}}_{h,j} = \left(\max\!\left(\tfrac{q_{84}-q_{16}}{2},\, 10^{-6}\right)\right)^2, \tag{13}$$

where $q_{84} - q_{16}$ is the model's own shipped predictive band of (3) at that cell — so the dial needs no units: $\tau = \frac{1}{2}$ makes the outside number exactly as informative as the model's own predictive density, $\tau \geq 0.999$ collapses to the hard pin (6), and $\tau \to 0$ is a no-op (τ is clamped to $[0.001, 0.999]$; the default is 0.7). Display-unit values are converted to level-space pins by the conventions of Section 4.3 (growth targets cumulate; any active long-run drift of Section 8 is netted out first, so the *displayed* line lands on the imported number). A cell the user has pinned by hand always wins over an imported one (`externalForecast.ts`).

In the interface (**External forecasts**, left panel): **Load WEO** imports the IMF World Economic Outlook real-GDP-growth and inflation forecast for the viewed economy, each annual figure applied softly to its four quarters; **Export template** writes an Excel file pre-filled with the model baseline, to be overwritten and re-imported; **Import forecasts...** accepts that file or a small JSON. Each source appears with its trust slider; imported points draw as teal diamonds with an error bar that grows as trust falls; the source's × (or zero trust) removes it. **Export session/Import session** bundle all countries' conditioning, targets, and imports into one JSON for hand-over. On a coupled economy an imported forecast also propagates cross-border, with one performance caveat documented in Section 5.7.

4.6 Scenario uncertainty bands

Pressing **Compute** with **Quick** or **Full** selected draws $N_b = 50$ or 200 predictive paths and bands the scenario. The default construction is deliberately simple and fast:

(i) for each band draw, sample a posterior draw $(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})$ from the $D = 500$ shipped draws (with replacement) and iterate the companion forward from the last observed state *with* innovation shocks $\eta_h \sim \mathcal{N}(0, \hat{\Sigma}^{(d)})$ — a full predictive draw carrying both parameter and shock uncertainty;

(ii) convert to display units and shift the whole path by the (draw-invariant) mean scenario delta of (12);

(iii) report the empirical 16th and 84th percentiles per cell.

The band is therefore the model's own predictive range *re-centred on the scenario path* — directly comparable to the baseline fan — and at the actively pinned quarters of a conditioned variable it is collapsed onto the line, since an imposed value carries no uncertainty. On the **% vs base** impact view the shading instead uses the smoothed conditional standard deviation from (11): it pinches at hard pins, shrinks partially at soft pins in proportion to (13), and tapers *into* a future pin — the honest conditional-variance picture.

Remark 2 (Why the default band is not the per-draw conditional cloud). Re-solving the conditional forecast draw by draw produces a genuinely conditional cloud, but with only a point or two pinned it collapses far too tightly under the estimated prior — a band the user reads as false confidence. The default recentring preserves the full predictive width; the per-draw conditional construction is available for coupled economies as the **Experimental** switch (bottom-left), which re-solves the entire cross-country system per draw (Section 5.8). Toggling it clears any band on screen, because bands built the two ways must not be read against each other.

Remark 3 (Long-run targets translate bands rigidly). A long-run anchor (Section 8) shifts the central path and every band quantile by the same deterministic amount, instantly and without recomputation — band *widths* are invariant by construction. Only pins change conditional uncertainty; targets move where the forecast is headed, not how well it is known.

5 The global model

Thirty-three of the deployed economies are coupled into a global VAR (GVAR) in the tradition of Pesaran et al. (2004) and Dees et al. (2007): the United States, the euro area as a single aggregate bloc, and thirty-one further economies, spanning roughly ninety percent of world GDP (the roster is in Appendix B). Two design choices organize everything in this section. First, *coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated BVAR, and cross-country spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Second, the United States is *fully endogenous*: it carries its own trade-weighted foreign variables and iterates inside the fixed point like every other bloc, so shocks abroad feed back into the US — the channel a classical frozen dominant unit cannot represent.

The euro area enters as one aggregate by necessity: a fully disaggregated euro GVAR is non-stationary (the coupling operator's spectral radius exceeds one, $\rho \approx 1.089$, in the disaggregated experiment), while aggregation restores contraction. The seventeen euro-area member states are therefore *standalone* models outside the coupled system, connected to it through a separate euro-area doorway (Section 5.6).

5.1 Star variables and trade weights

Every coupled bloc carries three trade-weighted *star* variables summarizing its foreign environment, the weak-exogeneity device of the GVAR literature: foreign demand (`ROWDEM`), foreign inflation (`ROWINF`), and foreign financial conditions (`ROWFIN`). At estimation time the stars are built as levels from the partners' *completed* panels (Section 5.2): with w_{cj} the trade weight of partner j in receiver c and $\ell_{j,t}$ the partner's stored $100\ln$ level,

$$\text{ROWDEM}_{c,t} = 4\sum_j \tilde{w}_{cj}\left(\ell^{\text{gdp}}_{j,t} - \ell^{\text{gdp}}_{j,t-1}\right), \qquad \text{ROWFIN}_{c,t} = \sum_j \tilde{w}_{cj}\, r_{j,t}, \tag{14}$$

(`ROWINF` as `ROWDEM` on price levels) — annualized growth for the two macro stars, the factor 4 of (2), and a level average of partner short rates $r_{j,t}$ for the financial star. The weights $\tilde{w}_{cj}$ are renormalized *per component and per quarter* over the partners actually present and eligible, so absent or excluded mass is reabsorbed rather than dropped: rate-less blocs (Switzerland, Norway) leave their partners' `ROWFIN`; Argentina's hyperinflation episode is excluded from partners' `ROWINF` and `ROWFIN` while its real demand stays in `ROWDEM`. (A deflate-by-subtraction rule exists for blocs reporting nominal GDP; since China was re-sourced to a real chain-linked volume index in the June-2026 re-estimate, the set of such blocs is currently empty.)

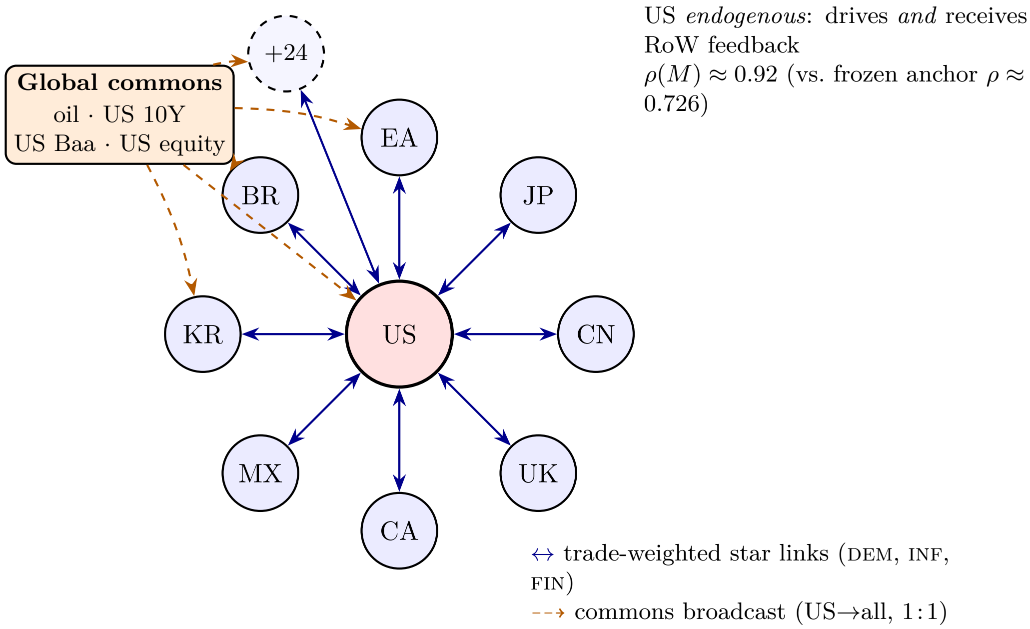

Figure 6: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 32 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 24 further blocs) through *bidirectional*, trade-weighted "RoW star" links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically (1 : 1) to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator's spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only) to $\rho(M) \approx 0.92$ at the full 33-bloc endogenous-US configuration (the value stored in the shipped `coupling_default.json.gz`; it moves slightly with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 5.1 and 5.3).

The bilateral weights come from the IMF's International Merchandise Trade Statistics (exports plus imports, 2021–2023 average, euro area as a single counterpart) and ship as one row per coupled bloc — the US row included — each summing to one with a zero diagonal (`weights_24bloc.json`; the file name is historical, the file lists all 33 blocs). The topology is *not* hardcoded: a bloc is a coupled member if and only if its specification carries the three `row`-tagged stars *and* it has a weights row, and the US is treated as endogenous if and only if its own model satisfies the same test. Absent either, the engine falls back to the legacy frozen-anchor solve unchanged.

5.2 Estimation: an EM fixed point in the completed stars

Before the formal statement, here is the idea in plain terms. To estimate any one country's model you need its *foreign environment* — what its trading partners' demand, inflation, and interest rates are doing. But each partner's path is itself the output of that partner's model, which needs *its* foreign environment, and so on around the world: everything depends on everything, so no country can be estimated on its own and first. The way out is to treat the unknown foreign environment as *missing data* and fill it in by iterating. Start from a rough guess of the foreign environment for every country; estimate each country's model against its guess (filling in the pandemic holes in the same step); then read off from the freshly-estimated countries what the foreign environment *actually* turned out to be, and re-estimate against that. Repeat. Each pass the guesses become more mutually consistent, and the loop stops when the environment each country *assumed* matches the one its partners *produce* — when the guesses have come true. That self-consistent set of country models is the estimate.

A concrete image helps: fifty country desks share one room, each forecasting its own economy but needing its neighbours' forecasts as inputs. Everyone pencils in a rough guess for the neighbours and produces a first forecast; the forecasts are passed back around the room and each desk redoes its own with the updated inputs; the numbers keep circulating until nobody's change any more. This is exactly the expectation–maximization logic of Dempster et al. (1977) with the "rest of the world" — and the pandemic gaps — as the missing data: an **E-step** that fills in the blanks given the current models, and an **M-step** that re-fits the models given the filled-in blanks, iterated to a fixed point. Two things are worth keeping straight. Nothing is accepted or rejected *across* countries: coherence is the loop's stopping rule, not an acceptance test, and the loop settles on one consistent set of paths rather than a distribution of draws. The genuine Bayesian sampling — the simulation smoother that draws the missing cells and the Metropolis step that updates the shrinkage hyperparameters — lives *inside* each country's single fit (Section 3.3), one level below the loop.

The stars cannot be built directly from observed data: the panels are unbalanced (blocs enter the sample at different dates), the pandemic macro cells are deliberately missing (Section 3.3), and several financial series start late. A star built only on complete quarters would truncate the sample to the common balanced span — exactly the information loss the data-augmentation design avoids. The estimator therefore iterates completion and re-estimation to a fixed point in the stars (in the spirit of Dempster et al., 1977):

- **E-step (completion).** For each bloc, one call to the data-augmentation sampler of Section 3.3 on the panel [domestic | stars] imputes every missing cell (COVID macro mask included) by the Durbin–Koopman simulation smoother; the posterior-mean completed panel is retained.

- **M-step (re-estimation).** The same call re-estimates the bloc's coefficients under the GLP prior on the completed data; the posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$ and the thinned draws are what ships.

- **Outer loop.** All blocs' stars are rebuilt via (14) from the *frozen* start-of-sweep completions, the blocs are re-estimated in parallel (a Jacobi sweep, order-independent), and the loop repeats until the stars stop moving: $\max_c \|x_c^{*(m)} - x_c^{*(m-1)}\|_\infty < \textsc{tol} = 0.03$ (COVID-masked macro-star cells excluded from the metric — their cross-iteration churn is imputation Monte-Carlo noise, with an observed floor near 0.13, so convergence in practice is a plateau at the noise floor), capped at $\textsc{maxit} = 12$ sweeps. A one-shot, no-stars US completion seeds the first

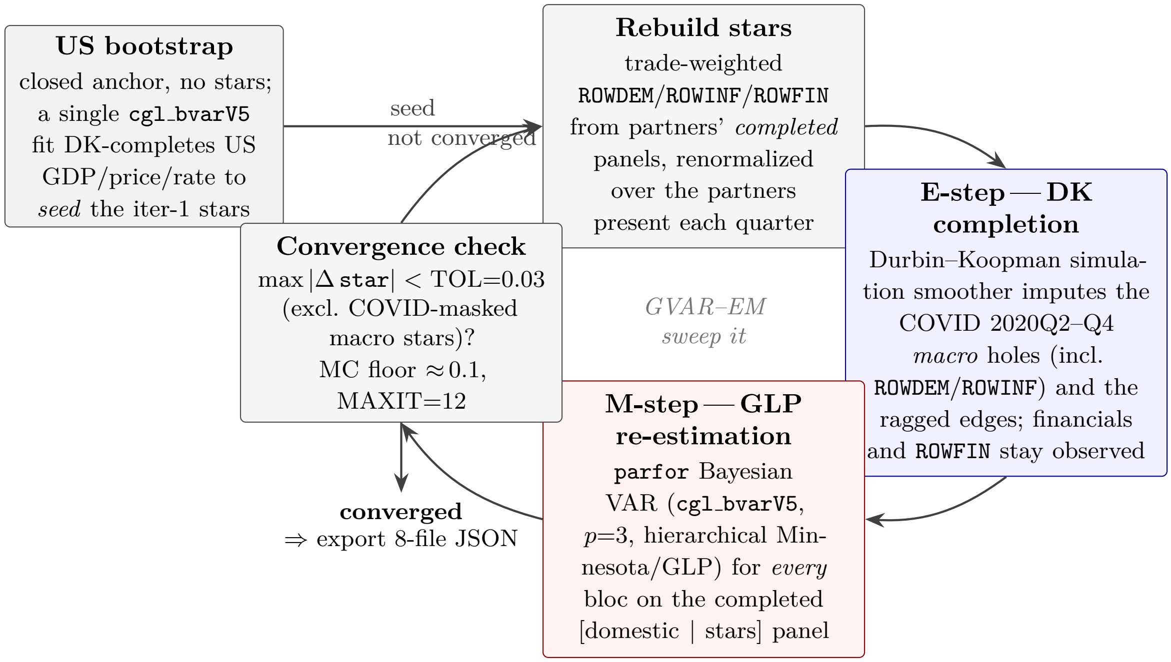

Figure 7: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the `ROWDEM/ROWINF/ROWFIN` stars from partners' completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc's single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and `ROWFIN` held observed) and the ragged, unbalanced-panel edges—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, $\max|\Delta\,\texttt{star}| < \mathrm{TOL} = 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is Monte-Carlo noise with floor ≈ 0.1), capped at $\mathrm{MAXIT} = 12$ iterations.

sweep.

Algorithm 2 states the loop (`pipeline/export_bvar_gvar24_loop.m`). Setting $\textsc{maxit} = 1$ recovers the single-pass mode used for the standalone blocs, whose stars are read off data once. Relative to the classical fixed-weight GVAR — stars built once from observed data on a balanced panel, one OLS pass per country — every departure here serves the unbalanced panel: model-based completion flowing *into* the stars, per-quarter renormalized weights, Bayesian shrinkage with the COVID-as-missing treatment, and the star fixed point itself. The out-of-sample evaluation of Section 11 races exactly this apparatus against its classical ancestor.

5.3 Solve-time coupling: a linear fixed point

At solve time the coefficients are frozen, and the key observation is that the conditional-forecast operator of Section 4 is *affine in the pinned values for a fixed pin pattern*: the filter and smoother

Algorithm 2 GVAR-EM estimation loop (`export_bvar_gvar24_loop.m`).

Input: per-bloc observed panels with `NaN` holes; trade weights; eligibility rules; TOL $= 0.03$, MAXIT $= 12$; fixed COVID window 2020Q2–Q4.

Output: per-bloc posterior means and draws; converged stars; the eight-file export.

$\triangleright$ Bootstrap. Complete the US panel with no stars (COVID macro cells masked); its posterior-mean completion seeds the first-sweep stars. Under US_ENDOGENOUS the US then joins the swept set with its own `ROW*_US` stars.

for $m = 1, \ldots,$ MAXIT:

  rebuild every receiver's stars (14) from the frozen start-of-sweep completed panels;

  parfor each bloc c: mask the COVID macro cells (domestic macro columns + `ROWDEM`, `ROWINF`; financials and `ROWFIN` stay observed); one `cgl_bvarV5` call $\Rightarrow$ completed panel + $(\hat{\beta}_c, \hat{\Sigma}_c)$ + draws;

  commit the new completions; mc $\leftarrow \max_c \|\Delta x^*_c\|_\infty$ (COVID macro-star cells excluded);

  if $m > 1$ **and** mc $<$ TOL: **break**.

Export each bloc to the eight-file JSON contract.

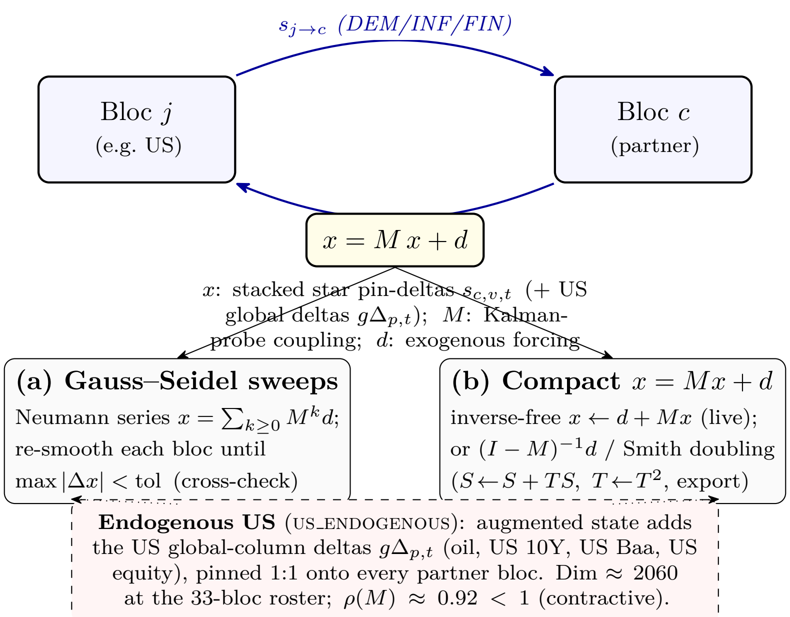

Figure 8: Solve-time cross-country coupling as the affine fixed point $x = M\,x + d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss–Seidel/Neumann sweeps through the Durbin–Koopman smoother, used as the cross-check, and (b) the compact M-state recursion, solved inverse-free by the Neumann fixed-point $x \leftarrow d + Mx$ on the production path (the explicit $(I - M)^{-1}d$ / Smith doubling are reserved for the small frozen-anchor case and the export-time precompute, since the in-browser $O(\text{dim}^3)$ inverse is prohibitive at dim $\approx$ 2060). With the US endogenous the augmented state has dimension $\approx$ 2060 and spectral radius $\rho(M) \approx 0.92 < 1$, so the iteration contracts and the two routes agree to $\sim 10^{-9}$.

are linear in the data. Each bloc's level forecast can therefore be written

$$x_c \;=\; a_c \;+\; \sum_v \sum_{s=0}^{H-1} K_c[v][s]\; \Delta^*_{c,v,s}, \tag{15}$$

where a_c is the bloc's forecast with its *own* pins active and its stars and global columns held at baseline, $\Delta^*_{c,v,s}$ is the deviation pinned on star (or global) column v at horizon s, and the kernel $K_c[v][s] \in \mathbb{R}^{H\times n}$ is the smoothed response to a unit bump of that single cell. The kernels are computed from one filter–smoother pass per bloc by value-independent delta recursions (`affineBlocKernels`, numerically identical to explicit probing to below 10^{-9} and an order of magnitude faster), and they depend only on the bloc's coefficients and its own-pin *pattern* — never on pin values or on partners — so they are cached and reused across every value change of an interactive drag.

The solver never rebuilds star *levels* from partner levels. It pins each receiver's star column at its own baseline plus the trade-weighted partner *deviation*:

$$\Delta^*_{c,\mathrm{DEM},t} = \sum_j \tilde{w}_{cj} \cdot 4\left[\left(x^{\mathrm{scen}}_{j,t} - x^{\mathrm{scen}}_{j,t-1}\right) - \left(x^{\mathrm{base}}_{j,t} - x^{\mathrm{base}}_{j,t-1}\right)\right], \qquad \Delta^*_{c,\mathrm{FIN},t} = \sum_j \tilde{w}_{cj}\left(r^{\mathrm{scen}}_{j,t} - r^{\mathrm{base}}_{j,t}\right), \tag{16}$$

read off the partners' gdp/price/rate columns (at $t = 0$ the shared last in-sample level cancels in the difference). Stacking all star deviations — and, with the US endogenous, the deviations of the four US-determined global columns (Section 5.4) — into a vector x, the deviation system is the linear fixed point

$$x = M\,x + d, \tag{17}$$

where d collects the exogenous forcing (each bloc's own pins evaluated at zero partner deviation, plus any long-run drift, Section 8.1) and M carries three coupling families: star $\leftarrow$ partner star (through the kernels and weights, equations (15)–(16), the US on both sides), the US-global definition rows (a global column's deviation as the US's own response to its star pins), and star $\leftarrow$ global (a partner's star response to its own global pin). At the deployed roster the state dimension is $33\times3\times20+4\times20 = 2060$, and the shipped operator has spectral radius

$$\rho(M) = 0.947 \;<\; 1 \tag{18}$$

(the value stored in the shipped coupling artifact, 0.946626, eigendecomposition-exact — before 2026-07-17 the field read 0.935044 from an under-converged power iteration on a complex dominant pair; it moves with each re-estimate), so the map is contractive, $(I - M)$ is nonsingular, and the fixed point is unique.

The fixed point is solved three equivalent ways, all implemented in `gvarSolve.ts`:

(a) **Gauss–Seidel sweeps**: re-condition each bloc in turn on the freshest partner deviations until the maximum level change falls below 10^{-4} (at most 12 sweeps) — the term-by-term Neumann sum $\sum_k M^k d$, and the reference the other routes are validated against;

(b) **closed form** $x = (I - M)^{-1}d$, using a precomputed inverse (below), followed by one conditional-forecast pass per bloc to reconstruct all free variables;

(c) **inverse-free Neumann iteration** $x \leftarrow d + Mx$ (tolerance 10^{-10} relative, capped at 2000 iterations) when the precomputed inverse is unusable — the in-browser $O(\dim^3)$ inverse at $\dim = 2060$ would take the better part of an hour and is never built live.

The routes agree to 10^{-6} or better in the validation battery. Divergence is *detected, not prevented*: if a future roster or degenerate pin set pushed $\rho(M) \geq 1$, the iterations' convergence flag trips, a console warning fires, and the result is not cached — but a path is still returned (Section 13.2).

The coupling artifact and the model signature. M depends on the models and the pin *pattern*, not the pin values, so for the default (empty) pattern both M and $(I - M)^{-1}$ are precomputed offline — the inverse by a BLAS solve in MATLAB (`app/scripts/invert_coupling.m`; a NumPy fallback ships alongside) — rounded to six significant figures, gzipped (`coupling_default.json.gz`, about 30 MB compressed), and decompressed in the browser at load. Hydrating it reduces the first complete solve from a cold build of many minutes to a few seconds; it is mandatory infrastructure, not an optimization. A stale artifact is refused by a *model signature*: per bloc, the dimensions (n_c, p_c, H_c) and three six-digit fingerprints of the posterior mean — the first equation's intercept $\hat{\beta}_{0,0}$, the far corner of the last lag row $\hat{\beta}_{n_c p_c,\, n_c - 1}$ (zero-indexed; recall the intercept is row zero), and the leading residual variance $\hat{\Sigma}_{0,0}$. Any re-estimation flips the signature, and the solver falls back to route (c). A pin on any coupled bloc changes the pattern and likewise bypasses the cached inverse; only the changed bloc's kernels are re-probed.

5.4 Global commons

Four US-determined series are carried by every bloc: the WTI oil price (`GLB_OIL`), the US 10-year yield (`GLB_USLR`), the US Baa corporate yield (`GLB_USBAA`), and US equity (`GLB_USEQ`). In the US model they are ordinary endogenous variables; every other bloc carries the same identifiers (matched by name, never by position) as weakly exogenous columns that the solver pins *by deviation*:

$$\widehat{\mathrm{GLB}}_{c,t} = \bar{\mathrm{GLB}}_{c,t} + \left(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \bar{\mathrm{GLB}}_{\mathrm{US},t}\right), \tag{19}$$

each bloc's own baseline plus the US deviation. Pinning the US *level* instead would inject the blocs' baseline gaps (tens of 100 ln units) as spurious shocks; the deviation convention gives a zero-scenario no-op exactly and an identical common shock under conditioning. With the US endogenous the global deviations are unknowns of (17) — this is precisely the second and third coupling family — so a partner shock that moves the US moves the commons, which feed back to every other bloc, without double counting. On screen the commons are conditionable on the US and read-only elsewhere.

5.5 Operating the global solve

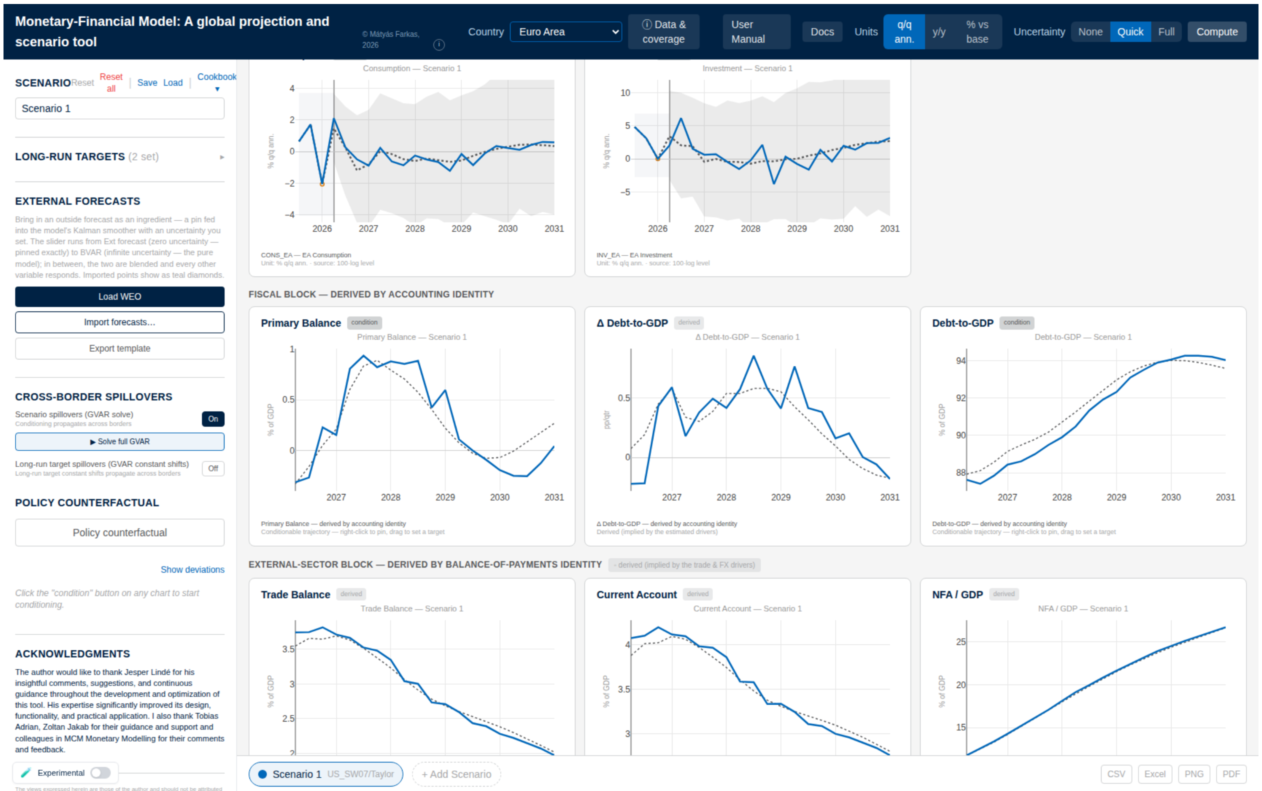

Figure 9: The euro-area view after Solve full GVAR on a US tightening scenario (spillovers On). The conditioning entered in the United States has propagated cross-border: euro-area consumption and investment deviate from baseline (dotted), and the derived satellite blocks — the fiscal block of §6 and the external-sector block of §7 — re-solve on the coupled paths. Switching countries after a full solve always shows the same converged system state.

By default a scenario moves only the economy on screen; partners stay frozen, and a status line says so. Cross-border transmission is explicit and two-tier:

- **Scenario spillovers (GVAR solve)** (left panel, **Cross-border spillovers**) switches on live propagation. While editing, the tool shows the *direct preview* — one cross-country hop through the cached kernels, no fixed-point feedback, sub-second — flagged by an amber **GVAR DIRECT preview** badge above the grid.
- **Solve full GVAR** (or **Compute**) runs the complete fixed point (17) — all feedback loops, including RoW→US — in a few seconds, and turns the badge blue (**GVAR full solve**), with the sweep count and solve time in the read-out. The full solve reflects the pins at the moment it runs; edit afterwards and press it again.

For small shocks preview and full solve are close (the second-round echoes are smaller than the first hop); for large shocks, or when the *receiving* country is the object of interest, run the full solve before quoting numbers.

Example 3 (A global shock reaching a partner)**.** *Load the **Cookbook** energy-supply recipe on the United States (**Load + restriction**): an oil-price spike with its documented supply-shock signature. Switch the **Country** dropdown to the euro area and set **Units** to **% vs base**: the euro-area GDP line is flat at zero — the shock has not been propagated. Return to the US, press **Solve full GVAR**, and switch back: euro-area GDP now shows the imported hit, at the 2026Q1 vintage building to roughly* -0.8 *percent of the baseline level by the end of the horizon. The difference between the flat line and the solved one is exactly the content of the coupled system: a country-only read cannot see a foreign shock at all.*

5.6 Euro-area members: one doorway

The euro members do not set their own interest rate, and the tool respects this: the area-wide instruments — the ECB policy rate, the short rate, the area corporate yield — are *endogenous* in the EA-aggregate bloc and enter each member's model as *weakly exogenous* columns, alongside the global commons and the member's foreign stars. Shocks reach a member through one doorway, by *deviation transmission*: after a full solve, the EA-aggregate deviation is mapped identifier-by-identifier onto the member's corresponding columns (policy rate onto policy rate, area yield onto area yield, area activity and prices onto the member's rest-of-euro-area aggregates), each pinned at the member's own baseline plus the area deviation, and the member's own estimated dynamics translate the area shock into its GDP, inflation, and yields. A member never updates silently: an amber note above its grid says when a US or EA scenario has not yet been passed through, and **Solve full GVAR** runs the pass.

The reverse direction exists for two national instruments. Pinning a member's ECB-rate column mirrors one-for-one into the area policy rate; and a member sovereign-spread scenario feeds the area long yield by the member's GDP weight,

$$\Delta\mathrm{YIELD10}_{\mathrm{EA},t} = \Delta\mathrm{Bund}_t + w_m\,\Delta\mathrm{spread}_{m,t}, \tag{20}$$

the two channels additive, with w_m the member's share in euro-area GDP (real Eurostat current-price weights, shipped in `ea_gdp_weights.json`; a missing weight suppresses the channel rather than inventing one). An Italian spread scenario thus moves the area yield about forty times more than an Estonian one.

5.7 Outside forecasts across borders

A hard external forecast (trust at 100%) is indistinguishable from a hand pin and rides the cached closed-form path. A *soft* one changes the effective filter gains through (13), which the value-independent kernels of (15) cannot represent; the solver therefore routes any request with a finite trust variance to the Gauss–Seidel path, threading the per-cell variances into every sweep's conditional forecast. Exact but slower — a few seconds across the panel — so an uncertain import propagates cross-border only on **Solve full GVAR** or **Compute**, never live while typing; the viewed economy still updates instantly.

5.8 Per-draw coupled bands

With the **Experimental** switch on, **Compute** re-solves the whole coupled system once per posterior draw (N_b of the D shipped draws, selected by deterministic stratification; parameter uncertainty only, no innovation shocks; only the viewed bloc's model and pins are rebuilt per draw, partners held at their means) and reports the empirical 16th/84th percentiles of the resulting conditional cloud. This band pinches naturally at every pin — and can collapse too tightly when only a point or two is pinned, which is why it is opt-in (Remark 2). Each draw runs the full Gauss–Seidel budget (tolerance 10^{-4}, 12 sweeps), so **Full** on a coupled bloc is the slowest computation in the tool.

5.9 Identification: what is assumed where

The separable estimate-then-couple design rests on stated assumptions. *Weak exogeneity of the stars*: each bloc is estimated conditional on its trade-weighted foreign aggregates, valid when no single partner dominates the weights — the granularity condition of Chudik and Pesaran (2016); the per-component renormalization is designed to preserve it. The EM loop relaxes the one-shot version of this assumption by iterating the conditioning information to mutual consistency, and the endogenous US removes the frozen-dominant-unit asymmetry altogether; what keeps the relaxed system well posed is the contraction (18). *Euro-area monetary identification*: policy is set for the area as a whole — endogenous in the aggregate, predetermined for each member — so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national experiments. *The scenario responses are conditional projections*: a multi-period pin bundles the direct conditioning with the coupled feedback and is not a structural impulse response (Section 11 keeps this distinction explicit).

6 Public finances

Twenty-three economies carry a fiscal block: the six majors (US, euro area, Japan, UK, Canada, China) and the seventeen euro-area member states (Appendix B). The design follows the stock-flow-consistent division of labour shared by the CBO's budget mechanics and the IMF's financial-programming and debt-sustainability frameworks (Escolano, 2010; International Monetary Fund, 2013): the BVAR forecasts a small set of fiscal *flow drivers* jointly with the macroeconomy, while the debt stock and the headline balances are *derived* by the budget identity. Debt is never a free VAR variable — the identity has a near-unit autoregressive root $(1+i)/(1+g)$, and small coefficient errors compound into divergent paths; imposing the identity by construction is the discipline the frameworks demand (`app/src/lib/compute/fiscalAccounting.ts`).

6.1 Estimated drivers

Four drivers enter the VAR as ordinary `lin` variables and inherit the full estimation machinery of Sections 3–4:

- `FREV_GDP`, `FPEXP_GDP`: general-government revenue and primary (non-interest) expenditure, percent of GDP, annual ratios;
- `EFFI`: the *effective* interest rate — net interest over lagged gross debt, percent per annum. The stock reprices only as maturing securities refinance, so this rate lags market yields and is estimated separately from them;
- `SFA`: the stock-flow adjustment, percent of GDP — the deficit–debt discrepancy (intragovernmental holdings, valuation effects, other below-the-line operations), constructed on history as the *exact* residual that closes the recursion below, with the pandemic quarters imputed rather than observed.

The three GDP-ratios carry the stationary `WN` own-lag prior of Section 3.2 — they are mean-reverting shares, and the random-walk center would let the projected ratio drift at the long end; under `WN` the US revenue ratio, for example, reverts to its in-sample mean of about $17\frac{1}{2}$ percent instead of freezing at its end-of-sample value. `EFFI` keeps the `RW` prior: a stock-average rate is genuinely persistent. Each fiscal bloc ships a `fiscal_meta.json` recording the column indices the recursion reads (`idx` = {`gdp`, `deflator`, `revenue`, `primaryExp`, `effRate`, `sfa`}), the seed debt ratio d_0, and the structural constant `sfaDefault` (the long-run SFA mean; e.g. US 1.75, EA 0.33, JP 2.23, UK 3.15, CA 3.19, CN 4.02 percent of GDP).

6.2 The debt recursion

Let d_t denote debt-to-GDP in percent, i_t^{eff} the effective rate in percent per annum, and g_t^n the *quarterly* nominal growth rate in decimal, read off the level-space forecast as

$$g_t^n \;=\; \tfrac{1}{100}\Big[\big(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\big) + \big(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\big)\Big], \tag{21}$$

the sum of the real-GDP and deflator log differences in the stored $100\ln$ space, divided by 100 (a bloc whose GDP series were nominal would omit the deflator term via the `gdpIsNominal` switch; no shipped bloc currently uses it). With the primary balance $pb_t = rev_t - pexp_t$ (percent of GDP, annual, positive = surplus), the debt stock rolls forward one quarter as

$$d_t \;=\; d_{t-1}\,\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n} \;-\; \frac{pb_t}{4} \;+\; \frac{\overline{sfa}}{4}, \qquad d_{-1} \equiv d_0, \tag{22}$$

exactly as coded: the annual rate quarterized by 400, the quarterly decimal growth entering directly, and one quarter's worth of the annual flow ratios accruing per step. The quarterly convention is pinned by a back-cast test that reconstructs the observed debt ratio from realized drivers. Two

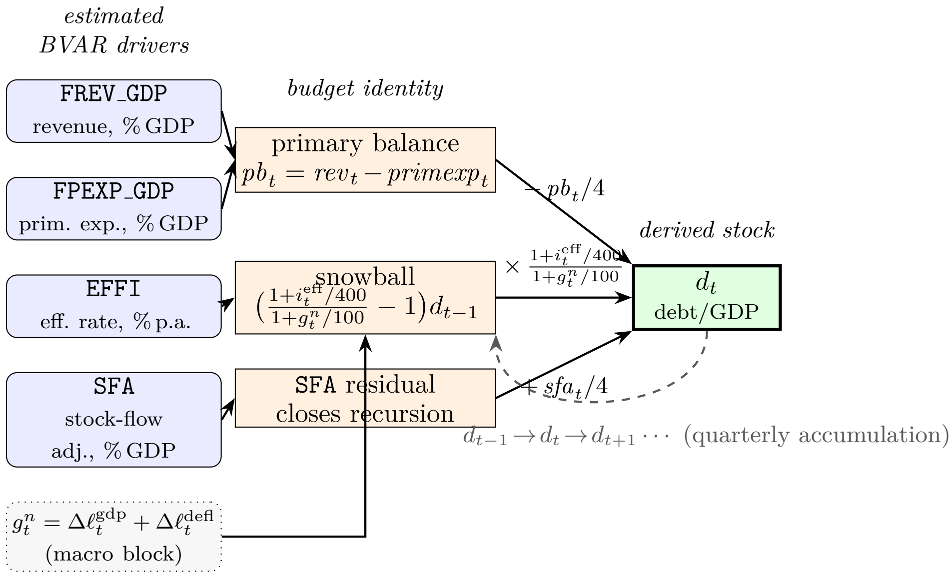

Figure 10: Fiscal stock-flow schematic. The four estimated BVAR drivers (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`; Section 6.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (§6.1); the effective rate `EFFI` and nominal growth g^n_t (21) form the interest–growth *snowball* (23); and the estimated `SFA` enters as the residual that closes the recursion. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1}\,(1{+}i^{\text{eff}}_t/400)/(1{+}g^n_t/100) - pb_t/4 + sfa_t/4$ (22), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 6.2).

derived flows complete the identity: the interest bill $int_t = i^{\text{eff}}_t\, d_{t-1}/100$ (annual, percent of GDP) and the deficit $def_t = int_t - pb_t$. Subtracting d_{t-1} from (22) isolates the automatic *snowball* from the discretionary balance,

$$\Delta d_t = \underbrace{\left(\frac{1+i^{\text{eff}}_t/400}{1+g^n_t} - 1\right)d_{t-1}}_{\text{snowball (interest–growth)}} - \frac{pb_t}{4} + \frac{\overline{sfa}}{4} \approx \frac{i^{\text{eff}}_t/400 - g^n_t}{1+g^n_t}\,d_{t-1} - \frac{pb_t}{4} + \frac{\overline{sfa}}{4}, \tag{23}$$

the quarterized $(i-g)$ form: debt rises mechanically whenever the quarterized effective rate exceeds quarterly nominal growth.

Remark 4 (The SFA in the roll-forward is the structural constant). Equation (22) deliberately uses the constant $\overline{sfa}$ = `sfaDefault`, *not* the forecast `SFA` column, even though the column is estimated and displayed with its own chart, dynamics, and bands. The forecast column responds to everything else in the VAR — including long-run anchors' cross-responses — and can be dragged to spurious extremes that would distort the derived debt path; the long-run mean is the economically correct closure for a projection. The `SFA` *chart* shows the model's forecast of the discrepancy; the debt *arithmetic* uses the stable constant. Conditioning the SFA chart is still meaningful — it moves the rest of the system through the estimated covariances — but it does not enter (22).

The recursion (21)–(23) is `computeFiscalBlock`, the only fiscal recursion wired to the charts:

it produces the baseline, scenario, and counterfactual debt paths and, run per posterior draw, the fiscal bands. It is *passive*: lagged debt enters every term, so a single forward sweep is exact, and there is no debt feedback onto the primary balance or the rate — the displayed balance is exactly $rev_t - pexp_t$ and the displayed rate exactly the BVAR forecast. Three derived charts are displayed, in a row labelled **Fiscal block — derived by accounting identity**: the **Primary Balance**, the per-quarter change Δ **Debt-to-GDP** (a pure identity output, not conditionable), and **Debt-to-GDP**. The remaining identity outputs (interest, deficit, overall balance, snowball) are computed internally but not charted.

Remark 5 (Timing of the debt ratio). The denominator convention follows GFSM 2014 and the Eurostat quarterly Maastricht series: d_t is the *end-of-quarter* gross debt stock divided by *contemporaneous rolling-annual* nominal GDP, $d_t = D_t / \sum_{s=t-3}^{t} Y_s$ — the four-quarter sum *ending in* quarter t, not lagged GDP. All fiscal ratios (`FREV_GDP`, `FPEXP_GDP`, debt/GDP) use the same contemporaneous four-quarter denominator, and d_0 is seeded at the last observed debt stock over the four quarters ending at that observation. The one deliberately *lagged* object is the effective rate, $\texttt{EFFI}_t = 400\, int_t / D_{t-1}$: interest accrues on the pre-existing stock, the financial-programming convention. In the roll-forward (22) the denominator's growth enters through the factor $(1 + i_t^{\text{eff}}/400)/(1 + g_t^n/100)$ with g_t^n the quarterly nominal-GDP growth of the same quarter; in sample, any wedge between quarterly growth and the rolling-sum growth is absorbed into the estimated `SFA` residual by construction, so the identity holds to machine precision on history.

6.3 Targeting a debt or balance path

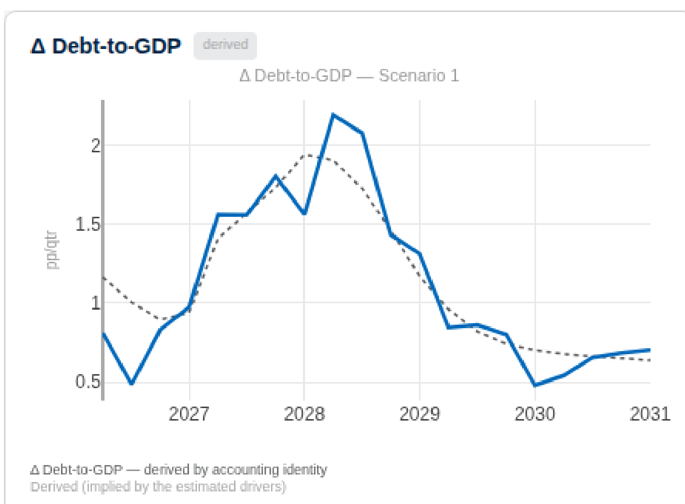

Figure 11: The derived debt block under the tightening scenario: the quarterly change in debt-to-GDP (solid) rises above baseline (dotted) as the snowball term absorbs the higher effective rate, then reverts. Debt panels are conditionable: right-click pins a point, dragging sets a target path, and the smoother back-solves the primary balance that delivers it (§6.3).

The recursion also runs backward. Given a target debt path $\{d_t^{\text{tgt}}\}$, the primary balance required at each quarter is the exact inverse of (22),

$$pb_t \;=\; 4\left[\frac{d_{t-1}^{\text{tgt}} + int_t/4}{1+g_t^n} \;-\; d_t^{\text{tgt}}\right] \;+\; \overline{sfa}, \tag{24}$$

with the lag taken along the target path and $int_t = i_t^{\text{eff}}\, d_{t-1}^{\text{tgt}}/100$ (`requiredPrimaryBalance`). The tool evaluates the effective rate and growth on the unconditional forecast, computes (24), and pins the combination $rev_t - pexp_t = pb_t$ as a linear-combination constraint (Section 4.2); the smoother finds the most plausible revenue/expenditure mix.

Example 4 (Drawing a debt path)**.** *On a fiscal economy, right-click the* ***Debt-to-GDP*** *chart where the path should pass — on this chart the dragged value is the absolute target,* 100 *meaning* 100 *percent of GDP — and shape the trajectory; right-click a quarter to pin or release it. A* ***Primary Balance*** *target set the same way is hit* exactly *(it is one linear constraint). A* ***Debt-to-GDP*** *target is hit* approximately*: the inversion* (24) *conditions on the model's unconditional rate and growth, which themselves respond a little to the implied consolidation, and the tool does not iterate on that feedback — expect a miss of up to a point or so of GDP over the horizon. For an exact decimal the instrument is the balance target; for a coherent consolidation scenario the debt target is the natural handle. Note that conditioning* any *fiscal variable implicitly conditions the derived row — a note next to the block header says so.*

Remark 6 (Fiscal bands are identity-consistent, and re-centred)*.* Each band draw pushes its own simulated drivers through (22), so the debt band widens exactly as fast as the underlying budget-and-growth uncertainty implies. Because the roll-forward compounds, the per-draw debt distribution is right-skewed: its median sits below the debt path computed at the mean drivers, which would leave the central line hugging the band's upper edge. At render the band is therefore re-centred — each quarter's band midpoint is shifted onto the displayed path, preserving the band's width and asymmetry (the fan-chart convention "central projection ± uncertainty"); at actively pinned quarters the band collapses onto the line. The same treatment applies to the external position below.

7 The external accounts

Three economies — the United States, the euro area, and China — carry an external block that derives the trade balance, the current account, and the net foreign asset (NFA) position from the estimated trade, price, and exchange-rate variables by the balance-of-payments identity (`app/src/lib/compute/externalAccounting.ts`). All three charts are read-only (**derived** badges): to move them, condition the drivers — exports, imports, the exchange rate, growth — and the external row follows.

With b_t the NFA-to-GDP ratio in percent, seeded at the latest published net international investment position ($b_{-1} \equiv nfa_0$: US -66.75, EA $+11.0$, CN $+20.35$ percent of GDP at the current vintage), the per-quarter identity as coded is

$$tb_t = 100\,\frac{X_t - M_t}{Y_t}, \qquad ca_t = tb_t + \frac{i^{\text{NFA}}\,b_{t-1}}{100}, \tag{25}$$

$$val_t = -\,\phi^{\text{FX}}\,b_{t-1}\,\frac{\Delta\ell_t^{\text{FX}}}{100}, \qquad b_t = \frac{b_{t-1}}{1+g_t^n} \;+\; \frac{ca_t}{4} \;+\; val_t, \tag{26}$$

where X_t, M_t, Y_t are the exponentiated export, import, and GDP levels (for the US and the euro area a real-trade-to-real-GDP ratio, validated against the published balance-of-payments ratios; for China a deflated real index with the shipped scale factors, monthly trade aggregated to quarterly), $\Delta\ell_t^{\text{FX}}$ is the quarterly change of the $100\ln$ effective exchange rate (positive = home appreciation, which shrinks the home-currency value of net foreign assets — hence the sign), and g_t^n is quarterly nominal growth as in (21). The stock update divides the inherited ratio by $(1+g_t^n)$ — the automatic denominator effect — adds one quarter of the annual current account, and adds the valuation term; the return on the stock enters through the income term of ca_t, computed on the undiluted lagged ratio (the second-order cross-terms this drops are noted in the code header).

Two inputs are stated modelling assumptions, not estimates, and are flagged as such in the shipped metadata: the return on net foreign assets $i^{\text{NFA}} = 3$ percent per annum, and the share of the position exposed to the exchange rate $\phi^{\text{FX}} = 0.4$. Treat the *level* of the income and valuation terms as indicative and their direction as the reliable part.

8 Long-run anchors

The long-horizon behaviour of the forecasts is inherited from the estimated VAR, whose level dynamics carry roots on or near the unit circle; the far end of a projection can therefore drift away from any value the user regards as a credible steady state — a two-percent inflation objective, a potential growth rate, a stable revenue ratio. The **Long-run targets** panel lets the user pin the terminal forecast of a chosen variable to a number, exactly and instantly. The mechanism is a deterministic counterfactual on the companion drift in the spirit of Villani (2009), implemented in `app/src/lib/compute/driftSetter.ts`.

The shipped consensus set. The tool ships a reviewed default anchor set (`default_anchors.json`; July-2026 transition-default revision: 35 anchors across 28 blocs) tuned under the now-default steady-state transition. Because that path phases the adjustment in rather than shifting the whole path, the set is anchored to *official* long-run values wherever the model shows a genuine terminal gap — central-bank inflation targets throughout (euro-area members at the ECB's two per cent, national targets elsewhere), medium-term potential growth where the raw path is clearly off, and the effective-rate / revenue / expenditure trio on fiscal blocs. This is a real gain of the transition shape: sixteen of these official-target anchors (Czechia, Norway, Portugal, Thailand, Russia, Sweden and more, all to two-to-four per cent) are rejected as harmful under the rigid shift, which would jump their near term, and all are clean under the transition. Anchors are admitted only if the sensitivity clears the guard of §8.1, and the shipped set is verified by the smell test (`anchor_smell.mjs`) and locked by a panel-wide non-explosiveness regression. Two guards still bite — both shape-independent, being properties of the estimated steady state rather than the path: a target that lands within the redundancy threshold of the model's own terminal is dropped, and one that would displace an unanchored headline variable by more than a point at the terminal is dropped or co-anchored. The euro-area primary-expenditure anchor stays removed for the second reason (its cross-drag drives the euro-area policy-rate terminal negative). A handful of high-inflation emerging markets — Colombia, Hungary, Türkiye, Israel, Malaysia — are shipped model-driven and flagged: their estimates link inflation and growth so tightly that pinning one drags the other several points at the steady state, which no univariate anchor and no choice of path can repair; these await re-sourcing or a joint re-estimation. Argentina (hyperinflation decay) and Japan (anchors disabled) also ship unanchored.

8.1 The drift setter

Write the deterministic part of the companion recursion (4) as $Y_{T+h} = T^h Y_T + S_h\, c_\alpha$ with

$$S_h = \sum_{k=0}^{h-1} T^k, \tag{27}$$

the partial sum of companion powers, built once per bloc (H matrix products, cached). A shift δc of the drift moves the forecast by exactly $\Delta Y_{T+h} = S_h\, \delta c$ — an algebraic identity, verified against the iterated recursion to 10^{-11}. We deliberately avoid the closed-form steady state $(I - \sum_\ell B_\ell)^{-1} c$: for near-unit-root level VARs that matrix is near-singular and the "steady state" a meaningless number of order 10^5, whereas the finite-horizon operator (27) is well posed. A target on variable j shifts only j's own equation, $\delta c = e_j\, \delta c_j$, and the coherent system-wide response is the j-th column, $\Delta Y_{T+h} = S_h[:, j]\, \delta c_j$. The scalar is chosen so the *displayed* terminal forecast lands on the target $\psi_j^\star$:

$$\delta c_j = \frac{\psi_j^\star - \psi_j^0}{\mathrm{sens}_j}, \qquad \mathrm{sens}_j = \begin{cases} 4\big(S_H[j,j] - S_{H-1}[j,j]\big), & \texttt{log}\ \text{variable}, \\ S_H[j,j], & \texttt{lin}\ \text{variable}, \end{cases} \tag{28}$$

with ψ_j^0 the current terminal display forecast — the slider's default position, so an unmoved slider is a byte-exact no-op. A variable whose own sensitivity is below the guard $|\mathrm{sens}_j| < 0.3$ has its slider disabled ("near-singular response"): forcing a target there would require an explosive drift.

Two paths to the target: steady-state transition (default) or rigid shift. Equation (28) fixes *how far* the terminal must move; a second choice — the **Path to target** toggle — fixes *how the adjustment is distributed across the horizon.* The constant-drift response $\Delta Y_{T+h} = S_h[:, j]\, \delta c_j$ places the *entire* effect from the first forecast quarter: because $S_1 = I$, a growth variable's displayed deviation at $h = 1$ is $4\, \delta c_j$ — the *full* terminal gap, applied at once and held flat. This **rigid shift** is exact at the terminal and band-coherent, but it *jumps* the near term at the origin and, for a large gap, can manufacture a near-term contraction or deflation. The shipped default is instead a

steady-state transition: the same terminal δc_j is phased in along the anchored variable's own estimated convergence profile

$$w_j(h) = \frac{S_h[j,j]}{S_H[j,j]}, \qquad w_j(1) \approx \tfrac{1}{S_H[j,j]} \approx 0, \quad w_j(H) = 1, \tag{29}$$

so the near term stays data-consistent (no origin jump) and the forecast *glides* to the target by the end of the horizon at the model's own mean-reversion speed. This is the reactive, no-re-estimation counterpart of a steady-state (mean-adjusted) prior in the spirit of Villani (2009): rather than translating the mean path by a constant, it lets the estimated dynamics carry the forecast from the observed edge to the long-run target. It is finite-horizon by construction — the identity $I - T^h = (I-T)\,S_h$ shows w_j is exactly the normalised $(I - T^h)$ operator — so it never touches the singular level steady state $(I - \sum_\ell B_\ell)^{-1}c$ that motivated (27). The terminal is identical to the rigid shift, so anchored targets are hit under either path; only the near-to-medium term differs. Each horizon is still a rigid translation, so band widths remain invariant (Remark 3). Cross-variable drag phases in at the source anchor's rate applied to the exact terminal cross-response.

Several targets: per-variable or joint. With two or more targets the cross-responses interfere — a drift on equation j moves every other anchored terminal through $S_H[k,j]$. The panel exposes a **Multiple-target solve** toggle:

- **Per-variable** (the default): each δc_k from its own diagonal sensitivity (28). Stable and bounded, but approximate — it ignores the cross-responses, so with several targets each terminal lands slightly off its stated value.

- **Joint**: solve the $K \times K$ display-unit system $M^{\mathrm{lr}}\,\delta c = \psi^\star - \psi^0$, whose diagonal is exactly (28), by Gaussian elimination with partial pivoting. Every anchored terminal then lands exactly — but a near-collinear target set (say two anchored interest rates the dynamics cannot separate in the long run) makes the system ill-conditioned: the exact solution buys terminal precision with huge offsetting drifts (measured up to $|\delta c| \approx 459$ where the diagonal needed ≈ 11) that wreck the path elsewhere. A singular system falls back to per-variable numerically; an ill-conditioned but solvable one is returned and *flagged* (the joint drift exceeding twice the per-variable maximum trips the flag), and the user is expected to switch back.

Per-variable is the shipped default. The near-term-implausibility that once accompanied a large gap — at the 2026Q1 origin, anchoring the US deflator (model terminal 4.3 against the 2.0 target) once displayed a -2.1 point disinflation shock at the first forecast quarter — was the signature of the *rigid shift*; the steady-state transition of (29) removes it, gliding the same series from 3.5 smoothly down to 2.0 with a first-quarter move of under a tenth of a point. The two toggles are orthogonal: the solve mode (per-variable / joint) sets *how far* each terminal moves; the path (transition / rigid) sets *how the move is phased in*.

Bands, fiscal forwarding, and visibility. The response ΔY is added identically to the central path and to every shipped band quantile — a rigid translation, widths provably invariant (Remark 3) — and, in level space, forwarded into the fiscal recursion (22), so the debt path responds coherently: a real-growth target owns the direction of the debt ratio through the $(i - g)$ snowball, which is why its row carries the tag **sets debt path**. Judgment is flagged: the affected chart title turns amber with a **judgement** badge stating target and estimate, and the grid-level status line counts the anchored variables. Every control starts at the model estimate, and clearing all targets restores the raw model byte-exactly.

Consensus preset. Blocs with publicly defensible long-run values carry an **Apply consensus anchors** preset — deliberately minimal (35 anchors across 28 economies at the July-2026 transition revision): potential growth and the inflation objective per economy, plus the effective-rate/revenue/expenditure trio for fiscal blocs; never policy rates, unemployment, or asset prices, whose collinear anchors would ill-condition the joint system. Each kept anchor was individually verified — under the shipped transition path — to fix a genuine terminal bias without displacing another headline variable. Where a preset ships it is on by default and fully reversible (**Clear**); the model-driven-and-flagged economies of the preceding paragraph ship none and stay entirely model-driven.

On a coupled bloc the target is marginal. The drift setter operates on one bloc's companion. On a GVAR economy it shifts that bloc's own long run; partners are not re-solved unless the second spillover switch, **Long-run target spillovers (GVAR constant shifts)**, is on. That channel is instant — the drift deviation enters the coupled system only through the forcing term d of (17), leaving M and its precomputed inverse untouched, and the receiving country's banner turns amber with **+ spillover from ...** naming the origin — judgment stays visible across borders.

8.2 The in-prior long-run prior

A second, offline mechanism exists: the "prior for the long run" (PLR) of Giannone et al. (2019), which augments the prior of Section 3.2 with dummy observations shrinking a set of long-run linear combinations Hy toward non-cointegration. It is a proper Bayesian prior — it changes the posterior and requires re-estimation — and is therefore part of the estimation pipeline, not the browser. A US head-to-head (run on an earlier, pre-expansion $n = 36$ US specification; the deployed bloc is $n = 40$) found the PLR acts mainly on the *bands*, tightening the GDP-growth 68% band by about 7 percent at unit tightness, while leaving the central long-run path essentially unchanged — the appropriate instrument for principled band tightening, not for pinning a terminal to a chosen number. The two mechanisms are complementary: to pin a long run interactively, use the drift setter; for disciplined bands, the PLR.

Remark 7 (A retired mechanism). An earlier release pinned long runs by entropic tilting (Robertson et al., 2005) — reweighting the posterior draws so the weighted terminal moment hits the target. It degenerated in practice: as the target moved away from the draw cloud the effective sample size

$1/\sum_d w_d^2$ collapsed toward one (a joint 13-constraint tilt reached ESS $\to 1$ at $D = 500$), leaving the anchored band resting on a handful of draws precisely where the user most wanted a firm answer. The drift setter replaced it wholesale — deterministic, exact at any target, instant — and the tilt is no longer wired to any interface control (its module remains in the repository, exercised only by its unit test).

9 Policy counterfactuals

A scenario asks what happens if the data come in differently; a policy counterfactual asks what a *deliberate policy change* would cause. The distinction is identification: the smoother's response to a pinned rate path is the estimated historical co-movement, which cannot tell a policy surprise from a response to something else. The counterfactual layer therefore borrows structure from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016), and overlays its identified responses on the scenario.

9.1 Impulse responses and shock inversion

The MMB catalogue in the tool lists 169 models; pre-computed impulse responses ship for 155 of them, each under up to nine interest-rate rules (Taylor, Smets–Wouters, CEE, Gerdesmeier–Roffia, Levin–Wieland–Williams, two Orphanides–Wieland variants, Coenen, Christiano–Motto–Rostagno). A model–rule pair is one JSON file of responses to a monetary (and, where the model has a fiscal sector, a government-spending) shock, with the rule embedded in the model's equilibrium when the responses were generated — the tool never evaluates a policy rule at run time. Rules without computed responses for the selected model are greyed out.

Let $h_r(s)$ denote the interest-rate response at horizon s to a unit monetary shock. Given a user rate adjustment Δr_t on a set of *constrained* quarters, the tool solves for the shock sequence by forward substitution on the lower-triangular Toeplitz system — a new shock only where the user pinned, zero elsewhere:

$$\varepsilon_t = \begin{cases} \dfrac{\Delta r_t - \sum_{j<t} h_r(t-j)\,\varepsilon_j}{h_r(0)}, & t \text{ constrained}, \\ 0, & \text{otherwise}, \end{cases} \tag{30}$$

so at unconstrained quarters the rate follows the decaying response of the earlier shocks — the model's rule, not a snap-back. The implied inflation and output responses are the convolutions $\Delta\pi_t = \sum_{j\le t} h_\pi(t-j)\,\varepsilon_j$ and $\Delta y_t = \sum_{j\le t} h_y(t-j)\,\varepsilon_j$; a parallel inversion runs on the fiscal shock for the spending sliders.

9.2 Re-conditioning and the response modes

The counterfactual is not an additive overlay of every variable: the DSGE deltas are imposed on the *key* variables and the BVAR fills in the rest. The **Macro response** selector offers two modes:

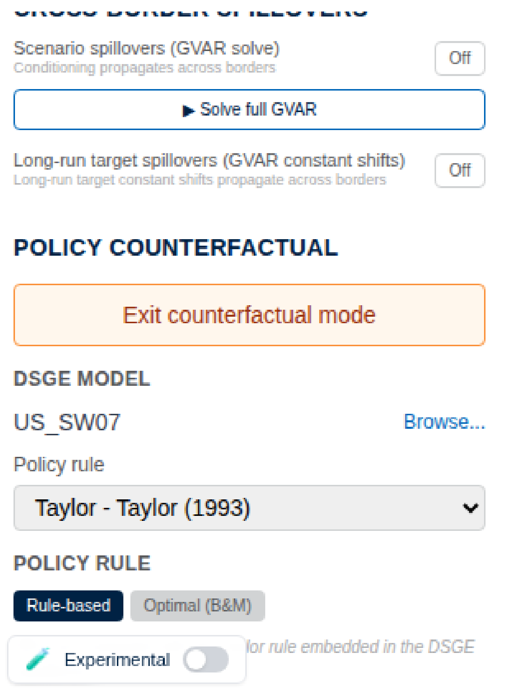

Figure 12: The counterfactual control rail: entering counterfactual mode selects the DSGE model (here US_SW07) and the policy rule — a named rule (Taylor 1993) or the optimal rule of §9.4 — whose impulse responses are layered on the BVAR scenario as described in §9.1.

- **DSGE/Taylor-rule (IRF-consistent)** (default): the rate is pinned at scenario plus Δr, inflation and output at scenario plus the DSGE deltas; all other variables are re-conditioned through the smoother, so the responses are consistent with both the identified transmission and the estimated covariances. One convention subtlety is handled per variable: the MMB inflation response is a *rate* (cumulated into the 100 ln price level, $\sum_{s\leq t} \Delta\pi_s/4$), the output-gap response a percent *level* deviation (applied directly, not cumulated), and a rate response an identity.
- **BVAR-endogenous (let adjust)**: only the rate path is pinned; everything else responds through the estimated co-movements, exactly as in a scenario. The two modes answer different questions — "what does this theory say the move would do" versus "what usually accompanies such a path" — and their disagreement is informative. If the selected pair has no computed responses the tool falls back to this mode and says so in an amber notice.

In both modes the displayed counterfactual is the scenario plus the smoother delta between the counterfactual pass and a reference pass with zero adjustment, converted to display units by the conventions of Section 4.3; a zero adjustment reproduces the scenario exactly.

9.3 Reading the result

The counterfactual draws as a dashed line in a distinct rust colour on every chart; the scenario line stays put. To isolate the policy effect, set **Units** to **% vs base**; in that view the **Level | Rate (IRF)** toggle selects between the cumulative effect on the *level* of each variable (the "% chg." reporting convention) and the quarter-by-quarter response in native units — the latter reproduces the MMB's own IRF plots exactly, since differencing the cumulated log-level deviation and scaling by four recovers the rate response. Uncertainty bands, when computed, belong to the scenario; the counterfactual is a single model-implied path.

Remark 8 (The rule choice is a first-order magnitude decision). The same one-quarter rate move can have an order-of-magnitude different effect across rules. In the shipped responses of the default US model (`US_SW07`, Smets and Wouters, 2007), the peak inflation response to a unit monetary shock under the model's own Smets–Wouters rule is about 32 times the response under the Taylor rule, the output response about 6 times. Treat pass-through magnitudes as model-and-rule dependent, never as a fact; flipping the model while the scenario stays put (each scenario tab carries its own model and rule) is the cheapest robustness check in the tool.

9.4 Optimal policy

Instead of hand-crafting the path, **Optimal (B&M)** computes it from the same sufficient statistics, following Barnichon and Mesters (2023): choose the shock sequence ε minimizing the quadratic loss

$$\mathcal{L} = \sum_{t=1}^{H} \Big[\lambda_\pi \big(\pi_t - \pi^\star\big)^2 + \lambda_y \big(y_t - y^\star\big)^2 \Big] \tag{31}$$

subject to the linear IRF mapping. Stacking the weighted Toeplitz operators, the minimizer is the Tikhonov-regularized least squares

$$\varepsilon^\star = \big(\tilde{\mathcal{H}}'\tilde{\mathcal{H}} + \alpha I\big)^{-1} \tilde{\mathcal{H}}'\,\tilde{b}, \qquad \tilde{\mathcal{H}} = \begin{pmatrix} \sqrt{\lambda_\pi}\,\mathcal{H}_\pi \\ \sqrt{\lambda_y}\,\mathcal{H}_y \end{pmatrix}, \quad \alpha = 10^{-6}, \tag{32}$$

with the implied rate path $\Delta r^\star = \mathcal{H}_r\,\varepsilon^\star$ handed to the IRF-consistent mode. One slider sets the preference weights ($\lambda_\pi = 5w$, $\lambda_y = 5(1-w)$); the $\pi^\star$ and $y^\star$ sliders set the targets. Unlike a hand-set adjustment, the optimal path pins *every* quarter — it is a full plan, not a one-off surprise.

A fiscal lever completes the block: twelve government-spending sliders scale the model's fiscal responses on top of the scenario, for the subset of models (fewer than half) that include a fiscal sector. For projecting the public finances themselves, the fiscal block of Section 6 is the instrument.

Table 1: Repository layout. Estimation in `pipeline/` produces a per-bloc JSON export; `port_gvar_exports.py` ports it into `app/public/data/bvar/`; the front end and its compute kernels consume that data and never re-estimate.

Top-level	Key contents
`pipeline/` (MATLAB + Python)	`export_bvar_gvar24_loop.m` (GVAR–EM estimator); `build_*.py` (FRED / Eurostat / BIS / ECB pulls, trade weights, fiscal drivers); `port_gvar_exports.py`; `_verify/` ground-truth checks.
`app/` (Next.js + compute)	`src/lib/compute/` (Durbin–Koopman, GVAR solve, fiscal accounting); `scripts/` (`precompute_gvar_coupling.mjs`, `verify_gvar24.mjs`, `invert_coupling.{m,py}`); `public/data/bvar/<CC>/` (the deployed JSON the app reads).
`docs/` (manual + runbooks)	`MFFM_User_Manual.tex`; `REESTIMATING_MODELS.md`, `ADDING_A_COUNTRY.md`, `ARCHITECTURE_FOR_AI.md`.

10 Data, vintages, and updates

10.1 Sources

No single provider supplies current quarterly data for the full roster, so the panels are assembled bloc by bloc from a small number of source families, wired by per-bloc build scripts (`pipeline/build_*.py`) and recorded, series by series, in a machine-readable coverage manifest (`coverage_manifest.json` under `app/public/data/`) from which the roster table of Appendix B is generated.

- **FRED** supplies the entire US bloc with exact series identifiers, the China bloc (built from verified real FRED series, with quarterly real activity anchored on a chain-linked volume index), the four global commons, and the BIS credit and residential-property series redistributed through FRED.
- **Eurostat** supplies the EU and EFTA economies: chain-linked volumes (`namq_10_gdp`), HICP, unemployment, interest rates, effective exchange rates, house prices; euro members carry the ECB policy and lending rates.
- **OECD quarterly national accounts and national sources** supply the remaining OECD members and emerging markets, with policy rates from national central banks or the BIS.
- **IMF WEO** is a back-fill of last resort: where a bloc's genuine quarterly series end short of the common estimation window, annual WEO projections are temporally disaggregated to fill the macro tail (proportional-Denton within-year profiling, observed quarters held fixed). WEO never overrides observed data, and every bloc whose live tail is WEO-derived is flagged in the manifest and in **Data & coverage**.
- **Derived series** — the stars, the effective interest rate, the stock-flow adjustment, the rest-of-euro-area aggregates — are computed, not pulled (Sections 5.1 and 6).

The **Data & coverage** dialog (top bar) lists, for the economy on screen, every variable's source, coverage span, last observed quarter, and caveats — the on-screen rendering of the manifest. Amber entries end before the panel origin and correspond exactly to the amber chart tails.

10.2 Vintage policy: estimate once, shift quarterly

The shipped panels are the product of two clearly separated steps. *Estimation build*: every bloc harmonized to a common 2025Q4 endpoint, on which all coefficients, draws, weights, and the coupling artifact were estimated and frozen. *Observed refresh*: the live panel then advances — currently to a default last-historical quarter of 2026Q1 — by pulling each newly published quarter from the same sources in the same transforms, with *no* re-estimation: the forecast re-anchors from the unchanged companion (`app/scripts/refresh-quarter.mjs`, fail-closed per bloc). Because the coupled system links economies row by row, the panel origin is *uniform*: at this writing every bloc's history ends at 2026Q1. Blocs whose sources lag reach the origin through the amber, flagged nowcast described next — never through fabricated observations. The one exception is Israel, whose statistics arrive with a long lag; its charts run on Israel's own calendar, about a year behind the panel (a documented row-offset approximation, flagged in **Data & coverage**).

Within a data round the not-yet-published quarters of the remaining series are *nowcast* by the coupled system itself: the panel is truncated to the last quarter at which every cell is observed, every observed cell in the ragged window is pinned at its published value, and one coupled solve (Section 5.3) delivers each missing cell's conditional expectation given everything already observed anywhere in the panel — the star channel carries the leaders' releases into the laggards' estimates. Observed cells are never altered; every imputed cell carries a provenance flag and draws amber (`nowcastCompletion.ts`). The label **NC** in the conditioning list is this nowcast quarter.

The **Monitor** page (top bar) shows what each data round did: per economy, the revision of the current-quarter estimate against what the previous vintage predicted for it, split by channel — **observed** (the economy's own release surprised) versus **cross-block** (the economy published nothing; the move is pure transmission of partners' releases). Source restatements of history are accepted as current best knowledge and logged in the same page. Once a year, after the previous calendar year is fully published for every economy, the whole system is re-estimated and the cycle restarts; only then do the estimated relationships change.

11 Evaluation

The system's forecasting performance is evaluated in a companion repository (`em-bayesian-gvar-eval`) under a protocol designed before the runs; this section summarizes the design and the realized results. Two questions are asked: does the estimation apparatus — Bayesian shrinkage, COVID-as-missing, the EM star fixed point, the endogenous US — earn its keep out of sample, and what does the endogenous US buy that a frozen anchor cannot deliver.

11.1 Design

Models. Nine forecast densities are scored: the proposed EM-Bayesian GVAR in its two variants (`emgvar_endo`, the full endogenous-US loop, and `emgvar_frozen`, the frozen-anchor ablation), six benchmarks estimated on the same truncated panels — a driftless random walk in the scored space, an AR(p) per series (p by AIC, iterated multi-step; Marcellino et al., 2006), a single-country BVAR with the stars dropped (the value-of-coupling ablation), the textbook fixed-weight OLS GVAR of Pesaran et al. (2004) and Dees et al. (2007) with no data augmentation (the headline classical competitor), a single-pass Bayesian GVAR (the value-of-EM ablation), and a Bayesian dynamic factor model — plus a sample-mean climatology sibling of the random walk.

Scheme. Pseudo-real-time recursive expanding window on the final vintage: quarterly origins 2010Q1–2024Q4 (60 raw origins; the four 2020 origins are dropped from the headline tables, leaving 56), horizons $h \in \{1, 2, 4, 8\}$, three targets per bloc (real-GDP growth and inflation in the 4Δ scored space, the short rate in levels), six headline blocs (US, EA, JP, UK, CA, CN). Look-ahead is excluded by construction: every panel is truncated to the origin before any transformation or star is built; trade weights are rolling, as-of-origin three-year averages (the shipped 2021–2023 weights are forbidden in the sweep); and the COVID mask is re-applied in-sample at every origin. The proposed variants require a longer warm-up and score at up to 52 origins. Predictive densities use 200 retained draws per origin with a stationarity backstop that discards explosive posterior draws.

Scoring. Point accuracy by relative RMSE against the random walk. Density accuracy by the continuous ranked probability score (CRPS), the primary metric, computed from the draw ensemble by the standard energy-form estimator

$$\widehat{\mathrm{CRPS}} = \frac{1}{m}\sum_{k=1}^{m}|X_k - y| - \frac{1}{2m^2}\sum_{k=1}^{m}\sum_{l=1}^{m}|X_k - X_l| \tag{33}$$

(Gneiting and Raftery, 2007), which degrades gracefully at modest draw counts; calibration by PIT histograms and interval coverage (Berkowitz, 2001). Because most model pairs are *nested*, predictive-ability inference uses the Giacomini and White (2006) conditional test as the headline (valid under nesting and correct for the expanding scheme), Clark and West (2007) for nested point comparisons, and Diebold and Mariano (1995) with the Harvey et al. (1997) correction only for non-nested pairs; multiplicity is controlled by Romano–Wolf step-down (Romano and Wolf, 2005), and the per-cell *model confidence set* (MCS) of Hansen et al. (2011) reports which models are statistically indistinguishable from the best.

11.2 Results

On *point* accuracy the univariate AR(p) and the factor model lead — a familiar macro-forecasting result — and the EM-GVAR does not win the horse race outright. The decisive evidence is in the density comparison and the MCS: the EM-Bayesian GVAR sits *inside* the 90% MCS for **inflation at every horizon** and for the **short rate at** $h \leq 4$ — statistically on par with the best model across the nominal block — and among the multivariate models the best inflation point forecast at $h = 1$ is `emgvar_endo` (relative RMSE 0.80, behind only the univariate AR at 0.64). It leaves the set only for GDP growth at $h \in \{1, 8\}$, where the univariate and factor models dominate. Two structural findings follow. First, the endogenous US is weakly preferred to the frozen anchor throughout the MCS — surviving where the frozen variant is eliminated (the short rate at $h = 1$) and eliminated later where both fall. Second, and central to the design's claim, the apparatus *decisively* beats its classical ancestor: the textbook OLS GVAR is eliminated from half of the twelve (target, horizon) MCS cells and carries badly miscalibrated densities (one-quarter-ahead 90%-interval coverage of 0.37–0.47 on the nominal and real blocks against the 0.90 target).

Two caveats are part of the record. The draw-based densities of the proposed variants are themselves over-confident at short horizons (realized 90% coverage 0.73–0.85): the forecast draws propagate parameter and shock uncertainty but not the smoother uncertainty of the COVID-imputed states. The CRPS and MCS compare score *differentials* and are robust to this; the under-coverage is a calibration caveat, not a ranking artifact. And the log predictive score is numerically unreliable on under-covered clouds (a parametric tail fit explodes on a handful of origins), so ranking rests on CRPS and the MCS only.

11.3 The endogenous-US contrast

What the solver computes is a *conditional-mean scenario response* — pin a path, read the coupled deltas via (17) — not a generalized impulse response or a variance decomposition; the evaluation keeps that distinction explicit. The realized contrast propagates a US +100bp policy-rate path through both topologies: under the frozen anchor the world-to-US direction is structurally absent — not small, absent — while the endogenous configuration returns a non-trivial RoW→US feedback, with sizeable, sign-heterogeneous gaps between the two topologies across partner blocs. These are conditional-projection deltas (the rate is pinned along the whole path, bundling the direct conditioning with the coupled feedback), so their *existence and direction* is the documented result; calibrated spillover *multipliers* await the structural extension (generalized IRFs and connectedness indices on the coupled system, Pesaran and Shin, 1998; Diebold and Yilmaz, 2012), which is constructible from the shipped exports but not part of the tool.

12 Implementation and reproducibility

12.1 Client-side computation

All matrix algebra, filtering, and simulation run in the browser in TypeScript on `Float64Array` storage (IEEE 754 double precision): LU decomposition with partial pivoting for F_t^{-1} (singularity threshold 10^{-15}), Cholesky factorization for shock draws (positive-definiteness threshold 10^{-14}, with a diagonal fallback), and row-major layouts throughout. Table 2 collects the numerical constants a replicator needs; each appears in exactly one place in the code.

Table 2: Numerical constants of the deployed engine.

Constant	Value	Role
σ_ε^2	10^{-12}	hard-pin measurement variance (6); soft-pin threshold
P_0	$10^{-7}I$	initial state covariance
LU pivot	10^{-15}	singularity threshold in F_t^{-1}
Cholesky	10^{-14}	positive-definiteness threshold
D	500	shipped posterior draws per bloc
N_b	50 / 200	band draws, **Quick** / **Full**
band quantiles	q_{16}, q_{84}	scenario bands (baseline adds q_{05}, q_{50}, q_{95})
trust default / clamp	0.7 / $[0.001, 0.999]$	soft-pin dial (13); ≥ 0.999 is hard
GS tolerance / sweeps	10^{-4} / 12	Gauss–Seidel coupled solve
Neumann tolerance / cap	10^{-10} rel. / 2000	inverse-free fixed point
$\rho(M)$ / dim	0.922 / 2060	shipped coupling artifact
artifact precision	6 sig. figures	all exported floats
sens guard	0.3	long-run slider disable (28)
B&M ridge α	10^{-6}	optimal-policy solve (32)
EM TOL / MAXIT	0.03 / 12	estimation star fixed point

Interactivity is engineered so the heavy work never blocks the mouse: no coupled solve fires during a drag (a signal bus outside the rendering state carries the drag flag); exactly one cheap direct-preview solve dispatches on release; the pinned dot is sourced from the click itself through an exact display-to-storage inverse, so it snaps to the cursor instantly while the line and bands reconcile; and each chart panel is memoized so a conditioning edit repaints one panel of the roughly thirty-five, not all of them. The full contract is asserted in the test suite.

12.2 Repository and replication

One repository holds the three concerns: `pipeline/` (MATLAB and Python: estimation and data assembly), `app/` (the front end and its compute kernels), `docs/` (this manual and the operational runbooks). The estimator writes a per-bloc JSON export that a porter script copies into `app/public/data/bvar/`; the deployed application reads only that JSON. Dependencies: a recent MATLAB with the Parallel Computing Toolbox (the bundled `BVAR/MFBVAR` code needs no further packages), Python 3 standard library for the data pulls (a FRED API key via `FRED_API_KEY`), and the pinned Node toolchain for the application.

The end-to-end path from raw data to a deployable build is six steps; the canonical commands are in `REESTIMATING_MODELS.md`:

1. **Build the data**: run the `pipeline/build_*.py` pulls to assemble the harmonized per-bloc panels.

2. **Estimate**: run `export_bvar_gvar24_loop.m` — the GVAR-EM loop of Algorithm 2 — which exports the eight-file contract per bloc.

3. **Port**: run `port_gvar_exports.py`, which copies the contract into the application, compacting floats to six significant figures.

4. **Regenerate the coupling**: run `scripts/precompute_gvar_coupling.mjs`, which assembles M, inverts $(I - M)$ offline, and writes the gzipped artifact of Section 5.3.

5. **Verify**: run `scripts/verify_gvar24.mjs` (artifact signature fresh; the shipped inverse inverts $(I-M)$; $\rho(M) < 1$ and matches the stored value; Gauss–Seidel agrees with the artifact-seeded closed form), then the type check and the test suite (`npm test`; the multi-second solver canaries run under `npm run test:heavy`).

6. **Deploy**: on a green gate, push `app/**` — a GitHub Action builds the static export and publishes it; the Action only copies files.

Because the model signature (Section 5.3) flips whenever any bloc is re-estimated, it is not possible to ship a coupling artifact that silently disagrees with its blocs; the regenerated per-economy tables of Appendix B enforce the same discipline on this document.

13 Final thoughts

The design bets on a separation of concerns. The baseline is computed once, offline, and shipped — the browser never estimates. The scenario layer imposes user paths without asking where the deviation comes from, in the tradition of Waggoner and Zha (1999); the counterfactual layer adds structural identification exactly where it is needed — the policy transmission — and lets the estimated covariances carry the rest, because no single DSGE model matches the empirical co-movement of thirty variables. The accounting blocks derive what must never be estimated freely. And every judgment device — pins, trust dials, anchors, counterfactuals — is built to be an exact no-op at its neutral setting, so the untouched screen is always the model, to the last digit.

13.1 Limitations

The dynamics are linear and time-invariant: coefficients are fixed at their posterior means for the deterministic engine, with no time-varying parameters, stochastic volatility, or regime switching — restrictive in periods of structural change. Innovations are Gaussian; crisis-period fat tails are not accommodated. Counterfactuals are point-identified under one model–rule pair at a time (the user can flip pairs, but the tool does not average over them), and the Lucas critique applies to the re-conditioned variables whose responses are governed by the estimated, policy-invariant-by-assumption covariances. The coefficients are frozen at the estimation vintage within the year: data revisions are accepted as knowledge and the origin advances, but the parameters see new data only at the annual re-estimation. Scenario responses — including the cross-country ones — are conditional projections, not causal impulse responses (Section 11.3).

13.2 Known issues

Specific, identified defects of the current deployment, documented in the interest of honesty for a live research tool:

1. **Divergence is detected, not prevented.** The coupled solve does not refuse to run if a future roster pushed $\rho(M) \geq 1$: the convergence flag trips, a console warning fires, and the result is not cached — but a (diverged) path would still be displayed. A hard spectral-radius backstop is the durable fix.

2. **The retired tilt module still ships.** The entropic-tilt code (Remark 7) is no longer reachable from the interface but remains in the repository with a latent defect (non-finite long-run moments of unstable draws would propagate through its softmax); it should be removed or finite-masked to prevent accidental reintroduction.

3. **Fiscal metadata is not regenerated by the porter.** The fiscal block reads its drivers by *positional* indices stored in `fiscal_meta.json`; a re-export that changes a bloc's variable order leaves those indices pointing at the wrong columns unless the metadata is rebuilt (a repair script exists; folding it into the porter is the durable fix).

4. **Mid-drag previews use the filtered variance.** The impact-view shading requests the smoothed variance (11) except while a drag is in flight, when the cheaper filtered variance is shown; the run-up to a future pin is momentarily overstated until release.

13.3 Directions

Natural extensions, in rough order of value: a hard contraction guard on the coupled solve; integrating the smoother uncertainty of the imputed states into the predictive draws (the documented source of the short-horizon under-coverage, Section 11.2); structural spillover objects — generalized IRFs and connectedness — on the coupled system; scenario priors for identification inside the BVAR (Farkas and Jakab, 2026); and mixed-frequency estimation to bring monthly indicators into the nowcast.

A Notation

Table 3: Notation. Symbols are local to their sections; the two uses of d (the GVAR forcing vector of Section 5, the debt ratio d_t of Section 6) never co-occur.

Symbol	Meaning
Estimation (Sections 3)	
y_t, n, p	bloc observables ($n \times 1$), variable count, lag order ($p = 3$)
c, B_ℓ, Σ	intercept, lag matrices, innovation covariance of (1)
$\hat{\beta}$	stored $(1 + np) \times n$ coefficients; row 0 the intercept
ℓ_t	stored level of a `log` variable, $100 \ln Y_t$
g_t	displayed q/q annualized growth, $4(\ell_t - \ell_{t-1})$
δ_i	own-lag prior mean: 1 (`RW`) or 0 (`WN`)
λ, μ	GLP tightness and sum-of-coefficients hyperparameters
D	shipped posterior draws (500)
State space and smoother (Section 4)	
$\alpha_t, T, c_\alpha, Z, Q, R$	companion state and system matrices (5)
$a_{t\|t-1}, P_{t\|t-1}$	predicted state mean and covariance
v_t, F_t, K_t	innovation, its variance, the Kalman gain
r_t, L_t, N_t, V_t	smoother recursions (10)–(11)
Y^*, H	pin matrix (NaN = free) and forecast horizon (20)
τ, R^{ext}	trust and soft-pin variance (13)
N_b	band draws (50 quick / 200 full)
Global model (Section 5)	
$w_{cj}, \tilde{w}_{cj}$	trade weight of partner j in bloc c; renormalized
x_c^*	bloc c's star columns (DEM, INF, FIN)
$K_c[v][s]$	affine response kernel (15)
x, M, d	deviation state, coupling operator, forcing of $x = Mx + d$
$\rho(M)$	spectral radius (shipped: 0.947)
Accounting blocks (Sections 6–7)	
d_t, d_0	debt-to-GDP (%) and its seed
i_t^{eff}, g_t^n	effective rate (% p.a.) and quarterly nominal growth (decimal)
$pb_t, \overline{sfa}$	primary balance; structural stock-flow constant
b_t, nfa_0	NFA-to-GDP (%) and its seed
$i^{\text{NFA}}, \phi^{\text{FX}}$	NFA return (3% p.a.) and FX-exposed share (0.4)
Anchors and counterfactuals (Sections 8–9)	
$S_h, \delta c_j, \text{sens}_j$	drift operator, drift shift, terminal sensitivity (28)
$\psi_j^\star, \psi_j^0$	target and current terminal display forecast
h_r, h_π, h_y	MMB impulse responses; ε_t the inverted shocks
$\lambda_\pi, \lambda_y, \alpha$	loss weights and ridge of (31)–(32)

B The deployed economies

The tables of this appendix are *generated* from the deployed data files and regenerated whenever the data change, so they cannot drift from the shipped models. The coverage matrix below (Table 4) is built by `build_coverage_table.py` from the coverage manifest; the roster (Table 5) and the per-economy variable tables that follow are built by `generate_manual_tables.py` — the roster from the same manifest and the per-bloc metadata, the variable tables from each bloc's `spec.json`.

The single grid in Table 4 answers, at a glance, which of the nine variable groups each of the 50 deployed blocs carries: a green check marks a group that is present in that bloc's shipped model, a dash marks a genuine absence. The last two columns record the last quarter shown and the primary source family; an italic tag on the source flags the handful of blocs whose most recent observations are model-extended (WEO-back-filled or under source review) rather than genuinely observed, so a reader can see immediately where the data are firm and where they are provisional.

Table 4: Country × variable-group coverage matrix for all 50 deployed blocs. Columns: **GDP** real activity; **Price** HICP/CPI; **Unemp** unemployment rate; **Pol** policy rate; **Long** 10-year/long government yield; **Cred** credit-to-GDP; **Lend** lending/short rate; **FX** nominal effective exchange rate; **Fisc** the four-variable fiscal block (`FREV`/`FPEXP`/`EFFI`/`SFA`). ✓ = carried; `--` = absent in that bloc. *Last obs* is the last quarter shown; an italic tag flags a tail that is model-extended rather than genuinely observed (*WEO* = WEO-back-filled, CN/IN; *review* = under source review, ID/KR/RU/TH; *frozen mirror* = carried from a frozen partner mirror, KR).

Bloc	**GDP**	**Price**	**Unemp**	**Pol**	**Long**	**Cred**	**Lend**	**FX**	**Fisc**	**Last obs**	**Source**
`AR`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`AT`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`AU`	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`BE`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`BR`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`CA`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`CH`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
`CL`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`CN`	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	FRED *WEO*
`CO`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`CR`	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`CZ`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
`DE`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
`DK`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
`EA`	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
`EE`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`ES`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`FI`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`FR`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`GR`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`HU`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
`ID`	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED *review*
`IE`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`IL`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2025Q1	OECD/FRED
`IN`	✓	✓	–	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *WEO*
`IS`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`IT`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
JP	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED
KR	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED *review,frozen mirror*
LT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
LU	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
LV	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
MX	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
MY	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
NL	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
NO	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
NZ	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
PL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
PT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
RU	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *review*
SA	✓	✓	✓	✓	✓	✓	–	✓	–	2026Q1	OECD/FRED
SE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
SI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
SK	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
TH	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED *review*
TR	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED
TW	✓	✓	✓	–	✓	✓	✓	–	–	2026Q1	OECD/FRED
UK	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
US	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	FRED
ZA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED

B.1 The deployed roster at a glance

Generated from the shipped `spec.json`, `fiscal_meta.json`, `external_meta.json`, the trade-weights file, and the coverage manifest (`coverage_manifest.json`); the table therefore reflects the deployed data exactly. n counts the variables of the estimated VAR (domestic, global, fiscal, and star columns included). *Coupled* marks membership of the cross-country fixed point (Section 5); the remaining blocs are standalone models. *Fiscal* and *External* mark the accounting blocks of Section 6. *Last obs.* is the bloc's last historical quarter.

Table 5: Deployed economies: dimension, coupling, accounting blocks, last observation.

Code	Economy	n	Coupled	Fiscal	External	Last obs.
`US`	United States	40	✓	✓	✓	2026Q1
`EA`	Euro area (aggregate)	25	✓	✓	✓	2026Q1
`JP`	Japan	25	✓	✓	—	2026Q1
`UK`	United Kingdom	27	✓	✓	—	2026Q1
`CA`	Canada	27	✓	✓	—	2026Q1
`CN`	China	22	✓	✓	✓	2026Q1
`AR`	Argentina	20	✓	—	—	2026Q1
`AT`	Austria	23	—	✓	—	2026Q1
`AU`	Australia	25	✓	✓	—	2026Q1
`BE`	Belgium	23	—	✓	—	2026Q1
`BR`	Brazil	24	✓	✓	—	2026Q1
`CH`	Switzerland	20	✓	—	—	2026Q1
`CL`	Chile	25	✓	✓	—	2026Q1
`CO`	Colombia	26	✓	✓	—	2026Q1
`CR`	Costa Rica	20	✓	✓	—	2026Q1
`CZ`	Czechia	25	✓	✓	—	2026Q1
`DE`	Germany	22	—	✓	—	2026Q1
`DK`	Denmark	26	✓	✓	—	2026Q1
`EE`	Estonia	23	—	✓	—	2026Q1
`ES`	Spain	23	—	✓	—	2026Q1
`FI`	Finland	23	—	✓	—	2026Q1
`FR`	France	23	—	✓	—	2026Q1
`GR`	Greece	22	—	✓	—	2026Q1
`HU`	Hungary	25	✓	✓	—	2026Q1
`ID`	Indonesia	24	✓	✓	—	2026Q1
`IE`	Ireland	23	—	✓	—	2026Q1
`IL`	Israel	25	✓	✓	—	2025Q1
`IN`	India	21	✓	—	—	2026Q1
`IS`	Iceland	21	✓	—	—	2026Q1
`IT`	Italy	23	—	✓	—	2026Q1
`KR`	Korea	25	✓	✓	—	2026Q1
`LT`	Lithuania	23	—	✓	—	2026Q1

Code	Economy	n	Coupled	Fiscal	External	Last obs.
LU	Luxembourg	23	—	✓	—	2026Q1
LV	Latvia	22	—	✓	—	2026Q1
MX	Mexico	26	✓	✓	—	2026Q1
MY	Malaysia	21	✓	—	—	2026Q1
NL	Netherlands	23	—	✓	—	2026Q1
NO	Norway	24	✓	✓	—	2026Q1
NZ	New Zealand	25	✓	✓	—	2026Q1
PL	Poland	25	✓	✓	—	2026Q1
PT	Portugal	23	—	✓	—	2026Q1
RU	Russia	17	✓	—	—	2026Q1
SA	Saudi Arabia	18	✓	—	—	2026Q1
SE	Sweden	26	✓	✓	—	2026Q1
SI	Slovenia	23	—	✓	—	2026Q1
SK	Slovakia	23	—	✓	—	2026Q1
TH	Thailand	26	✓	✓	—	2026Q1
TR	Türkiye	23	✓	✓	—	2026Q1
TW	Taiwan Province of China	20	✓	—	—	2026Q1
ZA	South Africa	25	✓	✓	—	2026Q1

B.2 Variables, transformations, and priors by economy

One block per economy, generated from the shipped `spec.json`. *Trans.*: `log` = the variable enters the VAR as $100 \ln(\cdot)$ (growth displayed); `lin` = the variable enters in levels. *Prior*: RW = random-walk own-lag prior mean ($\delta_i = 1$), WN = stationary white-noise own-lag prior mean ($\delta_i = 0$; Section 3.2). *Fin.*: the `isFinancial` tag — financial variables stay *observed* through the COVID window (Section 3.3). *Block*: dom. = domestic, glob. = US-determined global common, fisc. = fiscal driver, row = trade-weighted star, ext. = area-wide instrument carried as weakly exogenous (euro members). The dates give the *shipped* history window the application loads (the recent span the charts display and the state initializes from); the estimation sample is longer and ends at the 2025Q4 estimation vintage (Section 10.2).

Table 6: United States (US): $n = 40$, shipped history from 1998Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDPH	Real Gross Domestic Product	log	RW	—	dom.
CH	Real Personal Consumption Expenditures	log	RW	—	dom.
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW	—	dom.
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW	—	dom.
XH	Real Exports of Goods & Services	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
MH	Real Imports of Goods & Services	log	RW	—	dom.
GH	Real Government Consumption Expenditures & Gross Investment	log	RW	—	dom.
JGDP	Gross Domestic Product: Chain Price Index	log	RW	—	dom.
JC	Personal Consumption Expenditures: Chain Price Index	log	RW	—	dom.
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW	—	dom.
UI	CPI-U: All Items	log	RW	—	dom.
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW	—	dom.
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW	—	dom.
LANAGRA	All Employees: Total Nonfarm	log	RW	—	dom.
LR	Civilian Unemployment Rate: 16 yr +	lin	RW	—	dom.
IP	Industrial Production Index	log	RW	—	dom.
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW	—	dom.
HST	Housing Starts	log	RW	—	dom.
YPDH	Real Disposable Personal Income	log	RW	—	dom.
CSENT	University of Michigan: Consumer Sentiment	lin	RW	—	dom.
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW	✓	dom.
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
GLB_USLR	10-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	glob.
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW	✓	dom.
GLB_USBAA	Moodys Seasoned Baa Corporate Bond Yield	lin	RW	✓	glob.
FXTWM	Nominal Trade-Weighted Exch Value of US$ vs AFE Currencies	log	RW	✓	dom.
GLB_USEQ	Equity Price Index (OECD US share prices / S&P, public)	log	RW	✓	glob.
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW	✓	dom.
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW	✓	dom.
HP	House Price	log	RW	✓	dom.
LEV	Real nonfinancial business debt	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
FREV_GDP	Federal Receipts (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW	✓	fisc.
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN	—	fisc.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
ROWDEM_US	US RoW Demand	lin	RW	✓	row
ROWINF_US	US RoW Inflation	lin	RW	✓	row
ROWFIN_US	US RoW Fin	lin	RW	✓	row

Table 7: Euro area (aggregate) (EA): $n = 25$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EA	EA Real GDP	log	RW	—	dom.
CONS_EA	EA Consumption	log	RW	—	dom.
INV_EA	EA Investment	log	RW	—	dom.
GOV_EA	EA Gov Spending	log	RW	—	dom.
EXP_EA	EA Exports	log	RW	—	dom.
IMP_EA	EA Imports	log	RW	—	dom.
HICP_EA	EA HICP Index	log	RW	—	dom.
RATE_EA	EA 3M Interbank Rate	lin	RW	✓	dom.
YIELD10_EA	EA 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_EA	EA NEER	log	RW	✓	dom.
POLRATE_EA	EA ECB Deposit Facility Rate (policy)	lin	RW	✓	dom.
HPI_EA	EA Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	EA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EA	EA Credit to NFCs	log	RW	✓	dom.
ROWDEM_EA	EA RoW Demand	lin	RW	✓	row
ROWINF_EA	EA RoW Inflation	lin	RW	✓	row
ROWFIN_EA	EA RoW Fin	lin	RW	✓	row

Table 8: Japan (JP): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_JP	JP Real GDP	log	RW	—	dom.
CONS_JP	JP Consumption	log	RW	—	dom.
INV_JP	JP Investment	log	RW	—	dom.
GOV_JP	JP Gov Spending	log	RW	—	dom.
EXP_JP	JP Exports	log	RW	—	dom.
IMP_JP	JP Imports	log	RW	—	dom.
LR_JP	JP Unemployment Rate	lin	RW	—	dom.
CPI_JP	JP CPI Index	log	RW	—	dom.
RATE_JP	JP 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_JP	JP Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_JP	JP 10Y JGB Yield	lin	RW	✓	dom.
HP_JP	JP Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	JP Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	JP Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_JP	JP Credit to NFCs	log	RW	✓	dom.
LENDR_JP	JP Avg Contract Interest Rate on New Loans (% p.a.)	lin	RW	✓	dom.
ROWDEM_JP	JP RoW Demand	lin	RW	✓	row
ROWINF_JP	JP RoW Inflation	lin	RW	✓	row
ROWFIN_JP	JP RoW Fin	lin	RW	✓	row

Table 9: United Kingdom (UK): $n = 27$, shipped history from 1997Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_UK	UK Real GDP	log	RW	—	dom.
CONS_UK	UK Consumption	log	RW	—	dom.
INV_UK	UK Investment	log	RW	—	dom.
GOV_UK	UK Gov Spending	log	RW	—	dom.
EXP_UK	UK Exports	log	RW	—	dom.
IMP_UK	UK Imports	log	RW	—	dom.
IP_UK	UK Ind Production Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_UK	UK Unemployment Rate	lin	RW	—	dom.
CPI_UK	UK CPI Index	log	RW	—	dom.
RATE_UK	UK 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_UK	UK Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_UK	UK 10Y Gilt Yield	lin	RW	✓	dom.
HP_UK	UK Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_UK	UK NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	UK Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	UK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	UK Effective Rate (% p.a.)	lin	RW	✓	fisc.
SFA	UK Stock-Flow Adj (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_UK	UK Credit to NFCs	log	RW	✓	dom.
LENDR_UK	UK Effective Rate on New Loans to PN-FCs (% p.a., BoE CFMBJ82)	lin	RW	✓	dom.
ROWDEM_UK	UK RoW Demand	lin	RW	✓	row
ROWINF_UK	UK RoW Inflation	lin	RW	✓	row
ROWFIN_UK	UK RoW Fin	lin	RW	✓	row

Table 10: Canada (CA): $n = 27$, shipped history from 1998Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CA	CA Real GDP	log	RW	—	dom.
CONS_CA	CA Consumption	log	RW	—	dom.
INV_CA	CA Investment	log	RW	—	dom.
GOV_CA	CA Gov Spending	log	RW	—	dom.
EXP_CA	CA Exports	log	RW	—	dom.
IMP_CA	CA Imports	log	RW	—	dom.
IP_CA	CA Ind Production (Index 2015=100)	log	RW	—	dom.
LR_CA	CA Unemployment Rate	lin	RW	—	dom.
CPI_CA	CA CPI Index	log	RW	—	dom.
RATE_CA	CA 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CA	CA Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CA	CA 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CA	CA Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
NEER_CA	CA NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_CA	CA Credit to NFCs	log	RW	✓	dom.
LENDR_CA	CA Bank Lending Rate (effective business interest rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CA	CA RoW Demand	lin	RW	✓	row
ROWINF_CA	CA RoW Inflation	lin	RW	✓	row
ROWFIN_CA	CA RoW Fin	lin	RW	✓	row

Table 11: China (CN): $n = 22$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CN	CN Real GDP (quarterly, chain-linked SA volume index, base 2011Q1=100)	log	RW	—	dom.
CPI_CN	CN CPI	log	RW	—	dom.
RATE_CN	CN 3M Interbank Rate	lin	RW	✓	dom.
EXP_CN	CN Exports (USD)	log	RW	—	dom.
IMP_CN	CN Imports (USD)	log	RW	—	dom.
REER_CN	CN Real Effective FX	log	RW	✓	dom.
EQ_CN	Shanghai Share Prices	log	RW	✓	dom.
POLRATE_CN	CN Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CN	CN Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CN Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CN Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_CN	CN Credit to NFCs	log	RW	✓	dom.
CORP_CN	CN 10Y Government Bond Yield (sovereign proxy)	lin	RW	✓	dom.
ROWDEM_CN	CN RoW Demand	lin	RW	✓	row

ID	Description	Trans.	Prior	Fin.	Block
ROWINF_CN	CN RoW Inflation	lin	RW	✓	row
ROWFIN_CN	CN RoW Fin	lin	RW	✓	row

Table 12: Argentina (AR): $n = 20$, shipped history from 2004Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AR	AR Real GDP	log	RW	—	dom.
CONS_AR	AR Real Household Consumption	log	RW	—	dom.
INV_AR	AR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_AR	AR Real Government Consumption	log	RW	—	dom.
EXP_AR	AR Real Exports	log	RW	—	dom.
IMP_AR	AR Real Imports	log	RW	—	dom.
CPI_AR	AR CPI Index	log	RW	—	dom.
LR_AR	AR Unemployment Rate	lin	RW	—	dom.
RATE_AR	AR BADLAR Short-Term Rate	lin	RW	✓	dom.
NEER_AR	AR Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_AR	AR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_AR	AR Credit to NFCs	log	RW	✓	dom.
LENDR_AR	AR Lending Rate (IMF MFS162, % p.a.; extreme volatility)	lin	RW	✓	dom.
ROWDEM_AR	AR RoW Demand	lin	RW	✓	row
ROWINF_AR	AR RoW Inflation	lin	RW	✓	row
ROWFIN_AR	AR RoW Fin	lin	RW	✓	row

Table 13: Austria (AT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AT	AT GDP	log	RW	—	dom.
CONS_AT	AT CONS	log	RW	—	dom.
INV_AT	AT INV	log	RW	—	dom.
GOV_AT	AT GOV	log	RW	—	dom.
EXP_AT	AT EXP	log	RW	—	dom.
IMP_AT	AT IMP	log	RW	—	dom.
HICP_AT	AT HICP Index	log	RW	—	dom.
LR_AT	AT Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_AT	AT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_AT	AT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_AT	AT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_AT	AT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_AT	AT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_AT	AT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_AT	AT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	AT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_AT	AT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_AT	AT NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 14: Australia (AU): $n = 25$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AU	AU Real GDP	log	RW	—	dom.
CONS_AU	AU Real Household Consumption	log	RW	—	dom.
GOV_AU	AU Real Govt Consumption	log	RW	—	dom.
INV_AU	AU Real Investment (GFCF)	log	RW	—	dom.
EXP_AU	AU Real Exports	log	RW	—	dom.
IMP_AU	AU Real Imports	log	RW	—	dom.
CPI_AU	AU CPI All-items Index	log	RW	—	dom.
LR_AU	AU Unemployment Rate	lin	RW	—	dom.
NEER_AU	AU NEER Index	log	RW	✓	dom.
RATE_AU	AU Cash Rate (Interbank Overnight)	lin	RW	✓	dom.
YIELD10_AU	AU 10Y Govt Bond Yield	lin	RW	✓	dom.
HP_AU	AU Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
FREV_GDP	AU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	AU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	AU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_AU	AU Credit to NFCs	log	RW	✓	dom.
LENDR_AU	AU Business Lending Rate (RBA F7 outstanding large-business total, % p.a.)	lin	RW	✓	dom.
ROWDEM_AU	AU RoW Demand	lin	RW	✓	row
ROWINF_AU	AU RoW Inflation	lin	RW	✓	row
ROWFIN_AU	AU RoW Fin	lin	RW	✓	row

Table 15: Belgium (BE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BE	BE GDP	log	RW	—	dom.
CONS_BE	BE CONS	log	RW	—	dom.
INV_BE	BE INV	log	RW	—	dom.
GOV_BE	BE GOV	log	RW	—	dom.
EXP_BE	BE EXP	log	RW	—	dom.
IMP_BE	BE IMP	log	RW	—	dom.
HICP_BE	BE HICP Index	log	RW	—	dom.
LR_BE	BE Unemployment Rate	lin	RW	—	dom.
HPI_BE	BE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_BE	BE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_BE	BE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_BE	BE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB allbonds 10Y, %)	lin	RW	✓	ext.
RATE_BE	BE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_BE	BE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_BE	BE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	BE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
CREDIT_GDP_BE	BE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_BE	BE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 16: Brazil (BR): $n = 24$, shipped history from 2012Q2, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BR	BR Real GDP	log	RW	—	dom.
CONS_BR	BR Real Household Consumption	log	RW	—	dom.
INV_BR	BR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_BR	BR Real Government Consumption	log	RW	—	dom.
EXP_BR	BR Real Exports	log	RW	—	dom.
IMP_BR	BR Real Imports	log	RW	—	dom.
CPI_BR	BR CPI Index	log	RW	—	dom.
LR_BR	BR Unemployment Rate	lin	RW	—	dom.
NEER_BR	BR Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_BR	BR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_BR	BR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	BR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	BR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	BR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_BR	BR Credit to NFCs	log	RW	✓	dom.
LENDR_BR	BR Corporate Bank Lending Rate (BCB SGS 20718, % p.a.)	lin	RW	✓	dom.
ROWDEM_BR	BR RoW Demand	lin	RW	✓	row
ROWINF_BR	BR RoW Inflation	lin	RW	✓	row
ROWFIN_BR	BR RoW Fin	lin	RW	✓	row

Table 17: Switzerland (CH): $n = 20$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CH	CH GDP	log	RW	—	dom.
CONS_CH	CH CONS	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
INV_CH	CH INV	log	RW	—	dom.
GOV_CH	CH GOV	log	RW	—	dom.
EXP_CH	CH EXP	log	RW	—	dom.
IMP_CH	CH IMP	log	RW	—	dom.
HICP_CH	CH HICP Index	log	RW	—	dom.
LR_CH	CH Unemployment Rate	lin	RW	—	dom.
NEER_CH	CH NEER	log	RW	✓	dom.
POLRATE_CH	CH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CH	CH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_CH	CH Credit to NFCs	log	RW	✓	dom.
LENDR_CH	CH Corporate Lending Rate (SNB new investment loans, fixed rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CH	CH RoW Demand	lin	RW	✓	row
ROWINF_CH	CH RoW Inflation	lin	RW	✓	row
ROWFIN_CH	CH RoW Fin	lin	RW	✓	row

Table 18: Chile (CL): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CL	CL Real GDP	log	RW	—	dom.
CONS_CL	CL Real Household Consumption	log	RW	—	dom.
INV_CL	CL Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CL	CL Real Government Consumption	log	RW	—	dom.
EXP_CL	CL Real Exports	log	RW	—	dom.
IMP_CL	CL Real Imports	log	RW	—	dom.
CPI_CL	CL CPI Index	log	RW	—	dom.
LR_CL	CL Unemployment Rate	lin	RW	—	dom.
RATE_CL	CL 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CL	CL Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CL	CL 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CL	CL Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CL	CL NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CL	CL Credit to NFCs	log	RW	✓	dom.
ROWDEM_CL	CL RoW Demand	lin	RW	✓	row
ROWINF_CL	CL RoW Inflation	lin	RW	✓	row
ROWFIN_CL	CL RoW Fin	lin	RW	✓	row

Table 19: Colombia (CO): $n = 26$, shipped history from 2005Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CO	CO Real GDP	log	RW	—	dom.
CONS_CO	CO Real Household Consumption	log	RW	—	dom.
INV_CO	CO Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CO	CO Real Government Consumption	log	RW	—	dom.
EXP_CO	CO Real Exports	log	RW	—	dom.
IMP_CO	CO Real Imports	log	RW	—	dom.
CPI_CO	CO CPI Index	log	RW	—	dom.
LR_CO	CO Unemployment Rate	lin	RW	—	dom.
RATE_CO	CO 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CO	CO Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CO	CO 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CO	CO Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CO	CO NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CO	CO Credit to NFCs	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_CO	CO Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CO	CO RoW Demand	lin	RW	✓	row
ROWINF_CO	CO RoW Inflation	lin	RW	✓	row
ROWFIN_CO	CO RoW Fin	lin	RW	✓	row

Table 20: Costa Rica (CR): $n = 20$, shipped history from 2010Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CR	CR Real GDP (Industrial Production proxy)	log	RW	—	dom.
EXP_CR	CR Goods Exports (USD)	log	RW	—	dom.
IMP_CR	CR Goods Imports (USD)	log	RW	—	dom.
CPI_CR	CR CPI Index	log	RW	—	dom.
LR_CR	CR Unemployment Rate	lin	RW	—	dom.
RATE_CR	CR Monetary Policy Rate	lin	RW	✓	dom.
NEER_CR	CR NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CR	CR Credit to NFCs	log	RW	✓	dom.
LENDR_CR	CR Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CR	CR RoW Demand	lin	RW	✓	row
ROWINF_CR	CR RoW Inflation	lin	RW	✓	row
ROWFIN_CR	CR RoW Fin	lin	RW	✓	row

Table 21: Czechia (CZ): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CZ	CZ GDP	log	RW	—	dom.
CONS_CZ	CZ CONS	log	RW	—	dom.
INV_CZ	CZ INV	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
GOV_CZ	CZ GOV	log	RW	—	dom.
EXP_CZ	CZ EXP	log	RW	—	dom.
IMP_CZ	CZ IMP	log	RW	—	dom.
HICP_CZ	CZ HICP Index	log	RW	—	dom.
LR_CZ	CZ Unemployment Rate	lin	RW	—	dom.
YIELD10_CZ	CZ 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_CZ	CZ NEER	log	RW	✓	dom.
POLRATE_CZ	CZ Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CZ	CZ Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CZ Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CZ	CZ Credit to NFCs	log	RW	✓	dom.
LENDR_CZ	CZ NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_CZ	CZ RoW Demand	lin	RW	✓	row
ROWINF_CZ	CZ RoW Inflation	lin	RW	✓	row
ROWFIN_CZ	CZ RoW Fin	lin	RW	✓	row

Table 22: Germany (DE): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DE	DE GDP	log	RW	—	dom.
CONS_DE	DE CONS	log	RW	—	dom.
INV_DE	DE INV	log	RW	—	dom.
GOV_DE	DE GOV	log	RW	—	dom.
EXP_DE	DE EXP	log	RW	—	dom.
IMP_DE	DE IMP	log	RW	—	dom.
HICP_DE	DE HICP Index	log	RW	—	dom.
LR_DE	DE Unemployment Rate	lin	RW	—	dom.
HPI_DE	DE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_DE	DE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
POLRATE_DE	DE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_DE	DE 3M Rate	lin	RW	✓	ext.
ROWEA_GDP_DE	DE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_DE	DE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	DE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_DE	DE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_DE	DE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 23: Denmark (DK): $n = 26$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DK	DK GDP	log	RW	—	dom.
CONS_DK	DK CONS	log	RW	—	dom.
INV_DK	DK INV	log	RW	—	dom.
GOV_DK	DK GOV	log	RW	—	dom.
EXP_DK	DK EXP	log	RW	—	dom.
IMP_DK	DK IMP	log	RW	—	dom.
HICP_DK	DK HICP Index	log	RW	—	dom.
LR_DK	DK Unemployment Rate	lin	RW	—	dom.
RATE_DK	DK 3M Rate	lin	RW	✓	dom.
YIELD10_DK	DK 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_DK	DK NEER	log	RW	✓	dom.
POLRATE_DK	DK Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_DK	DK Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	DK Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	DK Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	DK Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_DK	DK Credit to NFCs	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_DK	DK Corporate Lending Rate (ECB MIR loans to NFCs, new business, % p.a.)	lin	RW	✓	dom.
ROWDEM_DK	DK RoW Demand	lin	RW	✓	row
ROWINF_DK	DK RoW Inflation	lin	RW	✓	row
ROWFIN_DK	DK RoW Fin	lin	RW	✓	row

Table 24: Estonia (EE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EE	EE GDP	log	RW	—	dom.
CONS_EE	EE CONS	log	RW	—	dom.
INV_EE	EE INV	log	RW	—	dom.
GOV_EE	EE GOV	log	RW	—	dom.
EXP_EE	EE EXP	log	RW	—	dom.
IMP_EE	EE IMP	log	RW	—	dom.
HICP_EE	EE HICP Index	log	RW	—	dom.
LR_EE	EE Unemployment Rate	lin	RW	—	dom.
HPI_EE	EE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_EE	EE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_EE	EE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_EE	EE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_EE	EE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_EE	EE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_EE	EE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	EE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EE	EE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_EE	EE ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 25: Spain (ES): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ES	ES GDP	log	RW	—	dom.
CONS_ES	ES CONS	log	RW	—	dom.
INV_ES	ES INV	log	RW	—	dom.
GOV_ES	ES GOV	log	RW	—	dom.
EXP_ES	ES EXP	log	RW	—	dom.
IMP_ES	ES IMP	log	RW	—	dom.
HICP_ES	ES HICP Index	log	RW	—	dom.
LR_ES	ES Unemployment Rate	lin	RW	—	dom.
HPI_ES	ES Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_ES	ES NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_ES	ES 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_ES	ES ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_ES	ES 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_ES	ES Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_ES	ES Rest-of-EA HICP	log	RW	—	row
FREV_GDP	ES Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ES Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_ES	ES Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_ES	ES NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 26: Finland (FI): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FI	FI GDP	log	RW	—	dom.
CONS_FI	FI CONS	log	RW	—	dom.
INV_FI	FI INV	log	RW	—	dom.
GOV_FI	FI GOV	log	RW	—	dom.
EXP_FI	FI EXP	log	RW	—	dom.
IMP_FI	FI IMP	log	RW	—	dom.
HICP_FI	FI HICP Index	log	RW	—	dom.
LR_FI	FI Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_FI	FI Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_FI	FI NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_FI	FI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FI	FI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_FI	FI 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_FI	FI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_FI	FI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FI	FI Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FI	FI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 27: France (FR): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FR	FR GDP	log	RW	—	dom.
CONS_FR	FR CONS	log	RW	—	dom.
INV_FR	FR INV	log	RW	—	dom.
GOV_FR	FR GOV	log	RW	—	dom.
EXP_FR	FR EXP	log	RW	—	dom.
IMP_FR	FR IMP	log	RW	—	dom.
HICP_FR	FR HICP Index	log	RW	—	dom.
LR_FR	FR Unemployment Rate	lin	RW	—	dom.
HPI_FR	FR Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_FR	FR NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_FR	FR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FR	FR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_FR	FR 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_GDP_FR	FR Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_FR	FR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FR	FR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FR	FR NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 28: Greece (GR): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_GR	GR GDP	log	RW	—	dom.
CONS_GR	GR CONS	log	RW	—	dom.
INV_GR	GR INV	log	RW	—	dom.
GOV_GR	GR GOV	log	RW	—	dom.
EXP_GR	GR EXP	log	RW	—	dom.
IMP_GR	GR IMP	log	RW	—	dom.
HICP_GR	GR HICP Index	log	RW	—	dom.
LR_GR	GR Unemployment Rate	lin	RW	—	dom.
NEER_GR	GR NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_GR	GR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_GR	GR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_GR	GR 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_GR	GR Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_GR	GR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	GR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	GR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_GR	GR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_GR	GR NFC Bank Lending Rate (ECB MIR new-business floating, %)	lin	RW	✓	dom.

Table 29: Hungary (HU): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_HU	HU GDP	log	RW	—	dom.
CONS_HU	HU CONS	log	RW	—	dom.
INV_HU	HU INV	log	RW	—	dom.
GOV_HU	HU GOV	log	RW	—	dom.
EXP_HU	HU EXP	log	RW	—	dom.
IMP_HU	HU IMP	log	RW	—	dom.
HICP_HU	HU HICP Index	log	RW	—	dom.
LR_HU	HU Unemployment Rate	lin	RW	—	dom.
YIELD10_HU	HU 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_HU	HU NEER	log	RW	✓	dom.
POLRATE_HU	HU Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_HU	HU Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	HU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	HU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	HU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	HU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_HU	HU Credit to NFCs	log	RW	✓	dom.
LENDR_HU	HU NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_HU	HU RoW Demand	lin	RW	✓	row
ROWINF_HU	HU RoW Inflation	lin	RW	✓	row
ROWFIN_HU	HU RoW Fin	lin	RW	✓	row

Table 30: Indonesia (ID): $n = 24$, shipped history from 2005Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ID	ID Real GDP (SA, constant prices)	log	RW	—	dom.
CONS_ID	ID Real Private Consumption (SA)	log	RW	—	dom.
INV_ID	ID Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_ID	ID Real Government Consumption (SA)	log	RW	—	dom.
EXP_ID	ID Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_ID	ID Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_ID	ID CPI (all items, index)	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
RATE_ID	ID Monetary Policy-related Rate (BI rate)	lin	RW	✓	dom.
NEER_ID	ID Nominal Effective Exchange Rate (BIS broad)	log	RW	✓	dom.
POLRATE_ID	ID Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ID	ID Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	ID Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ID Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ID Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	ID Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_ID	ID Credit to NFCs	log	RW	✓	dom.
LENDR_ID	ID IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_ID	ID RoW Demand	lin	RW	✓	row
ROWINF_ID	ID RoW Inflation	lin	RW	✓	row
ROWFIN_ID	ID RoW Fin	lin	RW	✓	row

Table 31: Ireland (IE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IE	IE GDP	log	RW	—	dom.
CONS_IE	IE CONS	log	RW	—	dom.
INV_IE	IE INV	log	RW	—	dom.
GOV_IE	IE GOV	log	RW	—	dom.
EXP_IE	IE EXP	log	RW	—	dom.
IMP_IE	IE IMP	log	RW	—	dom.
HICP_IE	IE HICP Index	log	RW	—	dom.
LR_IE	IE Unemployment Rate	lin	RW	—	dom.
HPI_IE	IE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_IE	IE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_IE	IE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IE	IE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_IE	IE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_GDP_IE	IE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IE	IE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IE	IE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IE	IE NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 32: Israel (IL): $n = 25$, shipped history from 2000Q1, last observation 2025Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IL	IL Real GDP	log	RW	—	dom.
CONS_IL	IL Real Household Consumption	log	RW	—	dom.
INV_IL	IL Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_IL	IL Real Government Consumption	log	RW	—	dom.
EXP_IL	IL Real Exports	log	RW	—	dom.
IMP_IL	IL Real Imports	log	RW	—	dom.
CPI_IL	IL CPI Index (All Items)	log	RW	—	dom.
LR_IL	IL Unemployment Rate	lin	RW	—	dom.
RATE_IL	IL 3M Treasury Bill Yield	lin	RW	✓	dom.
POLRATE_IL	IL Policy Rate (BoI)	lin	RW	✓	dom.
NEER_IL	IL NEER	log	RW	✓	dom.
HP_IL	IL Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	IL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	IL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	IL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_IL	IL Credit to NFCs	log	RW	✓	dom.
CORP_IL	IL Tel-Bond Corporate Bond Yield (5y govt + Tel-Bond spread, BoI)	lin	RW	✓	dom.
ROWDEM_IL	IL RoW Demand	lin	RW	✓	row
ROWINF_IL	IL RoW Inflation	lin	RW	✓	row
ROWFIN_IL	IL RoW Fin	lin	RW	✓	row

Table 33: India (`IN`): $n = 21$, shipped history from 2011Q4, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
`GDP_IN`	IN Real GDP (chain volume, SA)	`log`	RW	—	dom.
`CONS_IN`	IN Real Household Consumption (chain vol, SA)	`log`	RW	—	dom.
`INV_IN`	IN Real Gross Fixed Capital Formation (chain vol, SA)	`log`	RW	—	dom.
`GOV_IN`	IN Real Government Consumption (chain vol, SA)	`log`	RW	—	dom.
`EXP_IN`	IN Real Exports of Goods & Services (chain vol, SA)	`log`	RW	—	dom.
`IMP_IN`	IN Real Imports of Goods & Services (chain vol, SA)	`log`	RW	—	dom.
`CPI_IN`	IN CPI (all items, index)	`log`	RW	—	dom.
`RATE_IN`	IN Call Money / Immediate Interbank Rate	`lin`	RW	✓	dom.
`RATE3M_IN`	IN Short-term 3-Month Interbank Rate	`lin`	RW	✓	dom.
`YIELD10_IN`	IN Long-term 10Y Government Bond Yield	`lin`	RW	✓	dom.
`NEER_IN`	IN Nominal Effective Exchange Rate (BIS broad, index)	`log`	RW	✓	dom.
`POLRATE_IN`	IN Policy Rate (BIS CBPOL)	`lin`	RW	✓	dom.
`HP_IN`	IN Real Residential Property Price (BIS)	`log`	RW	✓	dom.
`GLB_OIL`	WTI crude oil, WTISPLC q-avg ($/bbl)	`log`	RW	✓	glob.
`GLB_USLR`	US 10Y Treasury yield (%)	`lin`	RW	✓	glob.
`GLB_USBAA`	US BAA corporate yield (%)	`lin`	RW	✓	glob.
`GLB_USEQ`	US equity (S&P 500)	`log`	RW	✓	glob.
`CREDIT_GDP_IN`	IN Credit to NFCs	`log`	RW	✓	dom.
`ROWDEM_IN`	IN RoW Demand	`lin`	RW	✓	row
`ROWINF_IN`	IN RoW Inflation	`lin`	RW	✓	row
`ROWFIN_IN`	IN RoW Fin	`lin`	RW	✓	row

Table 34: Iceland (`IS`): $n = 21$, shipped history from 2000Q2, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
`GDP_IS`	IS Real GDP	`log`	RW	—	dom.
`CONS_IS`	IS Real Private Consumption	`log`	RW	—	dom.
`INV_IS`	IS Real Investment	`log`	RW	—	dom.
`GOV_IS`	IS Real Govt Consumption	`log`	RW	—	dom.
`EXP_IS`	IS Real Exports	`log`	RW	—	dom.
`IMP_IS`	IS Real Imports	`log`	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
CPI_IS	IS CPI Index	log	RW	—	dom.
LR_IS	IS Unemployment Rate	lin	RW	—	dom.
RATE_IS	IS 3-Month Interbank Rate	lin	RW	✓	dom.
NEER_IS	IS NEER	log	RW	✓	dom.
POLRATE_IS	IS Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_IS	IS Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_IS	IS Credit to NFCs	log	RW	✓	dom.
LENDR_IS	IS Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_IS	IS RoW Demand	lin	RW	✓	row
ROWINF_IS	IS RoW Inflation	lin	RW	✓	row
ROWFIN_IS	IS RoW Fin	lin	RW	✓	row

Table 35: Italy (IT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IT	IT GDP	log	RW	—	dom.
CONS_IT	IT CONS	log	RW	—	dom.
INV_IT	IT INV	log	RW	—	dom.
GOV_IT	IT GOV	log	RW	—	dom.
EXP_IT	IT EXP	log	RW	—	dom.
IMP_IT	IT IMP	log	RW	—	dom.
HICP_IT	IT HICP Index	log	RW	—	dom.
LR_IT	IT Unemployment Rate	lin	RW	—	dom.
HPI_IT	IT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_IT	IT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_IT	IT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IT	IT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_IT	IT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_IT	IT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IT	IT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IT Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	IT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IT	IT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IT	IT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 36: Korea (KR): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_KR	KR Real GDP	log	RW	—	dom.
CONS_KR	KR Real Household Consumption	log	RW	—	dom.
INV_KR	KR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_KR	KR Real Government Consumption	log	RW	—	dom.
EXP_KR	KR Real Exports	log	RW	—	dom.
IMP_KR	KR Real Imports	log	RW	—	dom.
CPI_KR	KR CPI (all items, index)	log	RW	—	dom.
LR_KR	KR Unemployment Rate	lin	RW	—	dom.
RATE_KR	KR Money Market Rate (3-month, short-term)	lin	RW	✓	dom.
POLRATE_KR	KR Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_KR	KR Government Bond Yield (long-term)	lin	RW	✓	dom.
HP_KR	KR Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	KR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	KR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	KR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	KR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_KR	KR Credit to NFCs	log	RW	✓	dom.
CORP_KR	KR Corporate Bond Yield (3Y, AA-, BoK ECOS)	lin	RW	✓	dom.
ROWDEM_KR	KR RoW Demand	lin	RW	✓	row
ROWINF_KR	KR RoW Inflation	lin	RW	✓	row
ROWFIN_KR	KR RoW Fin	lin	RW	✓	row

Table 37: Lithuania (LT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LT	LT GDP	log	RW	—	dom.
CONS_LT	LT CONS	log	RW	—	dom.
INV_LT	LT INV	log	RW	—	dom.
GOV_LT	LT GOV	log	RW	—	dom.
EXP_LT	LT EXP	log	RW	—	dom.
IMP_LT	LT IMP	log	RW	—	dom.
HICP_LT	LT HICP Index	log	RW	—	dom.
LR_LT	LT Unemployment Rate	lin	RW	—	dom.
HPI_LT	LT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LT	LT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LT	LT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LT	LT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_LT	LT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LT	LT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LT	LT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LT	LT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_LT	LT ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 38: Luxembourg (LU): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LU	LU GDP	log	RW	—	dom.
CONS_LU	LU CONS	log	RW	—	dom.
INV_LU	LU INV	log	RW	—	dom.
GOV_LU	LU GOV	log	RW	—	dom.
EXP_LU	LU EXP	log	RW	—	dom.
IMP_LU	LU IMP	log	RW	—	dom.
HICP_LU	LU HICP Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_LU	LU Unemployment Rate	lin	RW	—	dom.
HPI_LU	LU Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LU	LU NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LU	LU 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LU	LU ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_LU	LU 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LU	LU Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LU	LU Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LU Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LU	LU Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_LU	LU ECB MIR Bank Lending Rate to NFCs (outstanding amounts, % p.a.)	lin	RW	✓	dom.

Table 39: Latvia (LV): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LV	LV GDP	log	RW	—	dom.
CONS_LV	LV CONS	log	RW	—	dom.
INV_LV	LV INV	log	RW	—	dom.
GOV_LV	LV GOV	log	RW	—	dom.
EXP_LV	LV EXP	log	RW	—	dom.
IMP_LV	LV IMP	log	RW	—	dom.
HICP_LV	LV HICP Index	log	RW	—	dom.
LR_LV	LV Unemployment Rate	lin	RW	—	dom.
HPI_LV	LV Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LV	LV NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LV	LV 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LV	LV ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
RATE_LV	LV 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LV	LV Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LV	LV Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LV Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LV Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LV	LV Credit to NFCs (% of GDP)	lin	RW	✓	dom.

Table 40: Mexico (MX): $n = 26$, shipped history from 2001Q4, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_MX	MX Real GDP (chain volume, SA)	log	RW	—	dom.
CONS_MX	MX Real Household Consumption (chain vol, SA)	log	RW	—	dom.
INV_MX	MX Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_MX	MX Real Government Consumption (chain vol, SA)	log	RW	—	dom.
EXP_MX	MX Real Exports (chain volume, SA)	log	RW	—	dom.
IMP_MX	MX Real Imports (chain volume, SA)	log	RW	—	dom.
CPI_MX	MX CPI (all items, index)	log	RW	—	dom.
LR_MX	MX Unemployment Rate	lin	RW	—	dom.
RATE_MX	MX Money Market Rate (short-term)	lin	RW	✓	dom.
POLRATE_MX	MX Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_MX	MX Government Bond Yield (long-term, S13)	lin	RW	✓	dom.
HP_MX	MX Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_MX	MX Nominal Effective Exchange Rate (index)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	MX Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	MX Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	MX Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
`SFA`	MX Stock-Flow Adjustment (% of GDP, ann.)	`lin`	WN	—	fisc.
`CREDIT_GDP_MX`	MX Credit to NFCs	`log`	RW	✓	dom.
`LENDR_MX`	MX Bank Lending Rate (IMF IFS FILR, % p.a.)	`lin`	RW	✓	dom.
`ROWDEM_MX`	MX RoW Demand	`lin`	RW	✓	row
`ROWINF_MX`	MX RoW Inflation	`lin`	RW	✓	row
`ROWFIN_MX`	MX RoW Fin	`lin`	RW	✓	row

Table 41: Malaysia (`MY`): $n = 21$, shipped history from 2015Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
`GDP_MY`	MY Real GDP (chain volume, SA)	`log`	RW	—	dom.
`CONS_MY`	MY Real Private Consumption (SA)	`log`	RW	—	dom.
`INV_MY`	MY Real Gross Fixed Capital Formation (SA)	`log`	RW	—	dom.
`GOV_MY`	MY Real Government Consumption (SA)	`log`	RW	—	dom.
`EXP_MY`	MY Real Exports of Goods & Services (SA)	`log`	RW	—	dom.
`IMP_MY`	MY Real Imports of Goods & Services (SA)	`log`	RW	—	dom.
`CPI_MY`	MY CPI Index (all items)	`log`	RW	—	dom.
`LR_MY`	MY Unemployment Rate	`lin`	RW	—	dom.
`RATE_MY`	MY Overnight Policy Rate (BNM)	`lin`	RW	✓	dom.
`NEER_MY`	MY Nominal Effective Exchange Rate	`log`	RW	✓	dom.
`POLRATE_MY`	MY Policy Rate (BIS CBPOL)	`lin`	RW	✓	dom.
`HP_MY`	MY Real Residential Property Price (BIS)	`log`	RW	✓	dom.
`GLB_OIL`	WTI crude oil, WTISPLC q-avg ($/bbl)	`log`	RW	✓	glob.
`GLB_USLR`	US 10Y Treasury yield (%)	`lin`	RW	✓	glob.
`GLB_USBAA`	US BAA corporate yield (%)	`lin`	RW	✓	glob.
`GLB_USEQ`	US equity (S&P 500)	`log`	RW	✓	glob.
`CREDIT_GDP_MY`	MY Credit to NFCs	`log`	RW	✓	dom.
`LENDR_MY`	MY IFS Lending Rate (% p.a.)	`lin`	RW	✓	dom.
`ROWDEM_MY`	MY RoW Demand	`lin`	RW	✓	row
`ROWINF_MY`	MY RoW Inflation	`lin`	RW	✓	row
`ROWFIN_MY`	MY RoW Fin	`lin`	RW	✓	row

Table 42: Netherlands (NL): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NL	NL GDP	log	RW	—	dom.
CONS_NL	NL CONS	log	RW	—	dom.
INV_NL	NL INV	log	RW	—	dom.
GOV_NL	NL GOV	log	RW	—	dom.
EXP_NL	NL EXP	log	RW	—	dom.
IMP_NL	NL IMP	log	RW	—	dom.
HICP_NL	NL HICP Index	log	RW	—	dom.
LR_NL	NL Unemployment Rate	lin	RW	—	dom.
HPI_NL	NL Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_NL	NL NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_NL	NL 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_NL	NL ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_NL	NL 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_NL	NL Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_NL	NL Rest-of-EA HICP	log	RW	—	row
FREV_GDP	NL Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_NL	NL Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_NL	NL NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 43: Norway (NO): $n = 24$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NO	NO GDP	log	RW	—	dom.
CONS_NO	NO CONS	log	RW	—	dom.
INV_NO	NO INV	log	RW	—	dom.
GOV_NO	NO GOV	log	RW	—	dom.
EXP_NO	NO EXP	log	RW	—	dom.
IMP_NO	NO IMP	log	RW	—	dom.
HICP_NO	NO HICP Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_NO	NO Unemployment Rate	lin	RW	—	dom.
NEER_NO	NO NEER	log	RW	✓	dom.
POLRATE_NO	NO Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_NO	NO Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	NO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NO	NO Credit to NFCs	log	RW	✓	dom.
LENDR_NO	NO NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_NO	NO RoW Demand	lin	RW	✓	row
ROWINF_NO	NO RoW Inflation	lin	RW	✓	row
ROWFIN_NO	NO RoW Fin	lin	RW	✓	row

Table 44: New Zealand (NZ): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NZ	NZ Real GDP	log	RW	—	dom.
CONS_NZ	NZ Real Consumption	log	RW	—	dom.
INV_NZ	NZ Real Investment	log	RW	—	dom.
GOV_NZ	NZ Real Govt Consumption	log	RW	—	dom.
EXP_NZ	NZ Real Exports	log	RW	—	dom.
IMP_NZ	NZ Real Imports	log	RW	—	dom.
CPI_NZ	NZ CPI Index	log	RW	—	dom.
LR_NZ	NZ Unemployment Rate	lin	RW	—	dom.
RATE_NZ	NZ Policy Rate (OCR)	lin	RW	✓	dom.
YIELD10_NZ	NZ 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_NZ	NZ Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_NZ	NZ NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	NZ Government Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	NZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NZ	NZ Credit to NFCs	log	RW	✓	dom.
LENDR_NZ	NZ business lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_NZ	NZ RoW Demand	lin	RW	✓	row
ROWINF_NZ	NZ RoW Inflation	lin	RW	✓	row
ROWFIN_NZ	NZ RoW Fin	lin	RW	✓	row

Table 45: Poland (PL): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PL	PL GDP	log	RW	—	dom.
CONS_PL	PL CONS	log	RW	—	dom.
INV_PL	PL INV	log	RW	—	dom.
GOV_PL	PL GOV	log	RW	—	dom.
EXP_PL	PL EXP	log	RW	—	dom.
IMP_PL	PL IMP	log	RW	—	dom.
HICP_PL	PL HICP Index	log	RW	—	dom.
LR_PL	PL Unemployment Rate	lin	RW	—	dom.
YIELD10_PL	PL 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_PL	PL NEER	log	RW	✓	dom.
POLRATE_PL	PL Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_PL	PL Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	PL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	PL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	PL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_PL	PL Credit to NFCs	log	RW	✓	dom.
LENDR_PL	PL NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_PL	PL RoW Demand	lin	RW	✓	row
ROWINF_PL	PL RoW Inflation	lin	RW	✓	row
ROWFIN_PL	PL RoW Fin	lin	RW	✓	row

Table 46: Portugal (PT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PT	PT GDP	log	RW	—	dom.
CONS_PT	PT CONS	log	RW	—	dom.
INV_PT	PT INV	log	RW	—	dom.
GOV_PT	PT GOV	log	RW	—	dom.
EXP_PT	PT EXP	log	RW	—	dom.
IMP_PT	PT IMP	log	RW	—	dom.
HICP_PT	PT HICP Index	log	RW	—	dom.
LR_PT	PT Unemployment Rate	lin	RW	—	dom.
HPI_PT	PT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_PT	PT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_PT	PT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_PT	PT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_PT	PT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_PT	PT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_PT	PT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	PT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_PT	PT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_PT	PT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 47: Russia (RU): $n = 17$, shipped history from 2003Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_RU	RU Real GDP (constant 2010 LCU, SA)	log	RW	—	dom.
CPI_RU	RU CPI Index (all-items)	log	RW	—	dom.
EXP_RU	RU Goods Exports (FOB, USD)	log	RW	—	dom.
LR_RU	RU Unemployment Rate	lin	RW	—	dom.
RATE_RU	RU Interbank Call-Money Rate	lin	RW	✓	dom.
NEER_RU	RU Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_RU	RU Policy Rate (BIS CBPOL)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
HP_RU	RU Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_RU	RU Credit to NFCs	log	RW	✓	dom.
LENDR_RU	RU Lending Rate (IMF MFS162 spliced w/ CBR loans-to-NFC up-to-1y, % p.a.)	lin	RW	✓	dom.
ROWDEM_RU	RU RoW Demand	lin	RW	✓	row
ROWINF_RU	RU RoW Inflation	lin	RW	✓	row
ROWFIN_RU	RU RoW Fin	lin	RW	✓	row

Table 48: Saudi Arabia (SA): $n = 18$, shipped history from 2010Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SA	SA Real GDP	log	RW	—	dom.
CONS_SA	SA Real Household Consumption	log	RW	—	dom.
INV_SA	SA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_SA	SA Real Government Consumption	log	RW	—	dom.
EXP_SA	SA Real Exports	log	RW	—	dom.
IMP_SA	SA Real Imports	log	RW	—	dom.
CPI_SA	SA CPI Index	log	RW	—	dom.
LR_SA	SA Unemployment Rate	lin	RW	—	dom.
NEER_SA	SA NEER	log	RW	✓	dom.
POLRATE_SA	SA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_SA	SA Credit to NFCs	log	RW	✓	dom.
ROWDEM_SA	SA RoW Demand	lin	RW	✓	row
ROWINF_SA	SA RoW Inflation	lin	RW	✓	row
ROWFIN_SA	SA RoW Fin	lin	RW	✓	row

Table 49: Sweden (SE): $n = 26$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SE	SE GDP	log	RW	—	dom.
CONS_SE	SE CONS	log	RW	—	dom.
INV_SE	SE INV	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
GOV_SE	SE GOV	log	RW	—	dom.
EXP_SE	SE EXP	log	RW	—	dom.
IMP_SE	SE IMP	log	RW	—	dom.
HICP_SE	SE HICP Index	log	RW	—	dom.
LR_SE	SE Unemployment Rate	lin	RW	—	dom.
RATE_SE	SE 3M Rate	lin	RW	✓	dom.
YIELD10_SE	SE 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_SE	SE NEER	log	RW	✓	dom.
POLRATE_SE	SE Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_SE	SE Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	SE Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	SE Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	SE Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_SE	SE Credit to NFCs	log	RW	✓	dom.
LENDR_SE	SE NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_SE	SE RoW Demand	lin	RW	✓	row
ROWINF_SE	SE RoW Inflation	lin	RW	✓	row
ROWFIN_SE	SE RoW Fin	lin	RW	✓	row

Table 50: Slovenia (SI): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SI	SI GDP	log	RW	—	dom.
CONS_SI	SI CONS	log	RW	—	dom.
INV_SI	SI INV	log	RW	—	dom.
GOV_SI	SI GOV	log	RW	—	dom.
EXP_SI	SI EXP	log	RW	—	dom.
IMP_SI	SI IMP	log	RW	—	dom.
HICP_SI	SI HICP Index	log	RW	—	dom.
LR_SI	SI Unemployment Rate	lin	RW	—	dom.
HPI_SI	SI Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SI	SI NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_SI	SI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
POLRATE_SI	SI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_SI	SI 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SI	SI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SI	SI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SI	SI Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_SI	SI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 51: Slovakia (SK): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SK	SK GDP	log	RW	—	dom.
CONS_SK	SK CONS	log	RW	—	dom.
INV_SK	SK INV	log	RW	—	dom.
GOV_SK	SK GOV	log	RW	—	dom.
EXP_SK	SK EXP	log	RW	—	dom.
IMP_SK	SK IMP	log	RW	—	dom.
HICP_SK	SK HICP Index	log	RW	—	dom.
LR_SK	SK Unemployment Rate	lin	RW	—	dom.
HPI_SK	SK Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SK	SK NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_SK	SK 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_SK	SK ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_SK	SK 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SK	SK Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SK	SK Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SK Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	SK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SK	SK Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_SK	SK ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 52: Thailand (TH): $n = 26$, shipped history from 2003Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TH	TH Real GDP (SA, constant prices)	log	RW	—	dom.
CONS_TH	TH Real Private Consumption (SA)	log	RW	—	dom.
INV_TH	TH Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_TH	TH Real Government Consumption (SA)	log	RW	—	dom.
EXP_TH	TH Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_TH	TH Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_TH	TH CPI (all items, index)	log	RW	—	dom.
LR_TH	TH Unemployment Rate	lin	RW	—	dom.
RATE_TH	TH Policy Rate (BOT)	lin	RW	✓	dom.
YIELD10_TH	TH 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_TH	TH Nominal Effective Exchange Rate (BIS broad)	log	RW	✓	dom.
POLRATE_TH	TH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TH	TH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	TH Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TH Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TH Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TH Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_TH	TH Credit to NFCs	log	RW	✓	dom.
LENDR_TH	TH IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_TH	TH RoW Demand	lin	RW	✓	row
ROWINF_TH	TH RoW Inflation	lin	RW	✓	row

ID	Description	Trans.	Prior	Fin.	Block
ROWFIN_TH	TH RoW Fin	lin	RW	✓	row

Table 53: Türkiye (TR): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TR	TR Real GDP	log	RW	—	dom.
CONS_TR	TR Real Household Consumption	log	RW	—	dom.
INV_TR	TR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_TR	TR Real Government Consumption	log	RW	—	dom.
EXP_TR	TR Real Exports	log	RW	—	dom.
IMP_TR	TR Real Imports	log	RW	—	dom.
CPI_TR	TR CPI (all items, index)	log	RW	—	dom.
LR_TR	TR Unemployment Rate	lin	RW	—	dom.
RATE_TR	TR Monetary Policy-related Rate	lin	RW	✓	dom.
POLRATE_TR	TR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TR	TR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	TR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_TR	TR Credit to NFCs	log	RW	✓	dom.
ROWDEM_TR	TR RoW Demand	lin	RW	✓	row
ROWINF_TR	TR RoW Inflation	lin	RW	✓	row
ROWFIN_TR	TR RoW Fin	lin	RW	✓	row

Table 54: Taiwan Province of China (TW): $n = 20$, shipped history from 1997Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TW	TW Real GDP	log	RW	—	dom.
CONS_TW	TW Real Household Consumption	log	RW	—	dom.
GOV_TW	TW Real Government Consumption	log	RW	—	dom.
INV_TW	TW Real Gross Fixed Capital Formation	log	RW	—	dom.
EXP_TW	TW Real Exports of Goods and Services	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
IMP_TW	TW Real Imports of Goods and Services	log	RW	—	dom.
CPI_TW	TW CPI (all items, index, 2021=100)	log	RW	—	dom.
LR_TW	TW Unemployment Rate	lin	RW	—	dom.
RATE_TW	TW CBC Discount Rate (policy)	lin	RW	✓	dom.
MMRATE_TW	TW Interbank Call-Loan Rate (money market)	lin	RW	✓	dom.
YIELD10_TW	TW 10-Year Government Bond Yield	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_TW	TW Credit to NFCs	log	RW	✓	dom.
LENDR_TW	TW 5-Major-Banks New Loan Rate (CBC EH45 total, % p.a.)	lin	RW	✓	dom.
ROWDEM_TW	TW RoW Demand	lin	RW	✓	row
ROWINF_TW	TW RoW Inflation	lin	RW	✓	row
ROWFIN_TW	TW RoW Fin	lin	RW	✓	row

Table 55: South Africa (ZA): $n = 25$, shipped history from 2002Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ZA	ZA Real GDP	log	RW	—	dom.
CONS_ZA	ZA Real Household Consumption	log	RW	—	dom.
INV_ZA	ZA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_ZA	ZA Real Government Consumption	log	RW	—	dom.
EXP_ZA	ZA Real Exports	log	RW	—	dom.
IMP_ZA	ZA Real Imports	log	RW	—	dom.
CPI_ZA	ZA CPI Index	log	RW	—	dom.
LR_ZA	ZA Unemployment Rate	lin	RW	—	dom.
RATE_ZA	ZA SARB Policy Rate (Repo)	lin	RW	✓	dom.
YIELD10_ZA	ZA 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_ZA	ZA Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_ZA	ZA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ZA	ZA Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	ZA Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ZA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ZA Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
`SFA`	ZA Stock-Flow Adjustment (% of GDP, ann.)	`lin`	WN	—	fisc.
`CREDIT_GDP_ZA`	ZA Credit to NFCs	`log`	RW	✓	dom.
`ROWDEM_ZA`	ZA RoW Demand	`lin`	RW	✓	row
`ROWINF_ZA`	ZA RoW Inflation	`lin`	RW	✓	row
`ROWFIN_ZA`	ZA RoW Fin	`lin`	RW	✓	row

C Glossary

BVAR

Bayesian vector autoregression; here estimated under the hierarchical Minnesota prior of Giannone et al. (2015) (Section 3).

GVAR, bloc, star

The coupled cross-country system; one economy's model; a bloc's trade-weighted foreign aggregate (Section 5).

Endogenous dominant unit / frozen anchor

The US as a full member of the fixed point, with its own stars and weights row, versus the legacy one-shot exogenous US.

Global commons

The four US-determined series (oil, US 10y, US Baa, US equity) carried by every bloc (Section 5.4).

COVID-as-missing

The 2020Q2–Q4 macro cells treated as missing and imputed inside estimation; financial columns stay observed (Section 3.3).

Nowcast (NC)

The model's estimate of a quarter that has happened but is not yet published; always drawn amber (Section 10.2).

Hard / soft pin

An exact conditioning value versus an outside forecast held with trust τ, i.e. measurement variance (13) (Sections 4.2, 4.5).

Drift setter (δc)

The long-run anchor mechanism: an exact constant shift of a variable's companion drift (Section 8).

SFA, EFFI

The stock-flow adjustment (deficit–debt discrepancy) and the effective interest rate on the debt stock (Section 6).

Snowball
The automatic interest–growth term of the debt identity (23).

Coupling artifact, model signature
The precomputed $(I - M)^{-1}$ and the fingerprint that refuses it when stale (Section 5.3).

CRPS, PIT, MCS
The continuous ranked probability score (33); the probability integral transform; the model confidence set (Section 11).

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